

The Mathematics Tower, Volume 3: The Upper Floors

Floors 26–37: From Differential Equations to Abstract Algebra

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On the Basement

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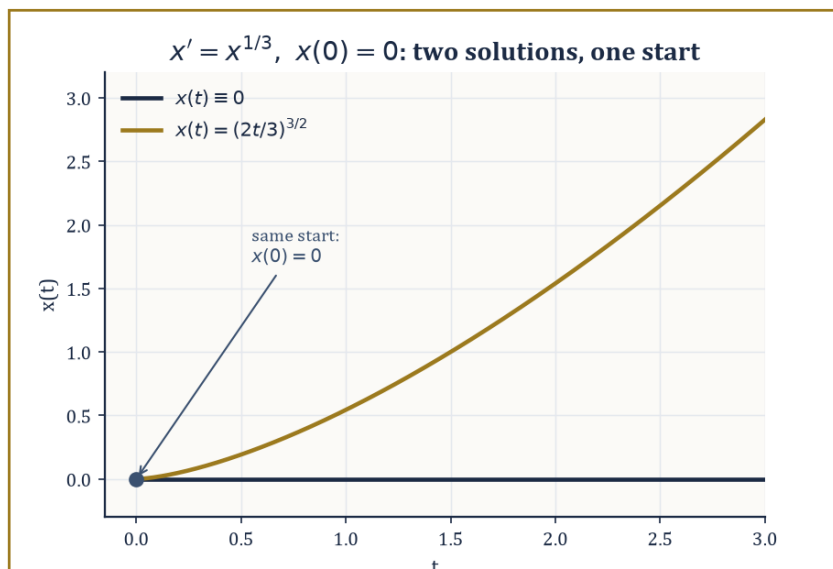
Floor 37 — Abstract Algebra

Floor 26: Differential and Integro-Differential Equations

Beyond the Elementary, and the Art of Equations That Remember the Past

Room 26.1: Ordinary Differential Equations Revisited

In Which the Same Old Problem Turns Out to Have Sharper Teeth



Two different solutions of $x' = x^{1/3}$ leaving the same starting point — Lipschitz continuity is the guard rail that was missing.

Numeric example — a fork in the road.

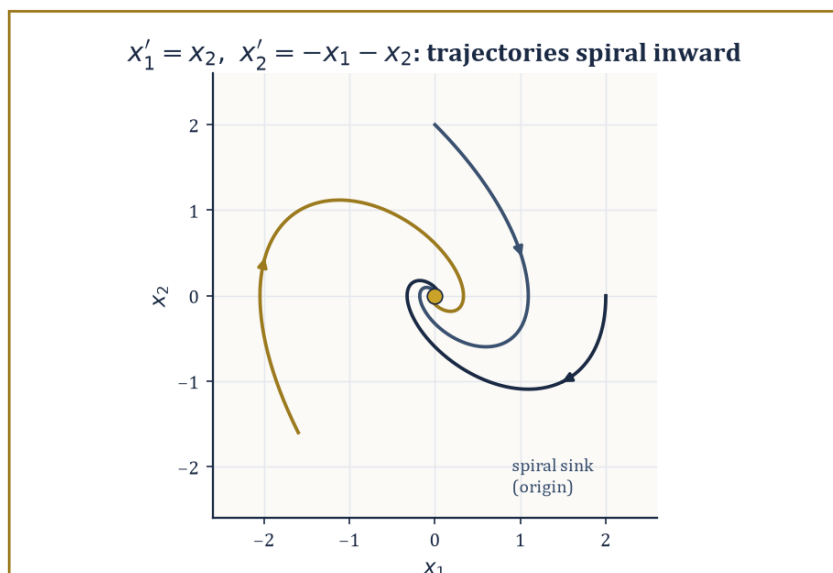
Numeric example — checking Abel's formula.

Numeric example — building an exponential from eigenvectors.

Numeric example — stability by alternation.

Room 26.2: Systems, Stability, and Lyapunov Functions

Energy, Judged Without Being Measured



Three trajectories of the damped spring system spiral into the origin — complex eigenvalues with negative real part, drawn rather than merely asserted.

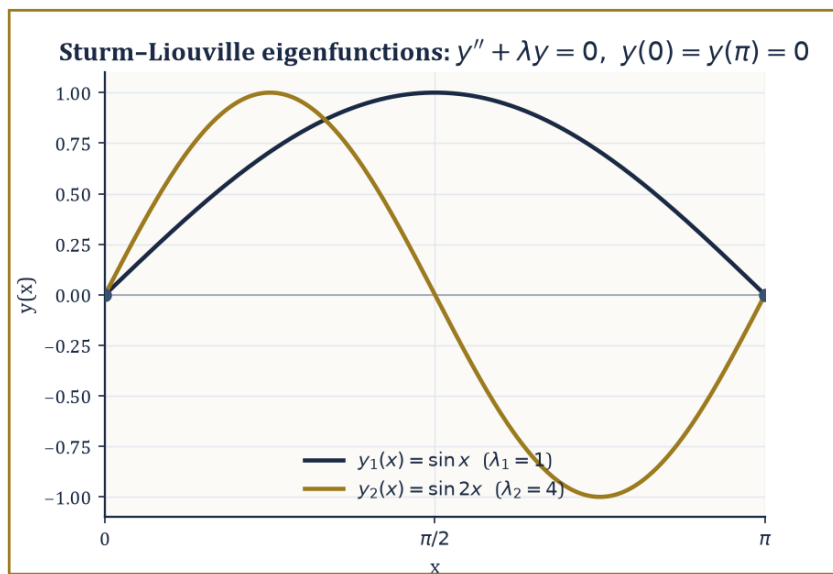
Numeric example — a spiral sink by eigenvalues.

Numeric example — succeeding where linearisation is mute.

Abstract example — the damped pendulum, rescued by LaSalle.

Room 26.3: Boundary-Value Problems and Sturm–Liouville Theory

Conditions at Both Ends, and the Infinite Orthogonal Choir Between Them



The first two eigenfunctions $\sin x$ and $\sin 2x$, pinned to zero at both ends and — as the worked example checks by hand — orthogonal on $[0, \pi]$.

Numeric example — a straightforward boundary-value problem.

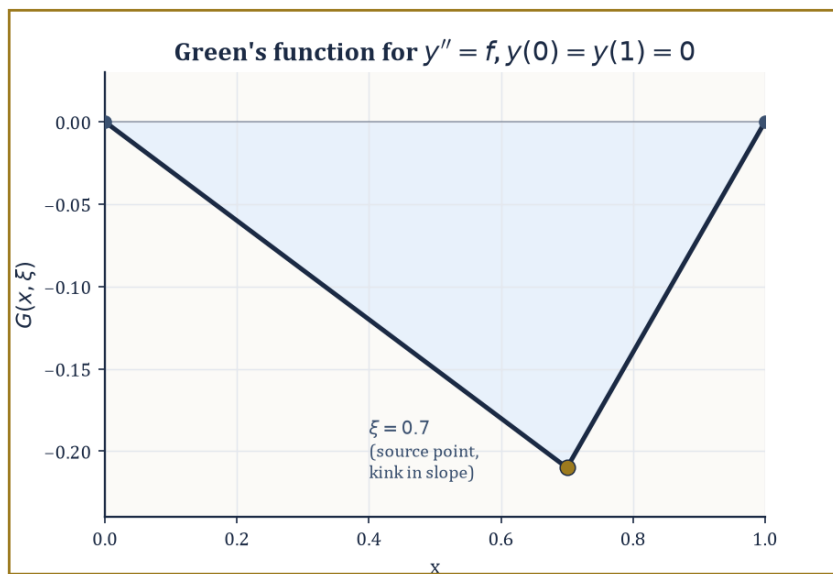
Numeric example — the simplest Sturm–Liouville problem, and orthogonality by hand.

Numeric example — the first Fourier sine coefficient of a sawtooth.

Numeric example — bounding an eigenvalue without solving the equation.

Room 26.4: Green's Functions for ODEs

Delta Functions, Jump Conditions, and the Ghost of a Point Force



The Green's function for $y'' = f$ on $[0, 1]$: continuous everywhere, but with a single polite kink in its slope at the source point ξ .

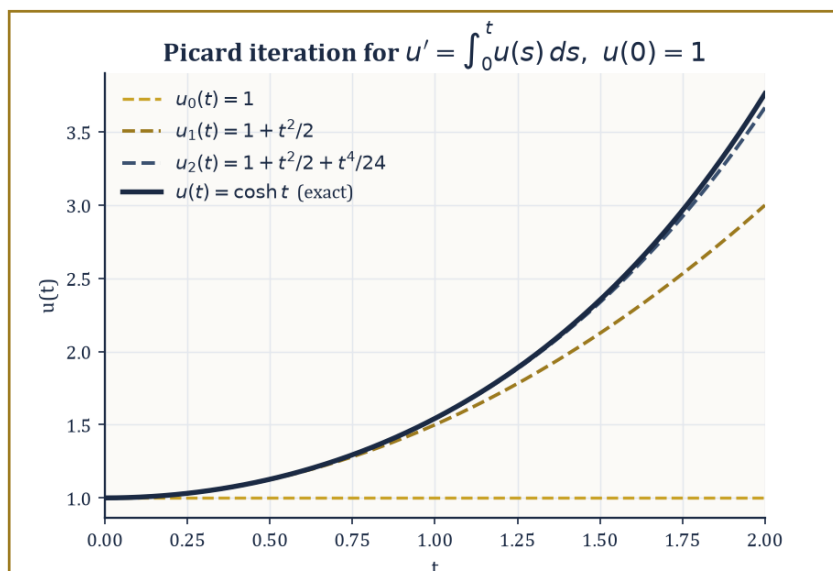
Numeric example — constructing and testing a Green's function.

Abstract example — the pattern for a higher-order operator.

Numeric example — reciprocity, checked with actual numbers.

Room 26.5: What Integro-Differential Equations Are

Equations That Keep a Diary



Successive Picard iterates u_0, u_1, u_2 visibly closing in on the exact solution $\cosh t$, one more term of its Taylor series at a time.

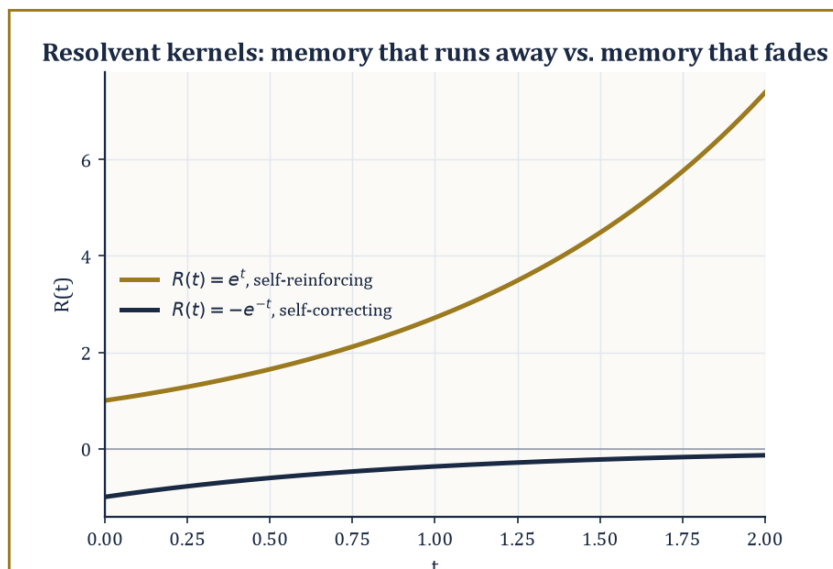
Numeric example — an equation that remembers everything.

Numeric example — Picard iteration building a Taylor series by hand.

Numeric example — the convolution theorem, checked both ways.

Room 26.6: Volterra Integro-Differential Equations

Causality, or: Why the Future Is Not Allowed to Vote



Flip the sign of one kernel and the resolvent stops exploding and starts decaying — the algebraic fingerprint of self-reinforcing versus self-correcting memory.

Numeric example — Laplace transform on a Volterra equation.

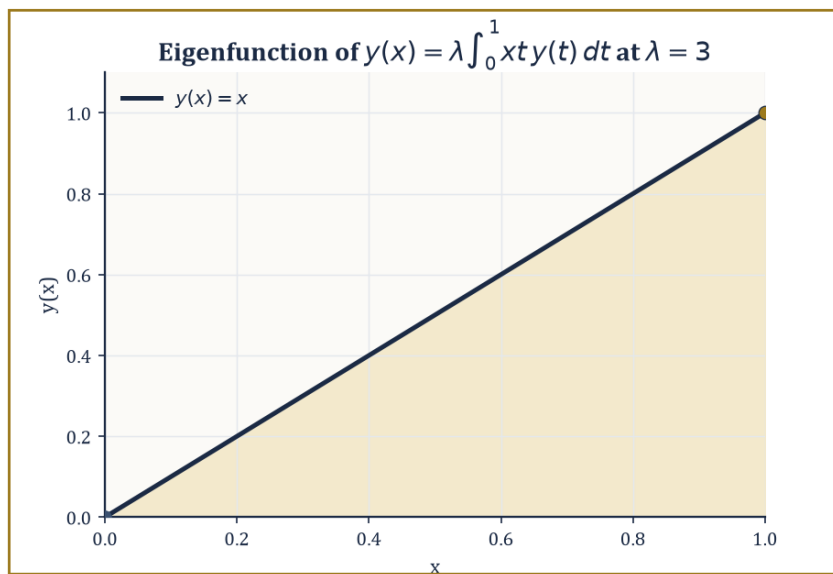
Numeric example — computing a resolvent kernel.

Numeric example — a self-correcting memory.

Worked example — viscoelastic stress relaxation.

Room 26.7: Fredholm Integro-Differential Equations and Eigenvalue Problems

No Causality Required: Every Point Consults Every Other Point at Once



The eigenfunction $y(x) = x$ for the degenerate kernel $K(x,t) = xt$, found by hand and checked to satisfy the equation at $\lambda = 3$.

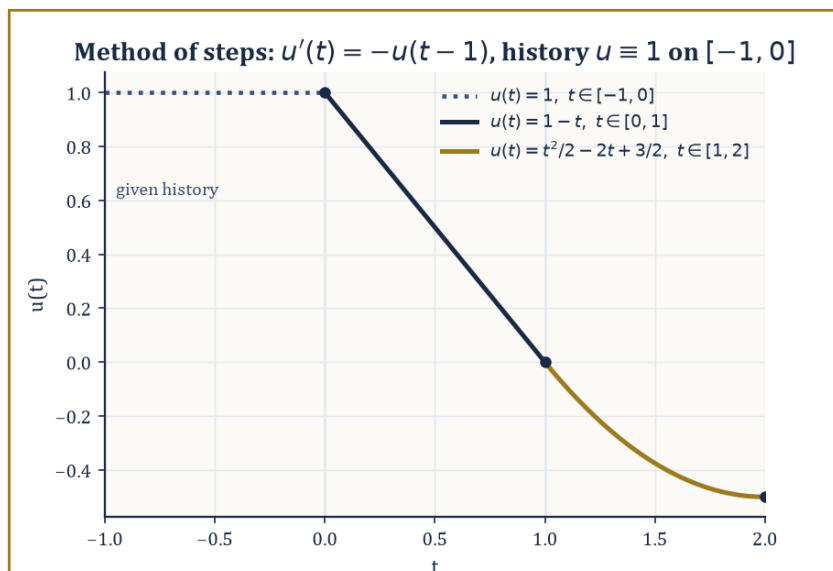
Numeric example — an eigenvalue by hand, degenerate kernel.

Numeric example — the Fredholm alternative at resonance.

Worked example — what the discretised problem looks like.

Room 26.8: Delay Equations and Equations with Memory

The State of the System Is Not a Point, It Is a Recent Autobiography



The method of steps in action: each new piece of the solution is built entirely from the piece before it, one delay-window at a time.

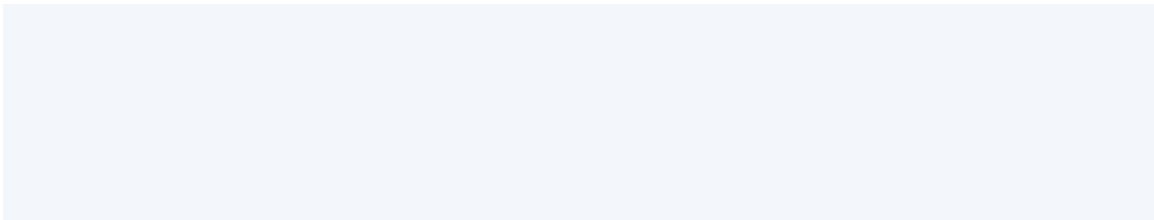
Numeric example — the method of steps.

Numeric example — sitting on the edge of stability.

Numeric example — a distributed delay smoothing a jump.

Room 26.9: Looking Ahead: From Integro-Differential Equations to PDEs and Functional Analysis

Packing for the Next Floor



Abstract example — two histories, one present value.

Numeric example — the price of forgetting how to forget.

Check Your Understanding — Floor 26

Answers are in the Answer Key at the back of the book — have a go before you look.

1.

See the answer →

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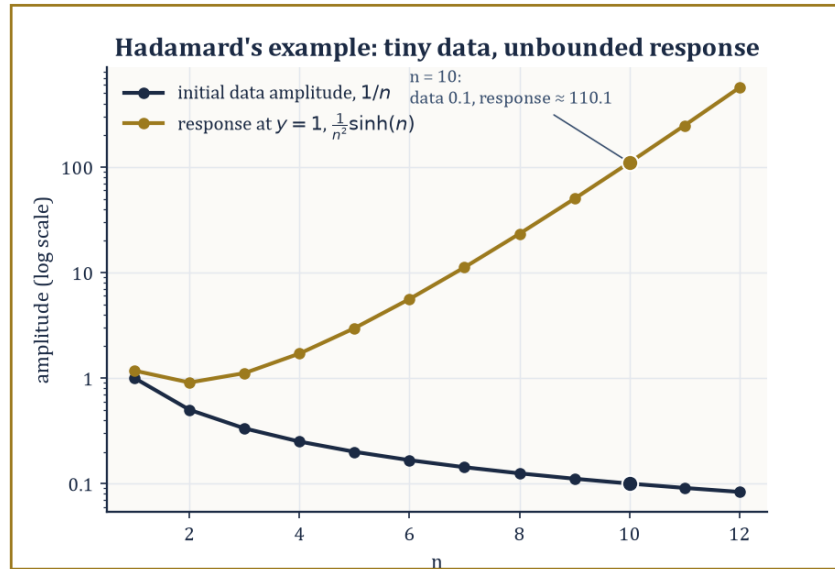
See the answer →

Floor 27: Partial Differential Equations

Waves, Diffusion, Equilibrium, and the Art of Equations in Several Variables

Room 27.1: What a Partial Differential Equation Actually Is

In which the unknown acquires several relatives to answer to



As n grows, Hadamard's initial data shrink to nothing while the corresponding solution explodes — ill-posedness made visible.

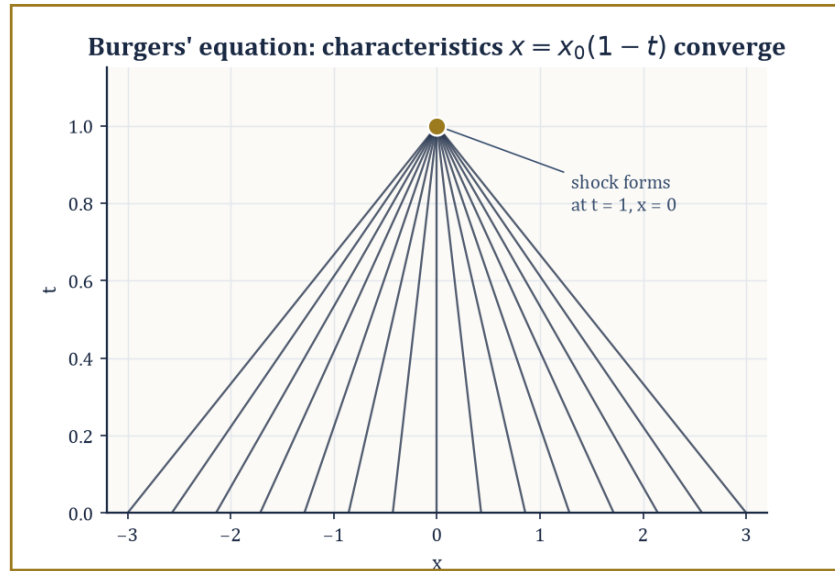
Numeric example — checking a candidate solution.

Numeric example — superposition succeeds, then fails.

Numeric example — Hadamard's ill-posed problem.

Room 27.2: First-Order Equations and the Method of Characteristics

Following information along the roads it travels



Every characteristic of the collapsing profile $u(x,0) = -x$ arrives at the same appointment: $x = 0, t = 1$.

Numeric example — linear transport.

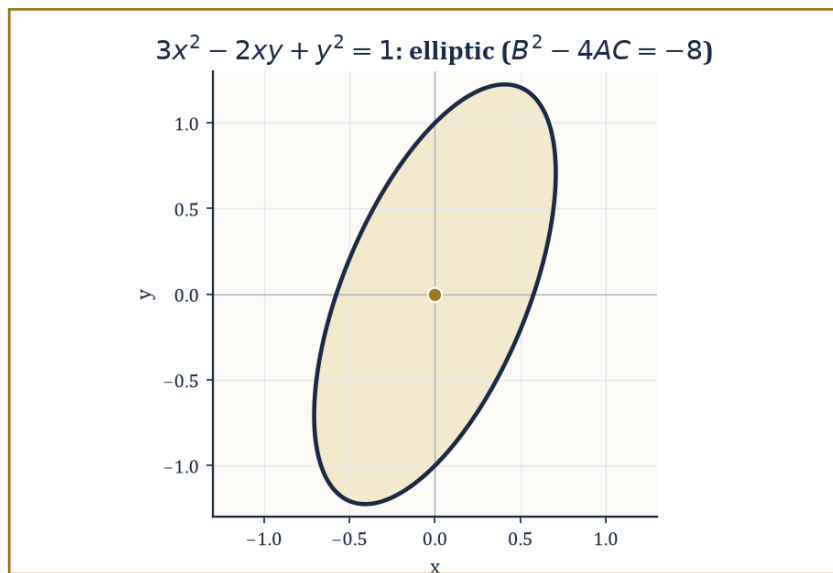
Numeric example — a smooth profile collapsing into a shock.



Numeric example — characteristics with a genuine source term.

Room 27.3: Classification of Second-Order Equations

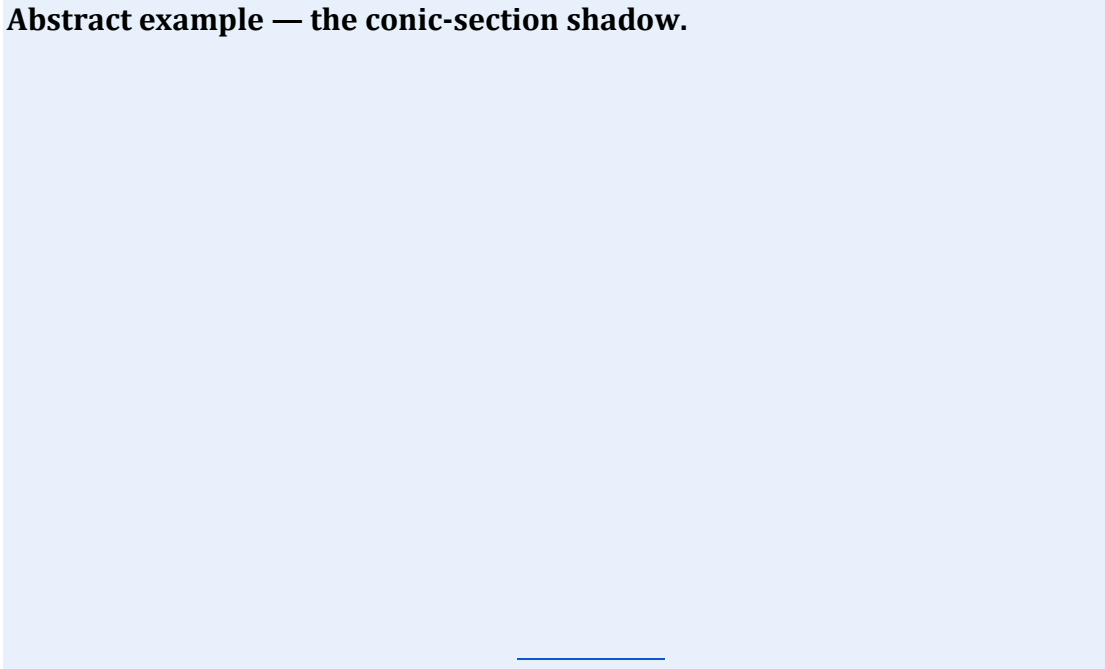
Elliptic, parabolic, hyperbolic — or, conic sections get their revenge



The principal symbol $3x^2 - 2xy + y^2 = 1$ draws an honest ellipse — the geometric fingerprint of an elliptic operator.

Numeric example — classifying the three prototypes.

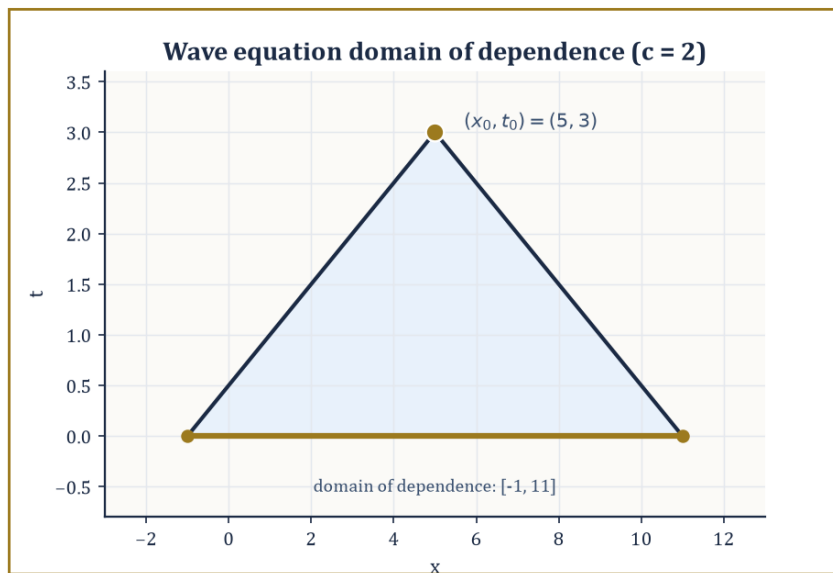
Abstract example — the conic-section shadow.



Numeric example — classification by eigenvalues in higher dimensions.

Room 27.4: The Wave Equation

d'Alembert solves everything twice, once going each way



The backward light cone from $(5, 3)$ touches the initial line at exactly $[-1, 11]$ — everything outside is none of its business.

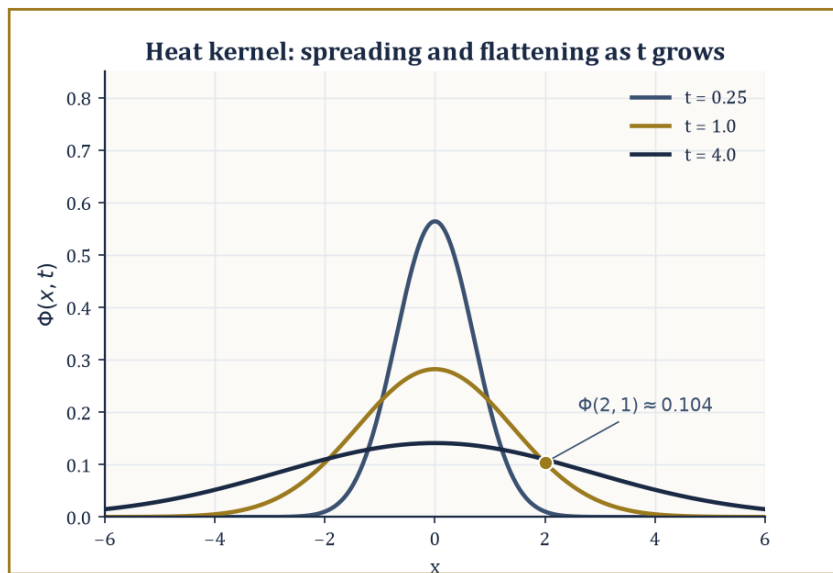
Numeric example — d'Alembert in action, twice.

Numeric example — domains of dependence and influence.

Abstract example — a sharp bang against a lingering ripple.

Room 27.5: The Heat Equation

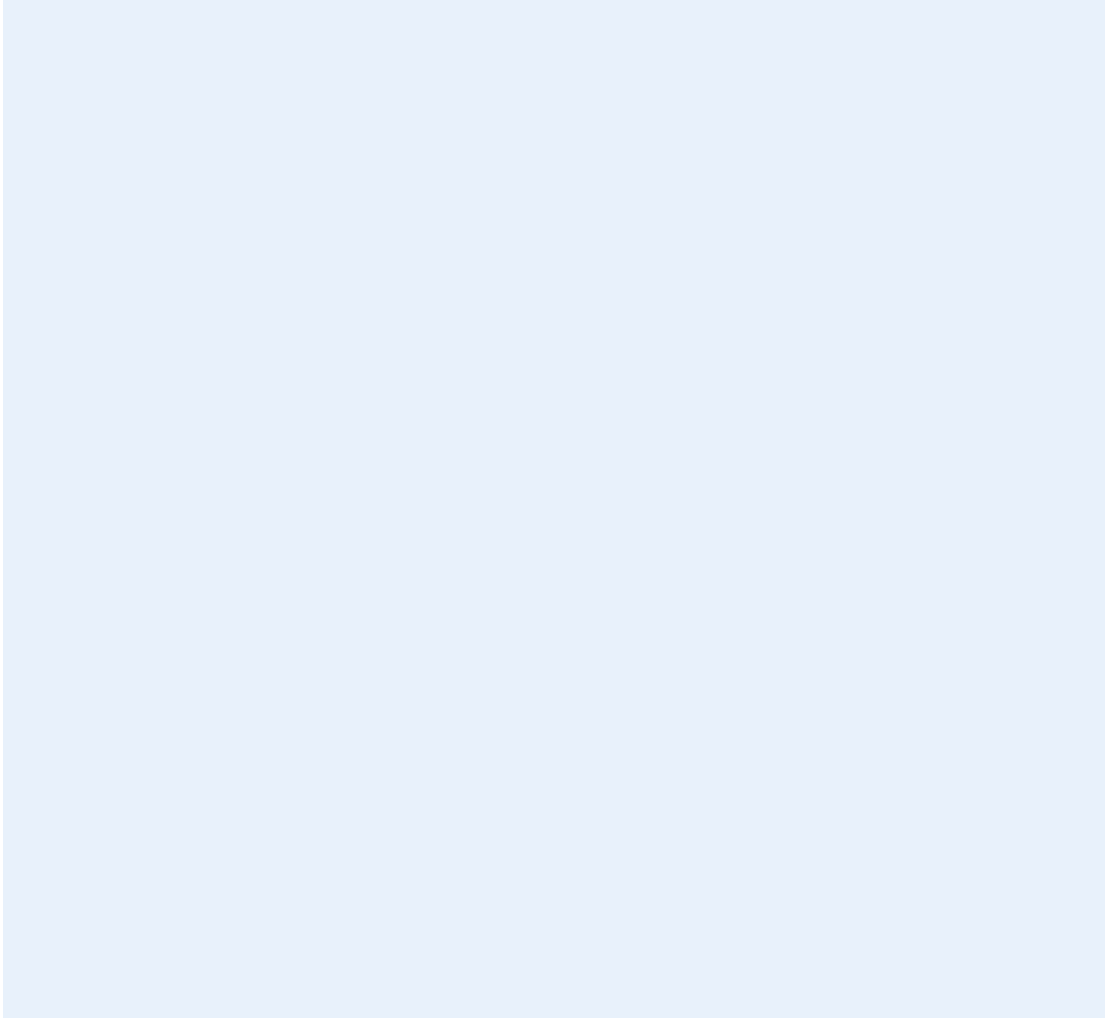
Where yesterday's sharp corners become today's smooth curves



The heat kernel at $t = 0.25, 1$, and 4 — same total area under each curve, ever more thinly spread.

Numeric example — evaluating the fundamental solution.

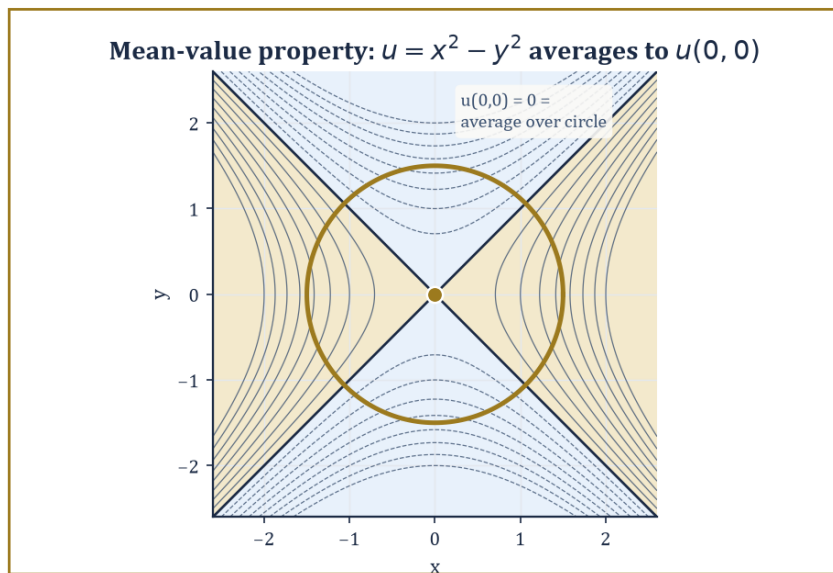
Numeric example — a Gaussian evolving into a wider, flatter Gaussian.



Numeric example — the maximum principle watching temperature decay.

Room 27.6: Laplace's Equation and Harmonic Functions

The mathematics of things that have settled down



The saddle $u = x^2 - y^2$ refuses to disagree with its own neighbourhood: its average over any circle centred at the origin is $u(0,0)$.

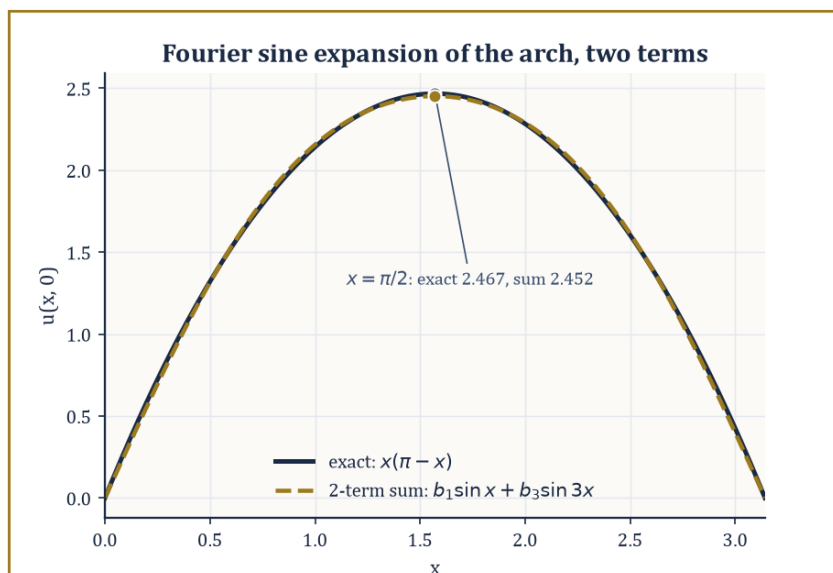
Numeric example — checking harmonicity directly.

Numeric example — the mean-value property by direct integration.

Numeric example — the Poisson kernel and a solved Dirichlet problem.

Room 27.7: Separation of Variables and Eigenfunction Expansions

Divide the problem, then add the pieces back up



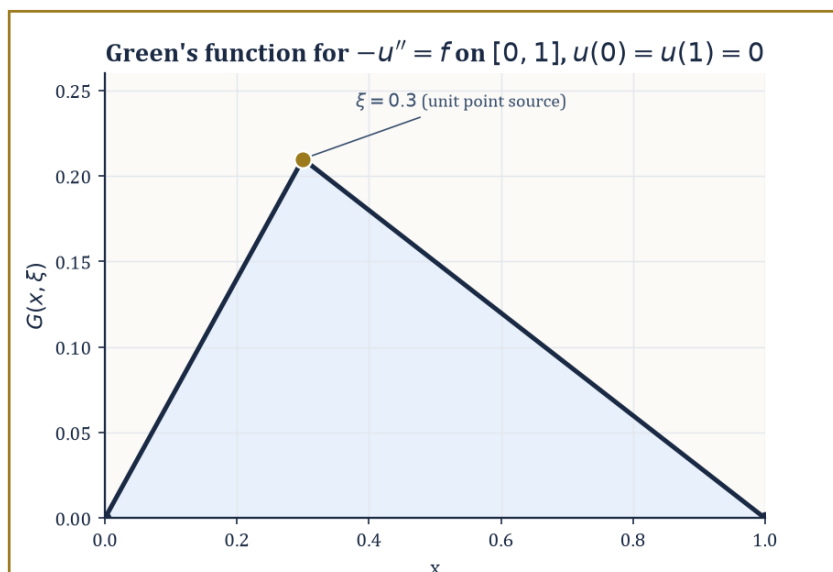
Just two terms of the sine expansion already trace the arch $x(\pi - x)$ to within 0.6% at its peak.

Numeric example — expanding a non-eigenfunction initial profile.

Abstract example — the fundamental note of a circular drum.

Room 27.8: Green's Functions, Maximum Principles, and Energy Methods

How to know a great deal about a solution you will never write down



The Green's function for $-u'' = f$ on $[0,1]$ is a tent pitched at the source point $\xi = 0.3$ — the response to a single unit push.

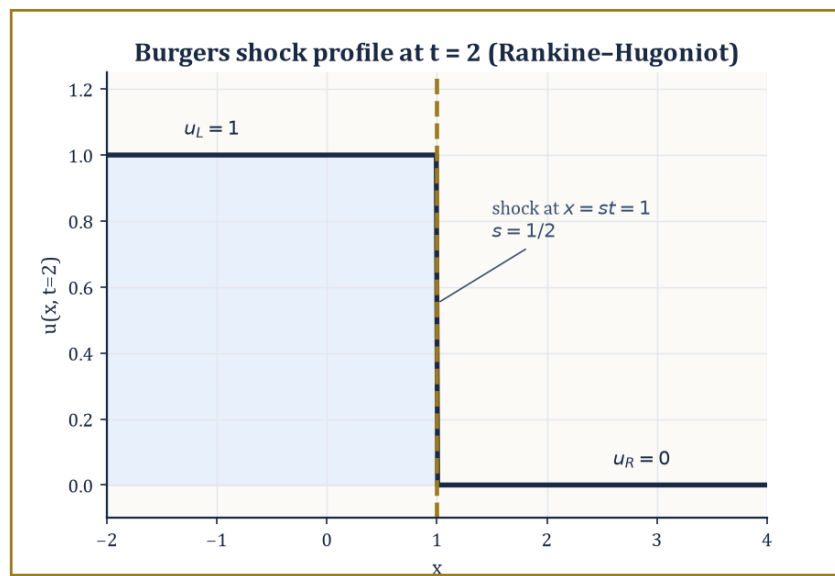
Numeric example — building and using a Green's function.

Numeric example — the maximum principle bounding an interior value.

Abstract example — uniqueness for Poisson's equation by the energy method.

Room 27.9: Looking Ahead: From Classical PDEs to Modern Theory and Applications

Everything above was the warm-up



Rankine-Hugoniot names the price of admission: the jump from $u = 1$ to $u = 0$ must travel at exactly speed $\frac{1}{2}$.

Numeric example — the speed of a shock, via Rankine-Hugoniot.

Numeric example — how large a Reynolds number gets.

Numeric example — one step of a finite-difference scheme, right at the edge of stability.

Check Your Understanding — Floor 27

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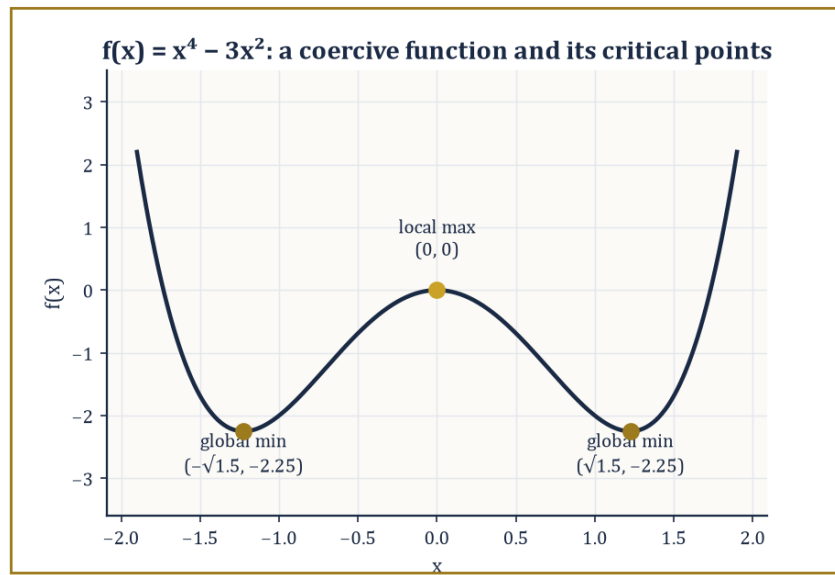
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Floor 28: Optimization & Calculus of Variations

Extrema, Constraints, and the Art of Finding the Best Path or the Best Function

Room 28.1: Finite-Dimensional Optimization: Unconstrained

In Which the Gradient Is Asked, Politely, to Vanish



$f(x) = x^4 - 3x^2$ is coercive: a middling local maximum at the origin flanked by two equally deep global minima, as the second-derivative test predicts.

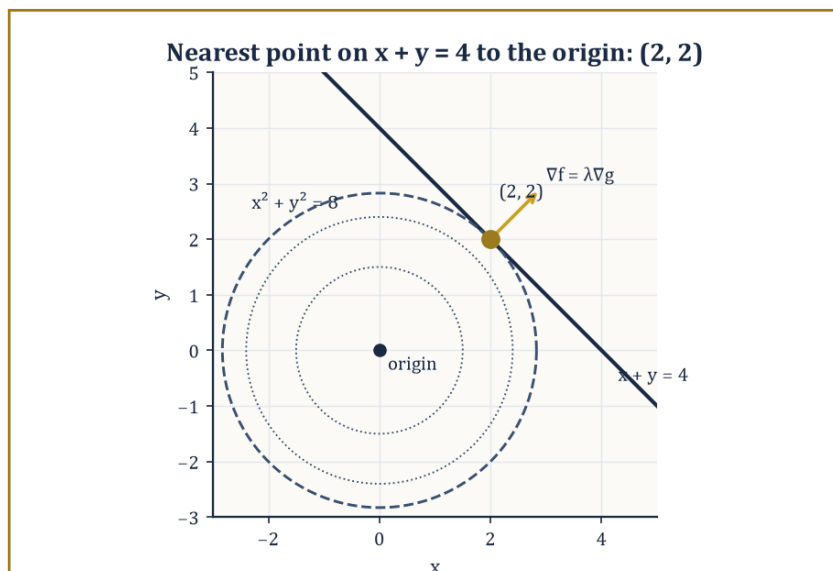
Numeric example — locating and classifying a critical point.

Numeric example — coercivity guarantees a global minimum.

Numeric example — three steps of gradient descent.

Room 28.2: Constraints and Lagrange Multipliers

Optimizing While Someone Is Watching



At the constrained optimum $(2, 2)$, the level circle $x^2 + y^2 = 8$ sits tangent to the line $x + y = 4$ — the geometric meaning of $\nabla f = \lambda \nabla g$.

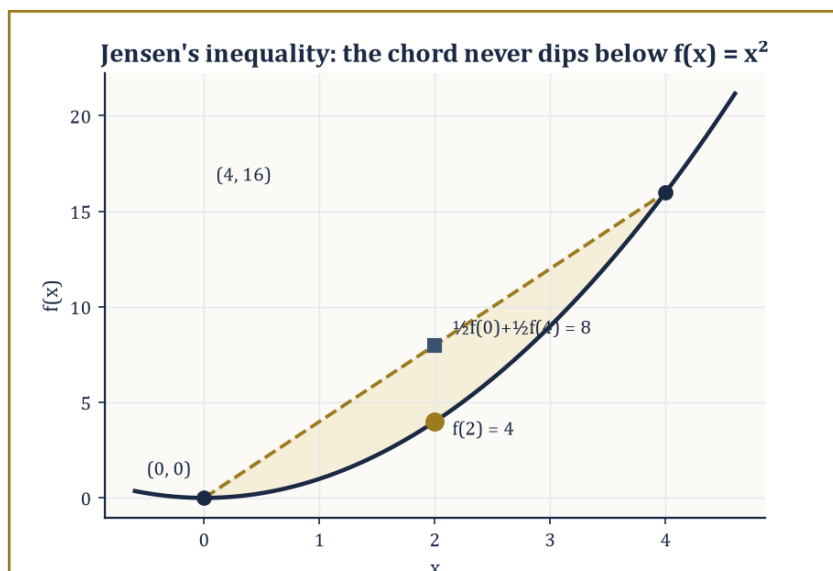
Numeric example — nearest point on a line.

Numeric example — two constraints, two multipliers.

Numeric example — an active inequality constraint.

Room 28.3: Convexity and Convex Optimization

The Dividing Line Between Tractable and Everything Else



Jensen's inequality made visible: the chord from $(0, 0)$ to $(4, 16)$ never dips below the graph of $f(x) = x^2$, with a comfortable gap of 4 to spare at the midpoint.

Numeric example — checking Jensen's inequality.

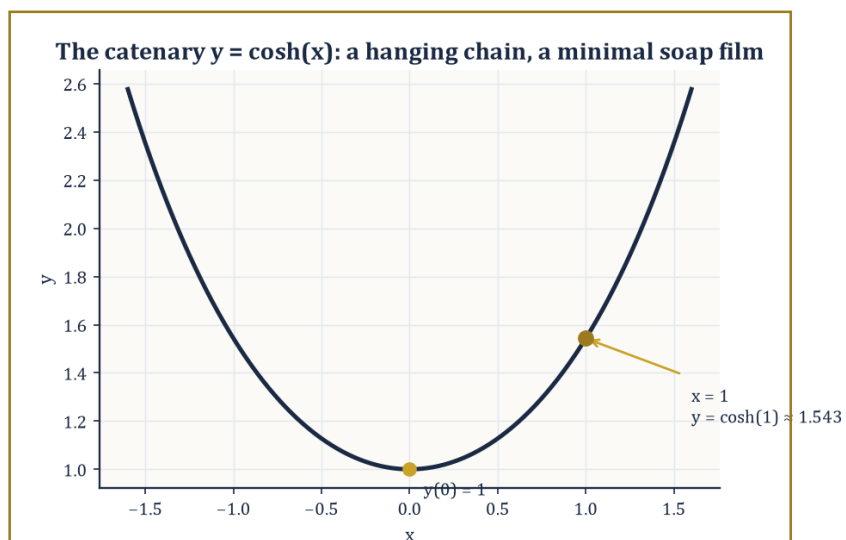
Numeric example — a local minimum that is not global.

Numeric example — a linear program and its dual.

Numeric example — a tie for convex, no tie for strictly convex.

Room 28.4: The Calculus of Variations: The Euler–Lagrange Equation

Where the Gradient Learns to Integrate by Parts



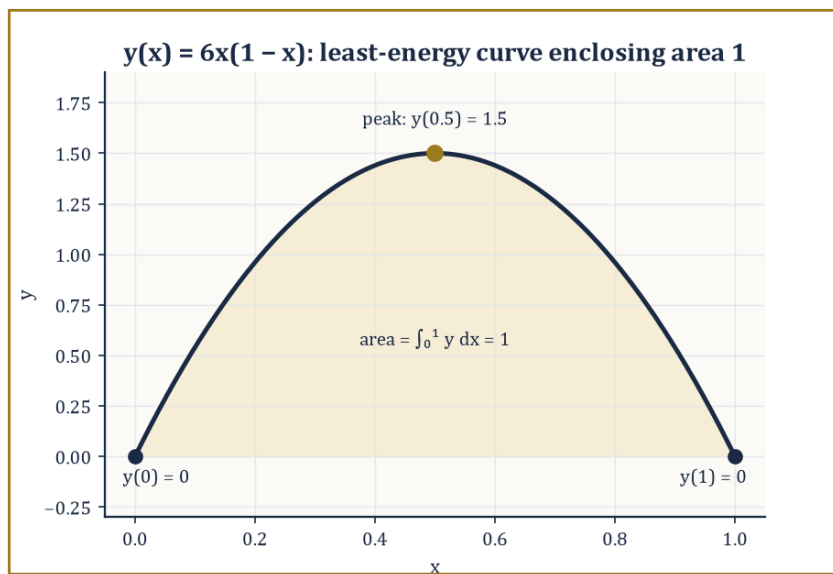
$y = \cosh(x)$: the curve the Beltrami identity hands back the moment you ask a soap film to mind its own curvature.

Numeric example — the shortest path is a straight line.

Abstract example — the Beltrami identity derives the catenary.

Room 28.5: First Integrals, Constraints, and Variable Endpoints

The Same Trick, With More Furniture



The parabola $y = 6x(1-x)$: the unique least-bending curve through $(0,0)$ and $(1,0)$ whose enclosed area is pinned at exactly 1.

Numeric example — minimum energy with a fixed total.

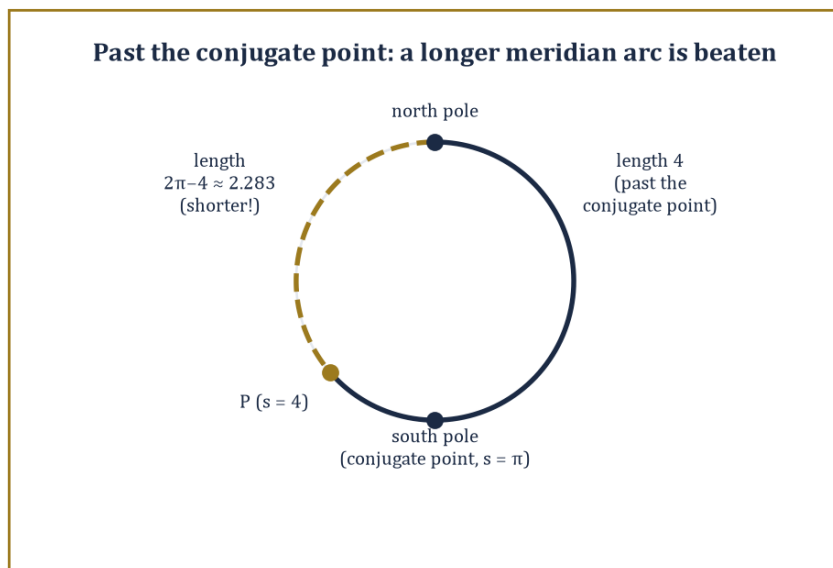
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Numeric example — a free right endpoint.

Abstract example — corner conditions and Snell's law.

Room 28.6: The Second Variation and Sufficient Conditions

Necessary Was Never Going to Be Enough



The length-4 meridian arc has already sailed past its conjugate point at the south pole — the dashed route the short way round is shorter still.

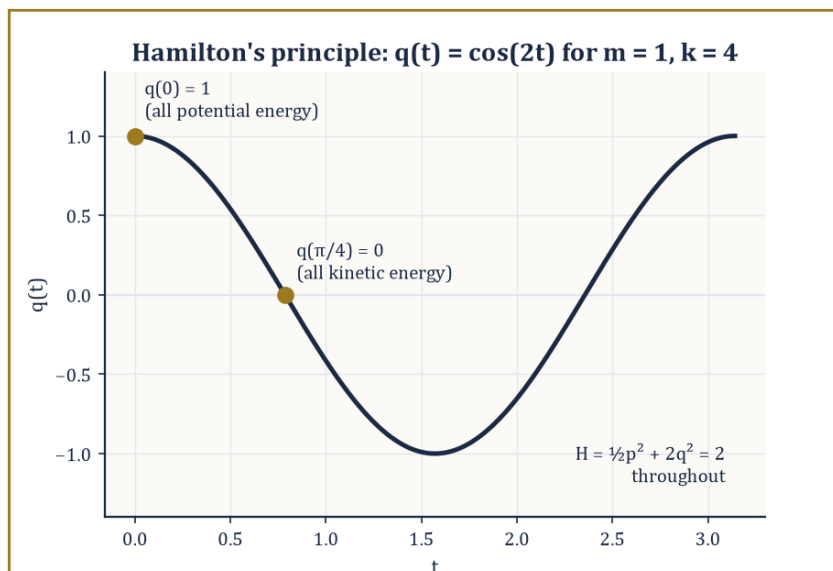
Abstract example — checking the Legendre and Jacobi conditions for a straight line.

Numeric example — conjugate points on the unit sphere.

Abstract example — the excess function for arc length.

Room 28.7: Hamilton's Principle and Classical Mechanics

The Universe, It Turns Out, Is Lazy



$q(t) = \cos(2t)$: the mass swaps potential energy for kinetic and back again, while $H = \frac{1}{2}p^2 + 2q^2$ stands guard at 2 the entire time.

Numeric example — the harmonic oscillator from Hamilton's principle.

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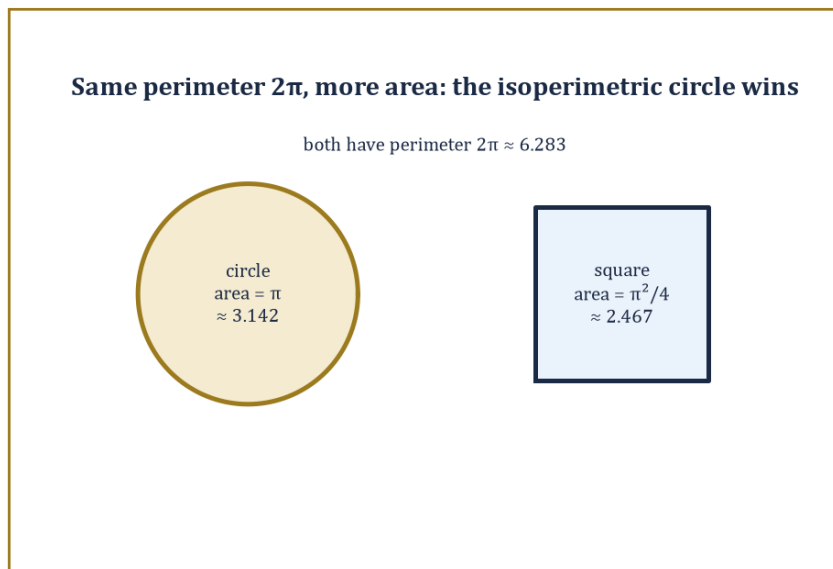
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Numeric example — energy conservation via the Hamiltonian.

Abstract example — translation symmetry conserves total momentum.

Room 28.8: Classical Applications: Geodesics, Minimal Surfaces, Isoperimetry

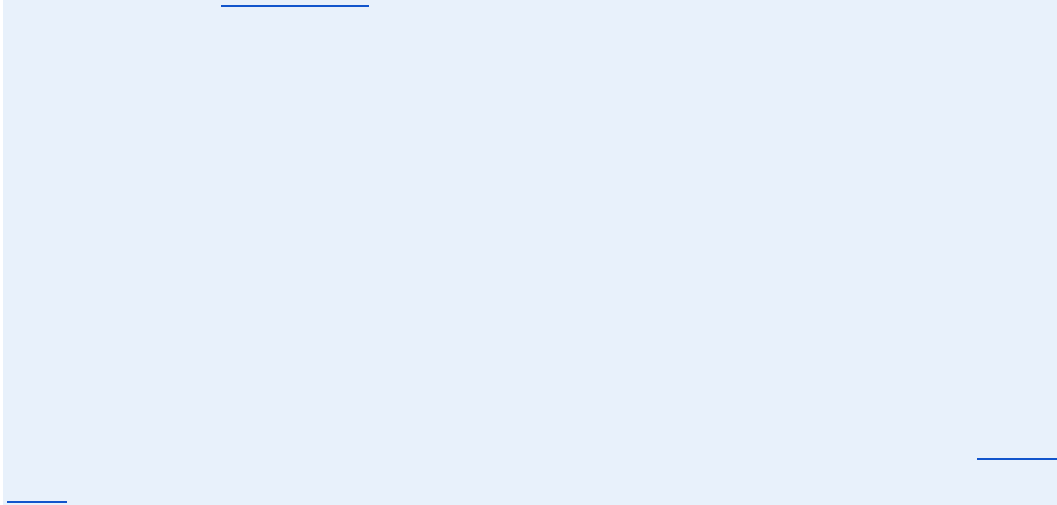
Straight Lines, Soap Films, and the Vindication of the Circle



Same fencing, more pasture: a circle and a square sharing perimeter 2π , with the circle enclosing roughly 27% more area — the isoperimetric inequality in a single picture.

Numeric example — geodesic distance on a cylinder.

Numeric example — checking that the catenoid is genuinely minimal.



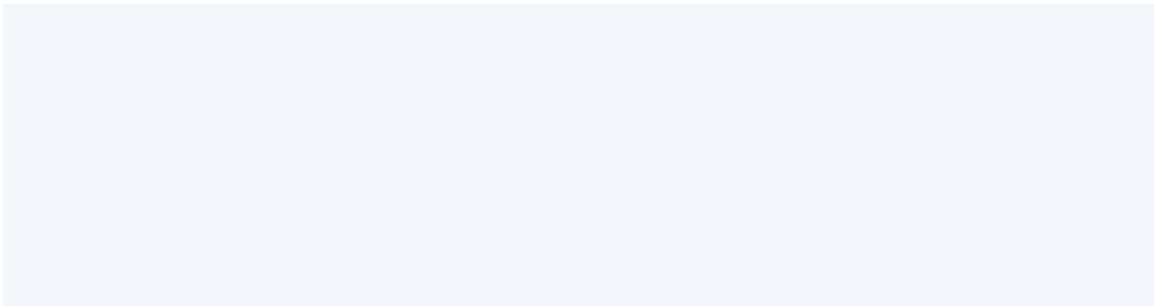
Numeric example — the circle beats the square.

Abstract example — the brachistochrone and the cycloid.

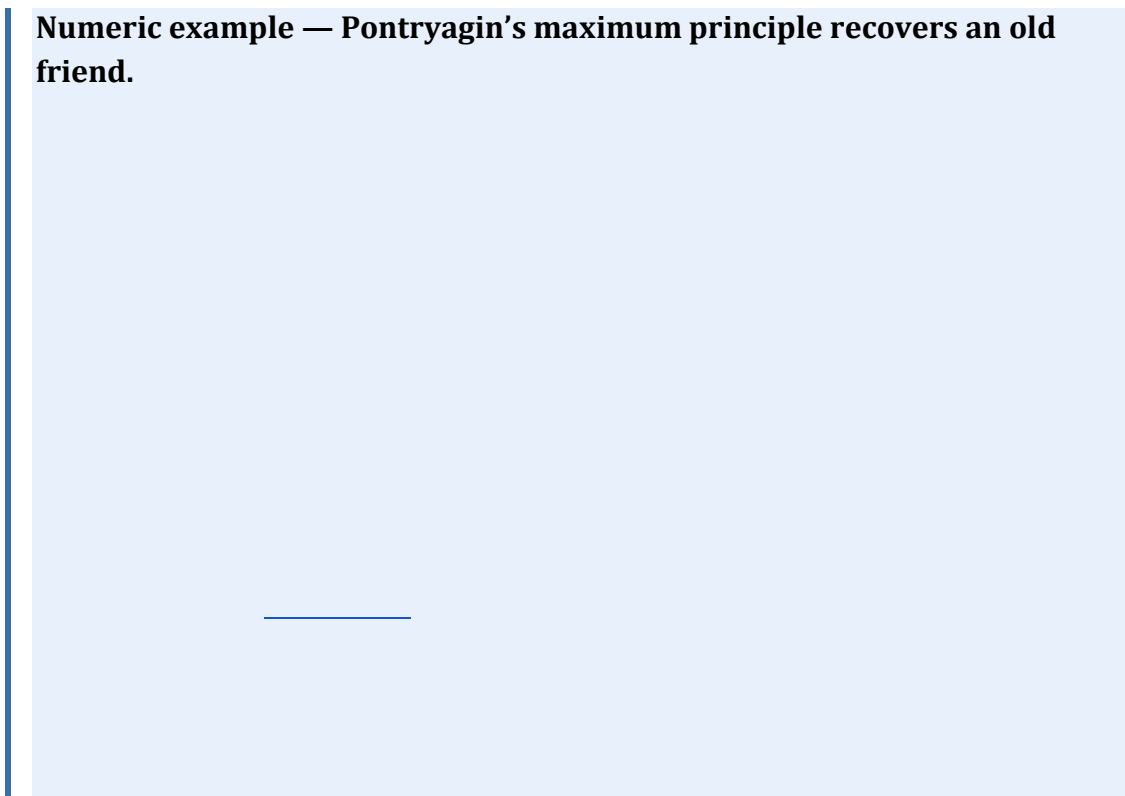
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Room 28.9: Looking Ahead: From Classical Variational Calculus to Optimal Control and Geometric Analysis

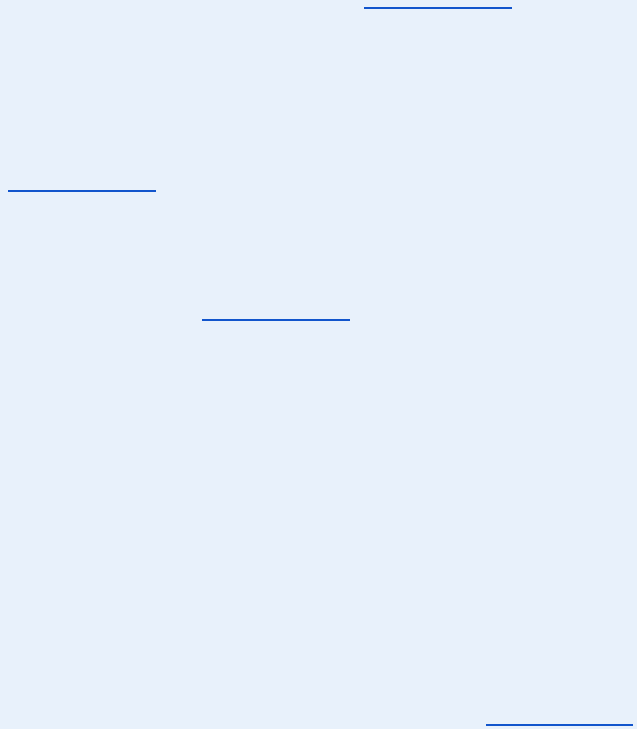
The View From the Top of This Particular Floor



Numeric example — Pontryagin's maximum principle recovers an old friend.



Abstract example — harmonic maps recover two old friends as special cases.



Abstract example — why a minimising sequence can fail to converge to a minimiser.

Check Your Understanding — Floor 28

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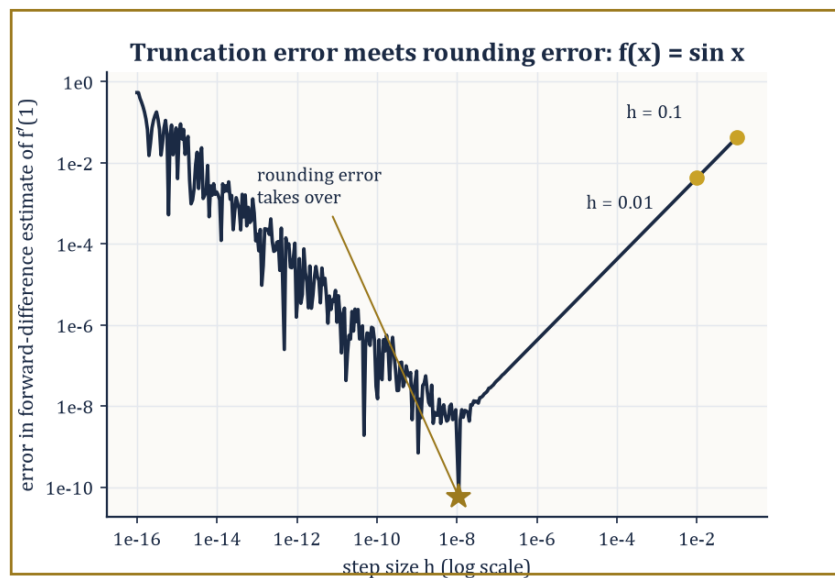
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Floor 29: Numerical Methods

Approximation, Discretization, and the Art of Computing What Cannot Be Solved Exactly

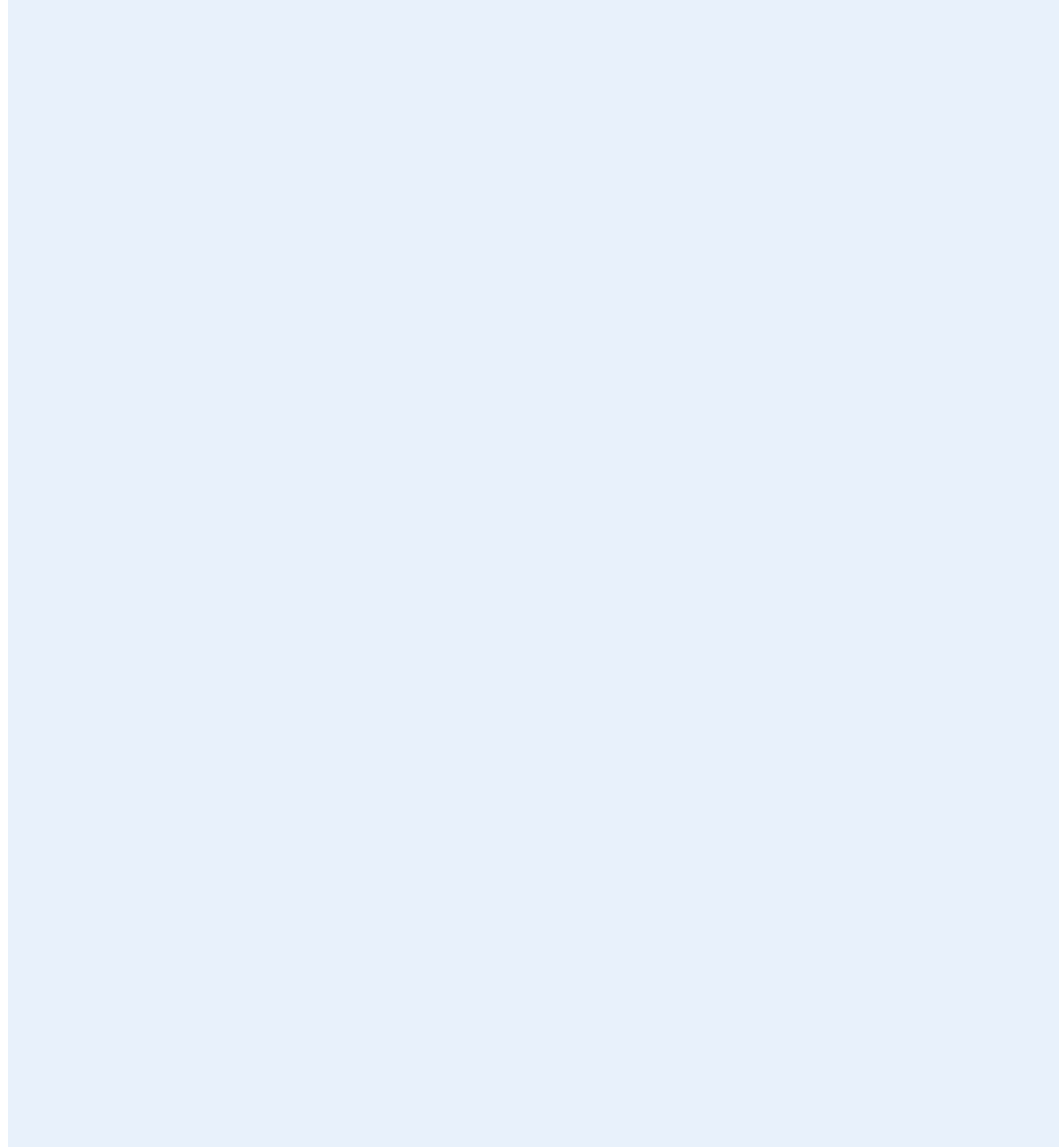
Room 29.1: What Numerical Methods Are

In which the infinite is replaced by the finite, and nobody is fooled about it



The forward-difference error for $f'(1) = \cos(1)$, plotted against step size: truncation error falling on the left, rounding error rising on the right, and the U's floor sitting right where the worked example warned it would.

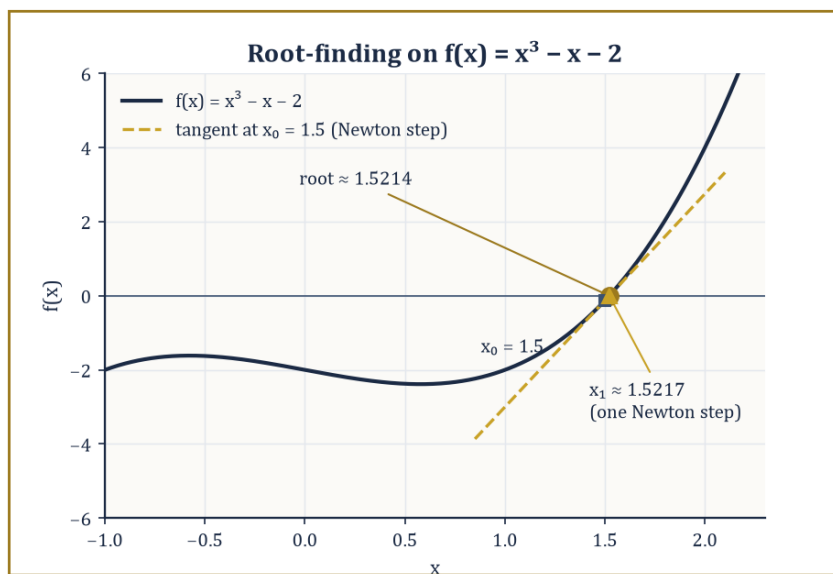
Numeric example — watching truncation error shrink, forward difference.



Numeric example — why the exponent in big-O matters more than the constant.

Room 29.2: Root-Finding and Nonlinear Equations

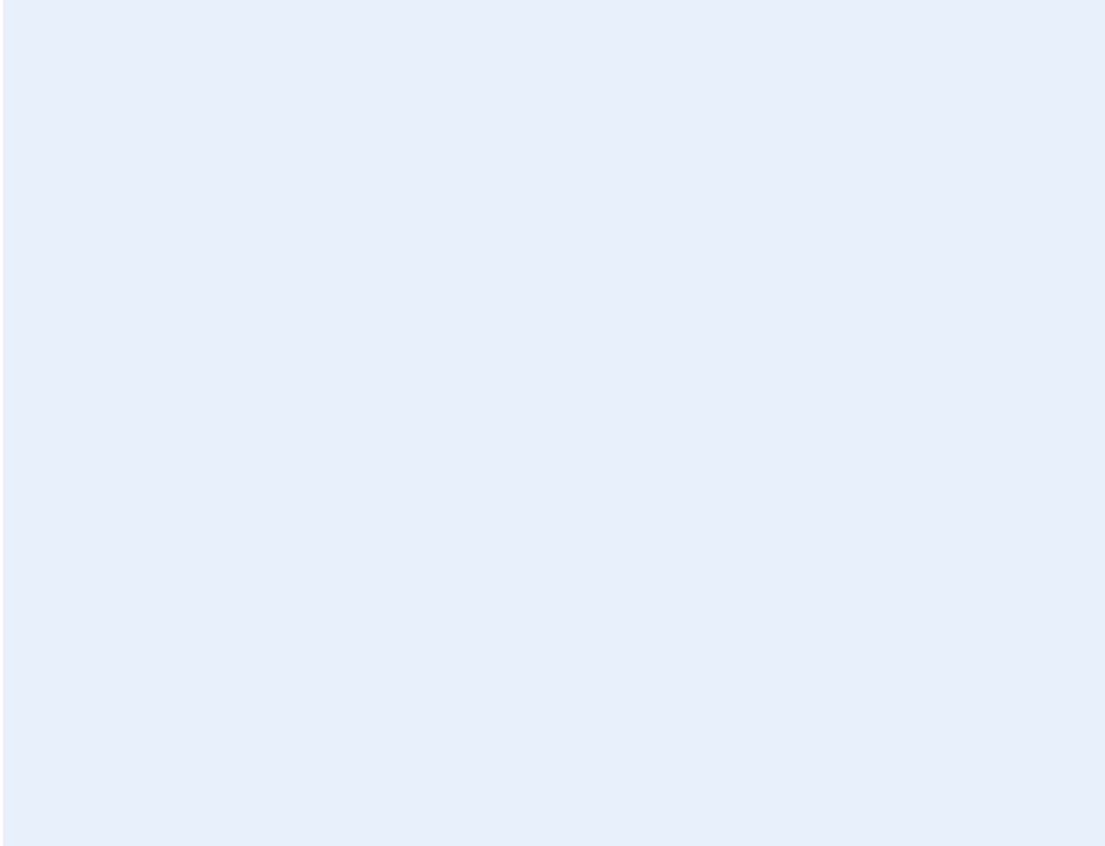
Bisection is slow and honest; Newton is fast and occasionally reckless



$f(x) = x^3 - x - 2$ and its root near 1.5214, with the Newton tangent from $x_0 = 1.5$ landing within 0.0003 of the answer in a single step.

Numeric example — bisection on $x^3 - x - 2$.

Numeric example — Newton on the same cubic — watch the digits double.



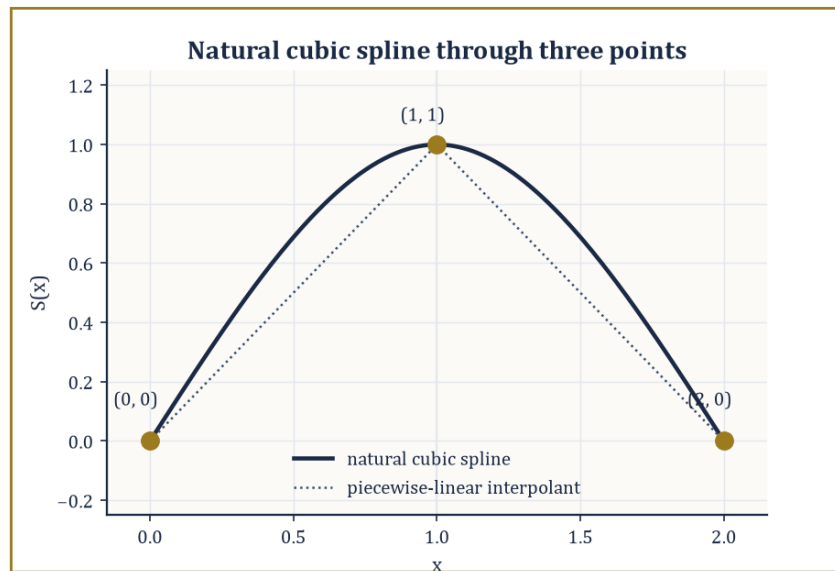
Numeric example — Newton's method on a 2×2 system.

Abstract example — the companion matrix of $x^3 - x - 2$.

Numeric example — Newton gone astray.

Room 29.3: Interpolation, Approximation, and Splines

Threading a curve through the data without letting it misbehave at the edges



The natural cubic spline through $(0,0)$, $(1,1)$, $(2,0)$, rising and falling with no kink at the knot — compared against the plain connect-the-dots path it improves on.

Numeric example — Lagrange interpolation through three points.

✓

✓

✓

Numeric example — five Chebyshev nodes.

Numeric example — least-squares line through four points.

Numeric example — a natural cubic spline through three points.

✓

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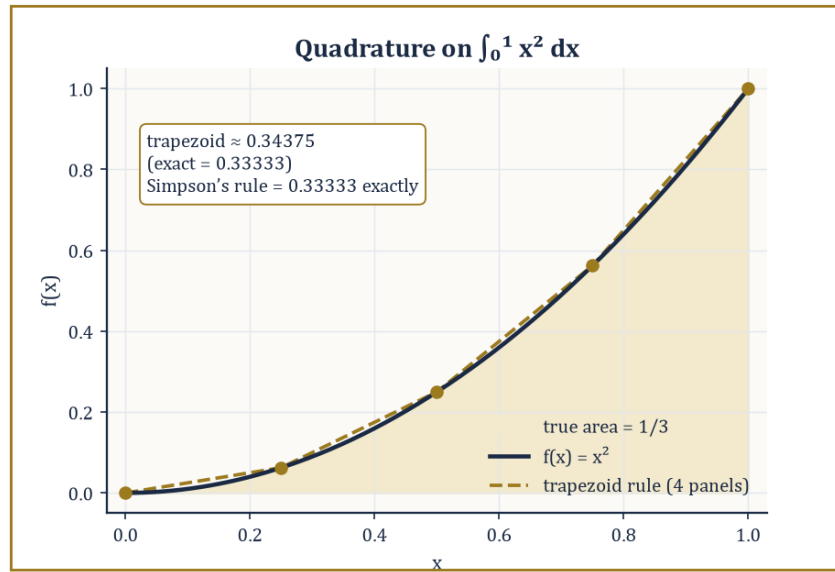
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Abstract example — Bernstein polynomials converging to x^2 .

Room 29.4: Numerical Differentiation and Integration

Differentiation punishes noise; integration, mercifully, forgives it



$\int_0^1 x^2 dx$: the trapezoid rule's straight segments overshoot the curve's true area, while Simpson's parabola-fitting gets the exact answer, $1/3$, from the same five points.

Numeric example — central difference beats forward difference.

Numeric example — Richardson extrapolation on the central difference.

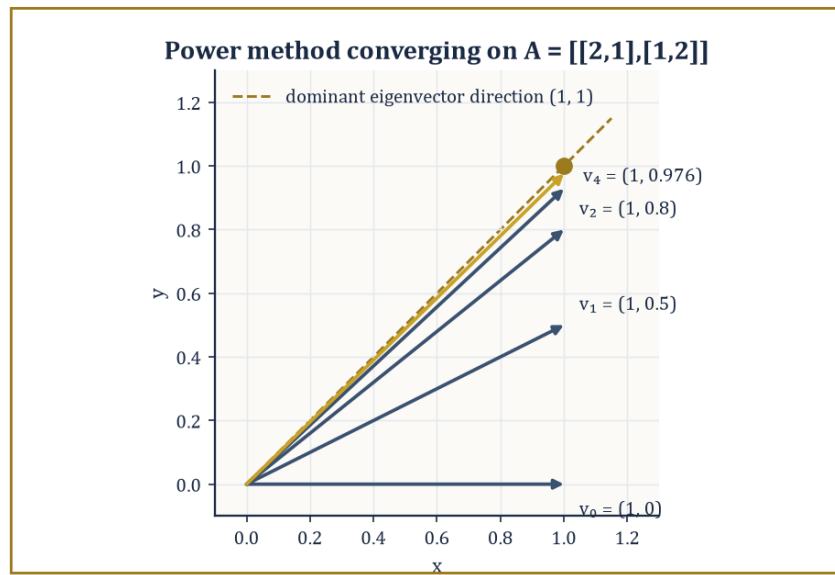
Numeric example — trapezoid vs Simpson on $\int_0^1 x^2 dx$.

Numeric example — two-point Gaussian quadrature.

Numeric example — the trapezoid rule's secret superpower on periodic integrands.

Room 29.5: Numerical Linear Algebra

Solving $Ax = b$, and finding out afterwards how much you should trust the answer



The power method's iterates v_0 through v_4 , started at $(1,0)$ and repeatedly multiplied by $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, swinging steadily into line with the dominant eigenvector direction $(1,1)$.

Numeric example — LU decomposition and a full solve.

✓
✓
✓

Numeric example — an ill-conditioned 2×2 system amplifying a tiny perturbation.

Numeric example — Jacobi iteration converging to a known answer.

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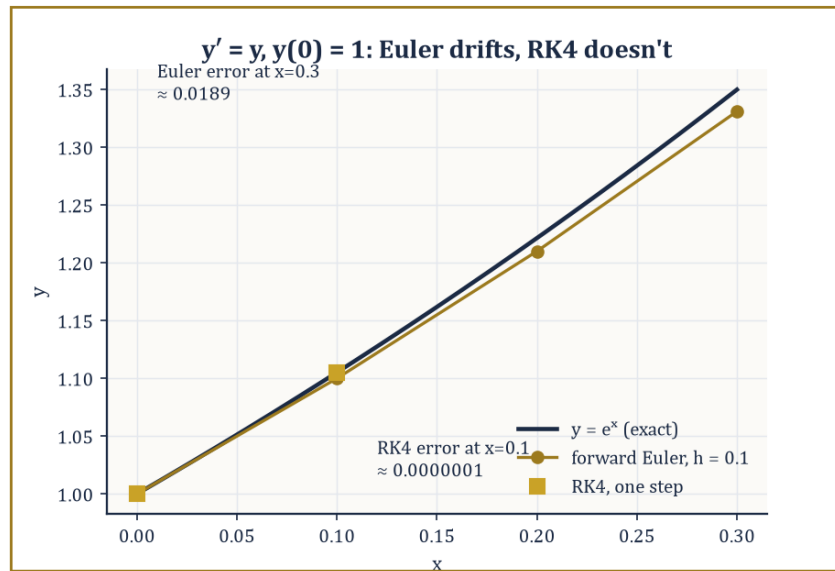
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Numeric example — the power method finding a dominant eigenvalue.

Numeric example — backward stability rescuing an alarming-looking answer.

Room 29.6: Numerical Methods for Ordinary Differential Equations

Marching a solution forward one careful step at a time — and knowing when the steps must be very careful indeed



$y' = y, y(0) = 1$: forward Euler's step points visibly drift below the true curve e^x , while a single RK4 step lands on it to six decimal places.

Numeric example — Euler's method drifting away from e^x .

Numeric example — one step of RK4 on the same problem.

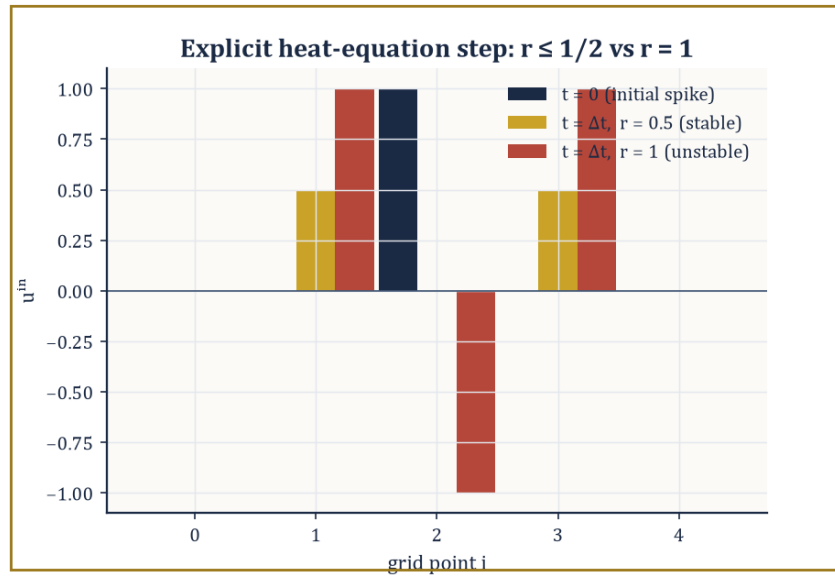
Numeric example — an explicit method blowing up on a stiff equation.

Numeric example — an implicit method handling the same equation gracefully.

Numeric example — adaptive step-size control in action.

Room 29.7: Numerical Methods for Partial Differential Equations

Explicit schemes are cheap and skittish; implicit schemes are expensive and imperturbable



One explicit step on the five-point temperature spike: $r = 0.5$ diffuses it smoothly and symmetrically, while $r = 1$ produces a negative temperature — nonsense that only the mesh ratio can be blamed for.

Numeric example — explicit heat equation, stable vs unstable mesh ratio.

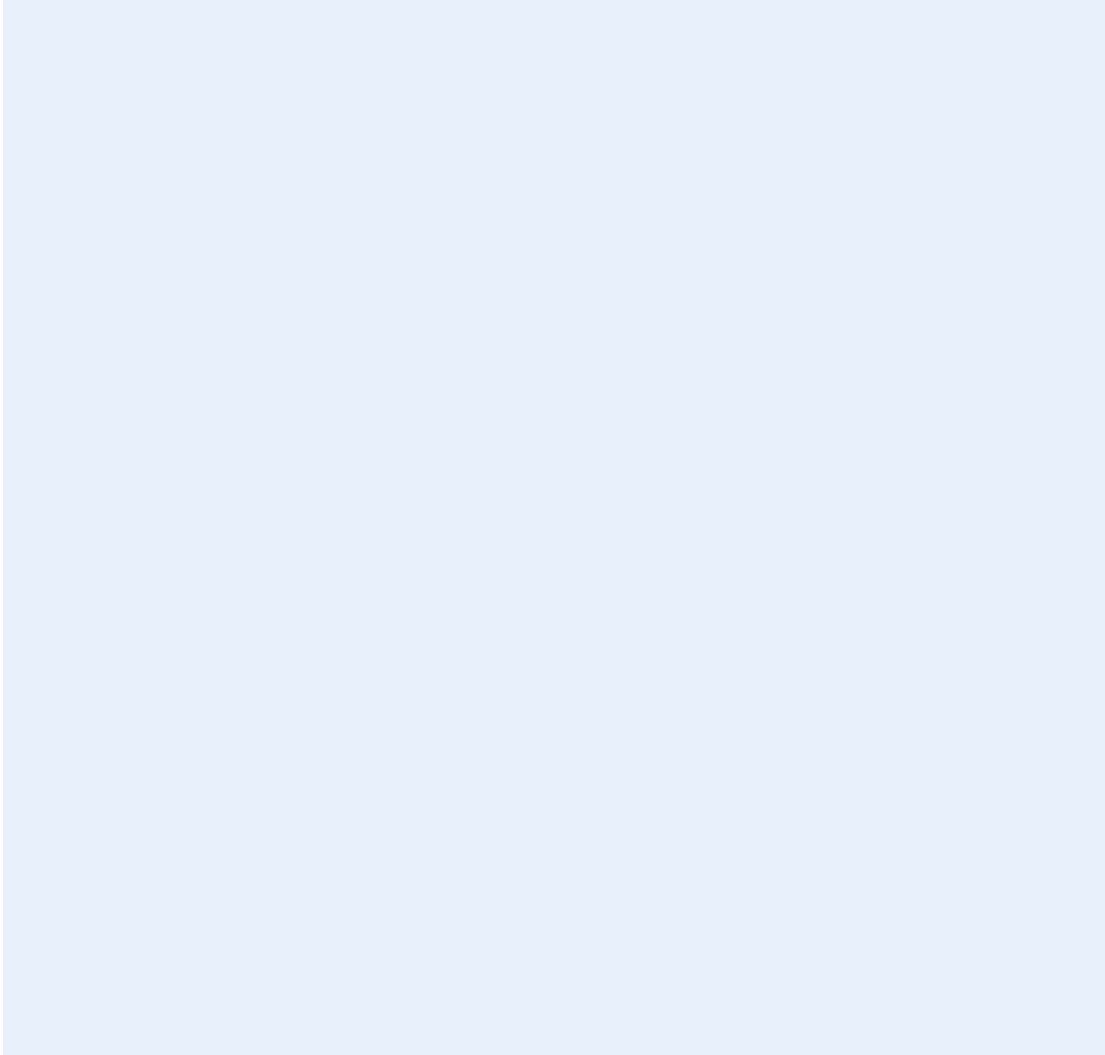
Numeric example — the implicit scheme tames the same $r = 1$.

Abstract example — a one-element-family finite-element solve.

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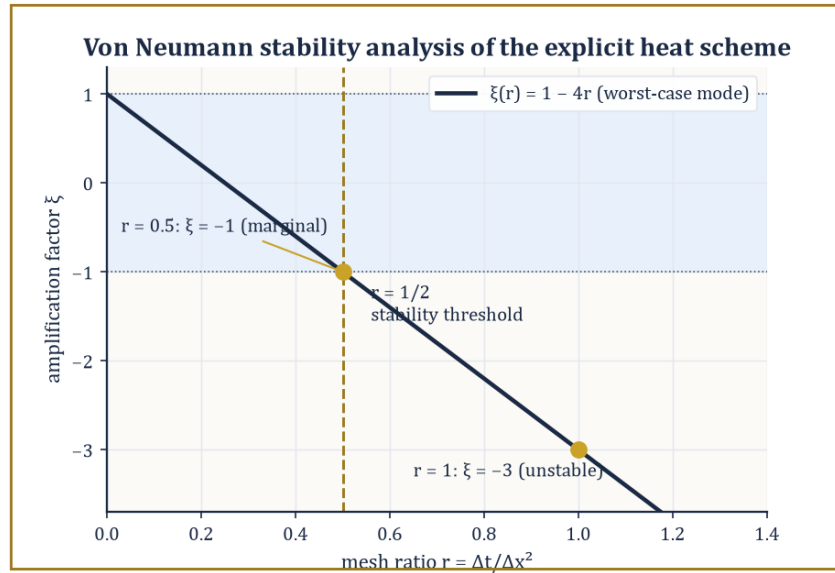
✓

Numeric example — algebraic decay versus spectral decay, side by side.



Room 29.8: Consistency, Stability, and Convergence

Lax's theorem: be truthful locally, be calm globally, and convergence follows for free



The worst-case amplification factor $\xi(r) = 1 - 4r$ for the explicit heat scheme, crossing out of the stable band $|\xi| \leq 1$ at the $r = 1/2$ threshold used, unproved, back in Room 29.7.

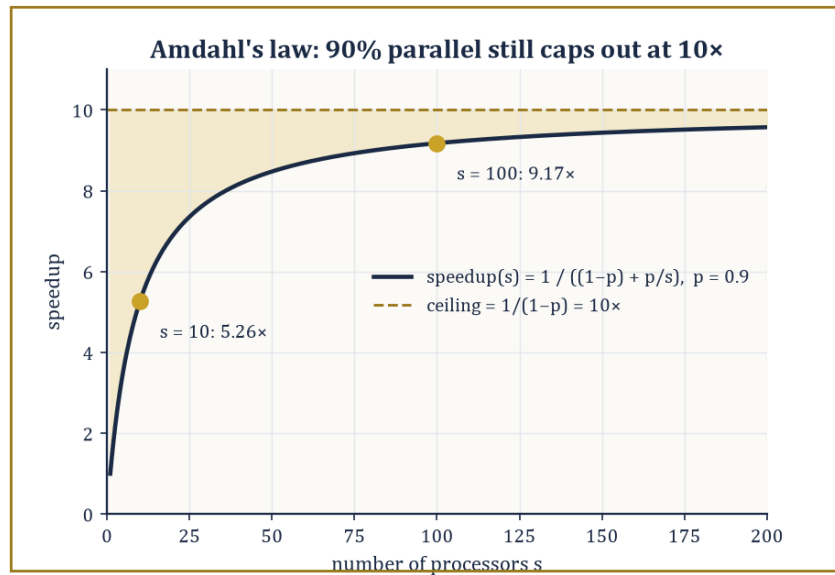
Worked example — von Neumann analysis derives the $r \leq 1/2$ threshold from Room 29.7.

Numeric example — Dahlquist's version, tied back to Room 29.6's stiff blow-up.

Numeric example — the CFL condition for advection.

Room 29.9: Looking Ahead: From Elementary Numerics to High-Performance and Adaptive Methods

The vocabulary is complete; the machine room, mercifully, keeps expanding



Amdahl's law at $p = 0.9$: speedup climbs fast at first, then flattens toward a hard ceiling of $10\times$, no matter how many processors you throw at the stubborn 10% that refuses to parallelize.

Numeric example — where adaptivity would have paid off.

Numeric example — Amdahl's law and the ceiling on speedup.

Numeric example — the diminishing returns of Monte Carlo sampling.

Check Your Understanding — Floor 29

Answers are in the Answer Key at the back of the book — have a go before you look.

1.

See the answer →

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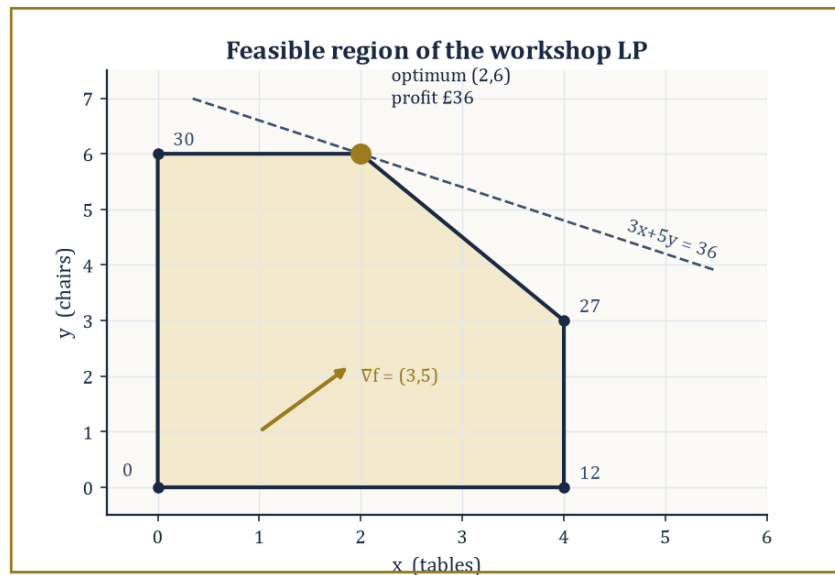
Floor 30: Optimization

Algorithms, Duality, and the Art of Finding Better Answers Efficiently



Room 30.1: Modelling and the Landscape of Optimization Problems

In Which We Learn That the Hardest Part Is Admitting What You Want



The workshop's feasible pentagon, with the objective's gradient and its level line through the optimum at (2,6) — the vertex the whole floor keeps returning to.

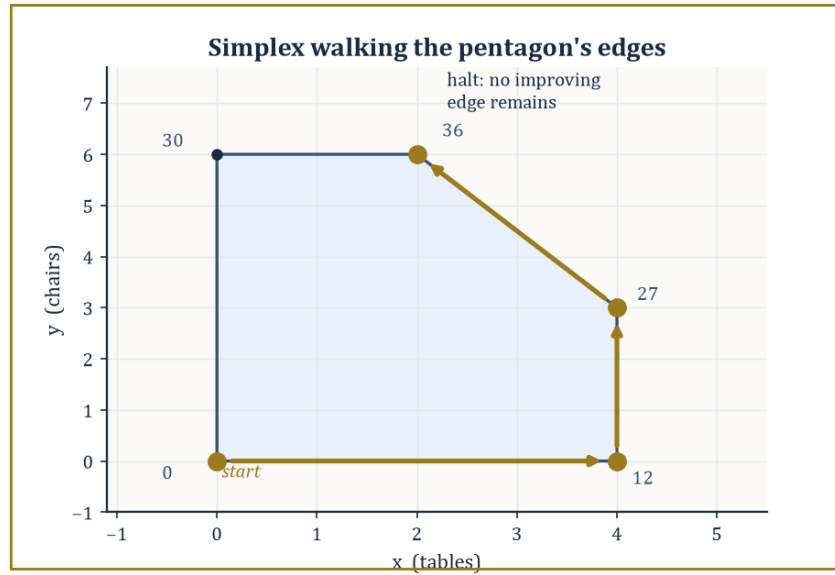
Numeric example — setting up and solving a small factory model.

Numeric example — the same workshop, three disguises.

Numeric example — unbounded and infeasible siblings.

Room 30.2: Linear Programming

Walking the Edges of a Very Well-Behaved Polyhedron



Simplex takes three improving edges — $(0,0)$ to $(4,0)$ to $(4,3)$ to $(2,6)$ — then halts, having visited only four of the pentagon's five vertices.

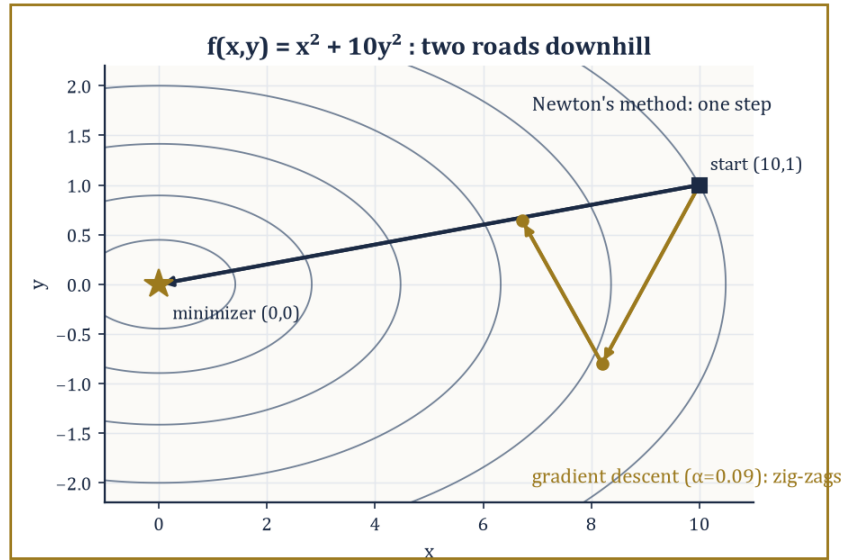
Numeric example — simplex walking the workshop's pentagon.

Numeric example — the workshop's dual, solved by complementary slackness.

Numeric example — a transportation problem, integral for free.

Room 30.3: Unconstrained Nonlinear Optimization

Downhill, but Cleverly



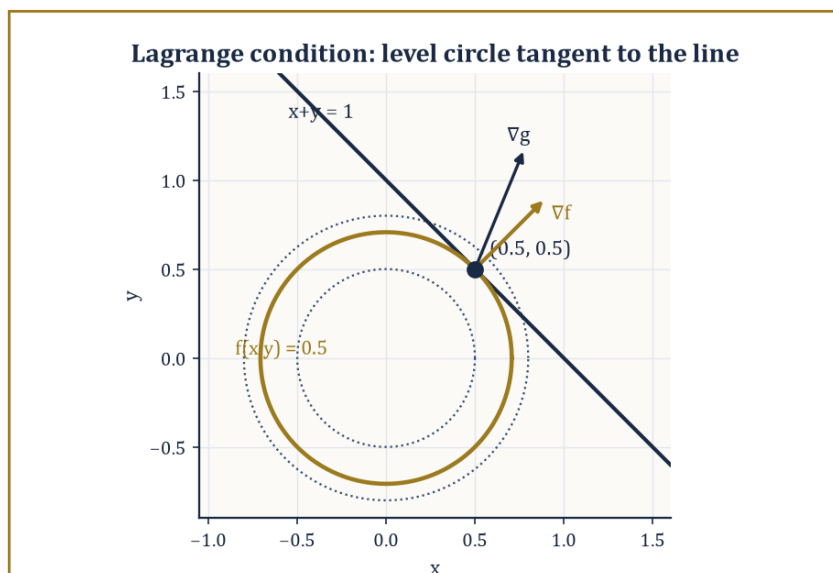
On the elongated bowl x^2+10y^2 , gradient descent zig-zags across the narrow valley while Newton's method walks straight to the minimiser in a single step.

Numeric example — Newton versus gradient descent on an ill-conditioned bowl.

Numeric example — quasi-Newton curvature from gradient differences alone.

Room 30.4: Constrained Nonlinear Optimization

Downhill, Without Falling Off the Map



At the constrained minimum $(0.5, 0.5)$, the level circle of f just touches the line $x+y=1$ — tangency is what $\nabla f = \lambda \nabla g$ looks like.

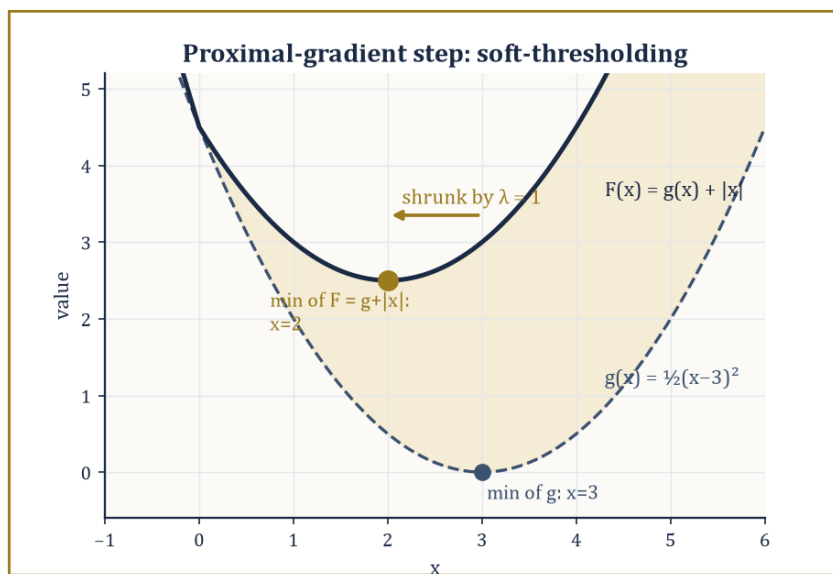
Numeric example — penalty and barrier methods homing in from opposite sides.

Numeric example — solving an equality-constrained problem via the Lagrangian.

Abstract example — a constraint qualification quietly failing.

Room 30.5: Convex Optimization Algorithms

Where Local Really Does Mean Global



Adding the ℓ_1 penalty $|x|$ to $g(x)$ drags the minimiser from $x=3$ to $x=2$ — soft-thresholding made visible.

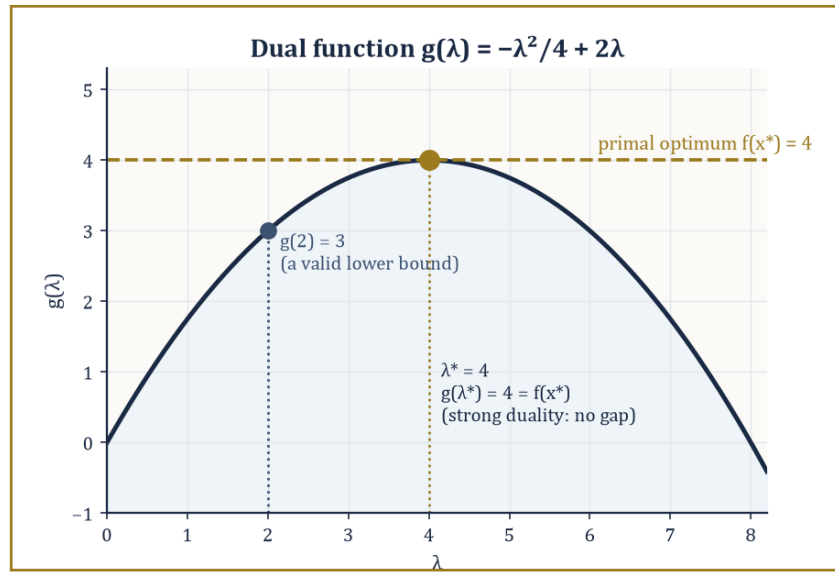
Numeric example — one proximal-gradient step solves a sparse problem exactly.

Numeric example — how the iteration budgets compare.

Abstract example — one ADMM cycle splitting a joint problem in two.

Room 30.6: Duality in Practice

How to Prove You've Won Without Playing Every Move



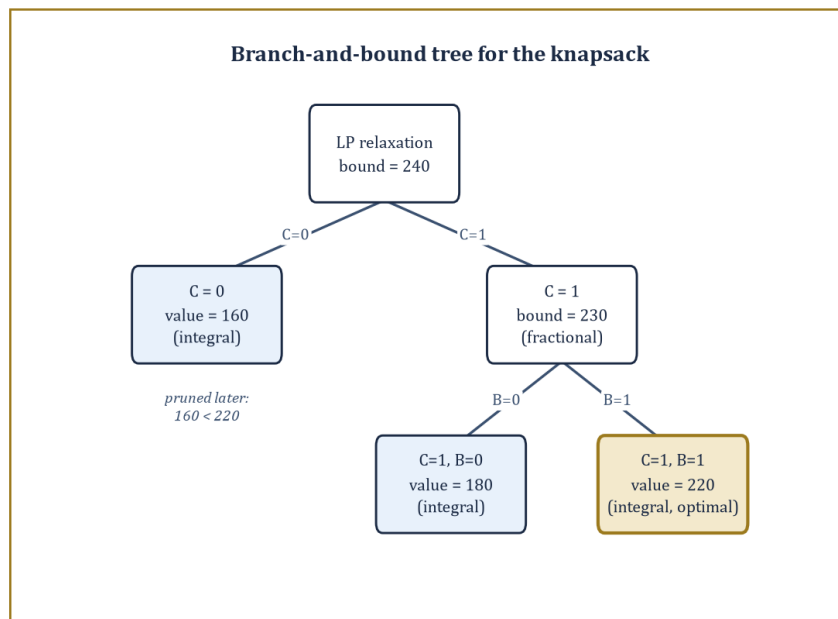
The dual function $g(\lambda)$ is always concave; here its peak at $\lambda^=4$ meets the primal optimum, closing the duality gap to zero.*

Numeric example — weak and strong duality for a one-variable convex program.

Numeric example — an LP-relaxation bound for a small knapsack problem.

Room 30.7: Integer and Combinatorial Optimization

Where the Continuous World's Tricks Mostly Stop Working



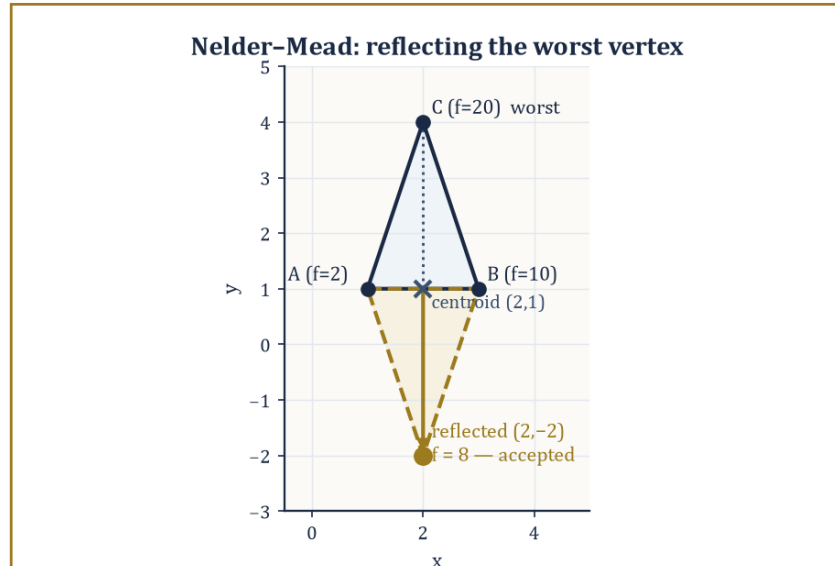
Four subproblems, not eight combinations: the knapsack's branch-and-bound tree, each bound doing the pruning the exhaustive search never needed.

Numeric example — branch-and-bound solving the Room 30.6 knapsack.

Numeric example — how fast the exact TSP search space explodes.

Room 30.8: Derivative-Free, Stochastic, and Large-Scale Methods

When You Can't (Or Shouldn't) Compute the Gradient



Nelder-Mead reflects the worst vertex C through the centroid of A and B , landing at $f=8$ and replacing C outright — no derivative in sight.

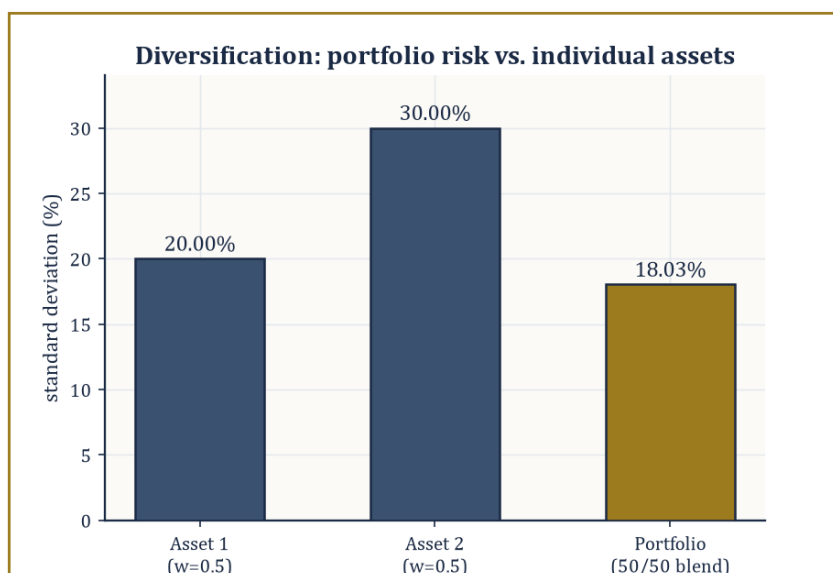
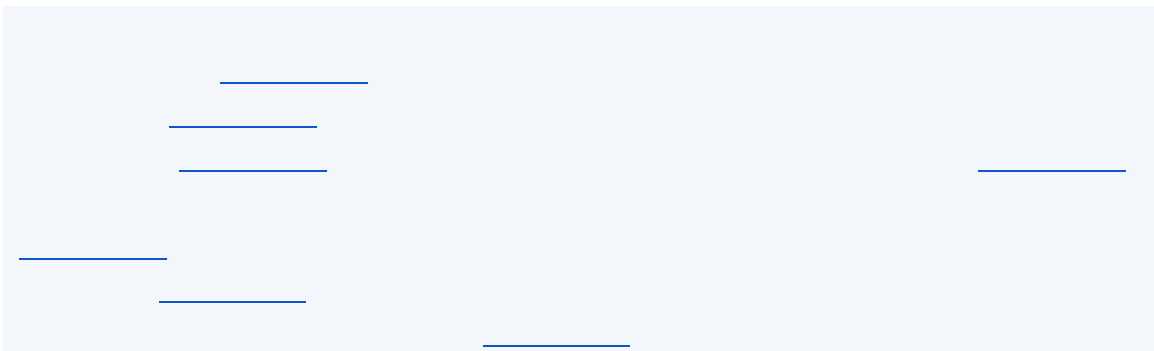
Numeric example — one Nelder-Mead reflection step.

Numeric example — one full-batch step versus two stochastic steps.

Numeric example — coordinate descent solving a separable problem in one cycle.

Room 30.9: Looking Ahead: From Algorithms to Advanced Optimization and Applications

We Have Learned to Compute the Best



An equal blend of two uncorrelated assets carries less risk than either asset alone — diversification, computed rather than merely asserted.

Numeric example — a two-asset portfolio, minimising risk for a target return.

Check Your Understanding — Floor 30

Answers are in the Answer Key at the back of the book — have a go before you look.

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See the answer →

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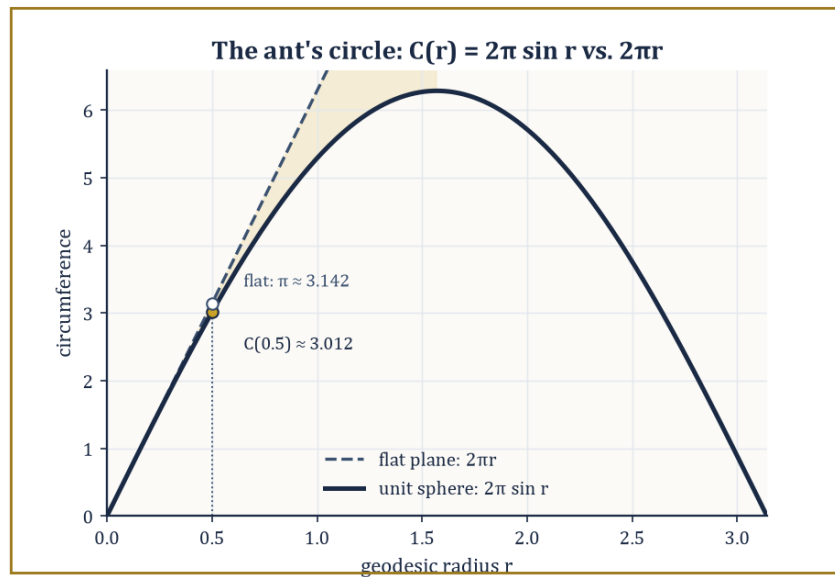
See the answer →

Floor 31: Differential Geometry

Curves, Surfaces, Curvature, and the Art of Geometry by Calculus

Room 31.1: What Differential Geometry Is

In which we discover that geometry can be done without ever standing outside it



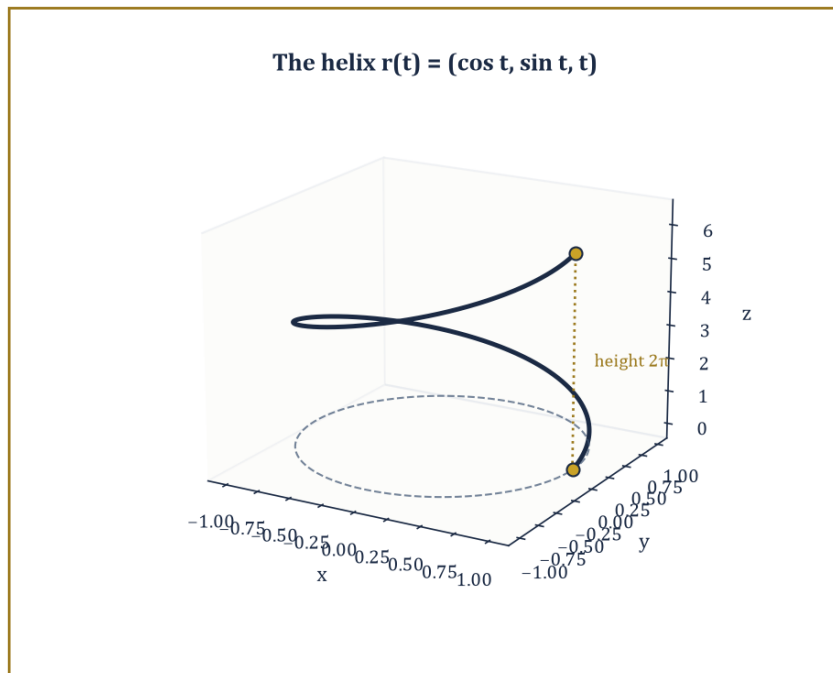
The ant's rope traces $C(r) = 2\pi \sin r$ on the sphere, falling short of the flat-plane prediction $2\pi r$ — a shortfall measured entirely from within.

Numeric example — The ant's shortfall.

Abstract example — The flight to Vancouver.

Room 31.2: Curves in the Plane and in Space

Where speed becomes irrelevant and shape becomes everything



One full turn of the helix $\mathbf{r}(t) = (\cos t, \sin t, t)$, climbing a height of 2π while circling the unit circle below.

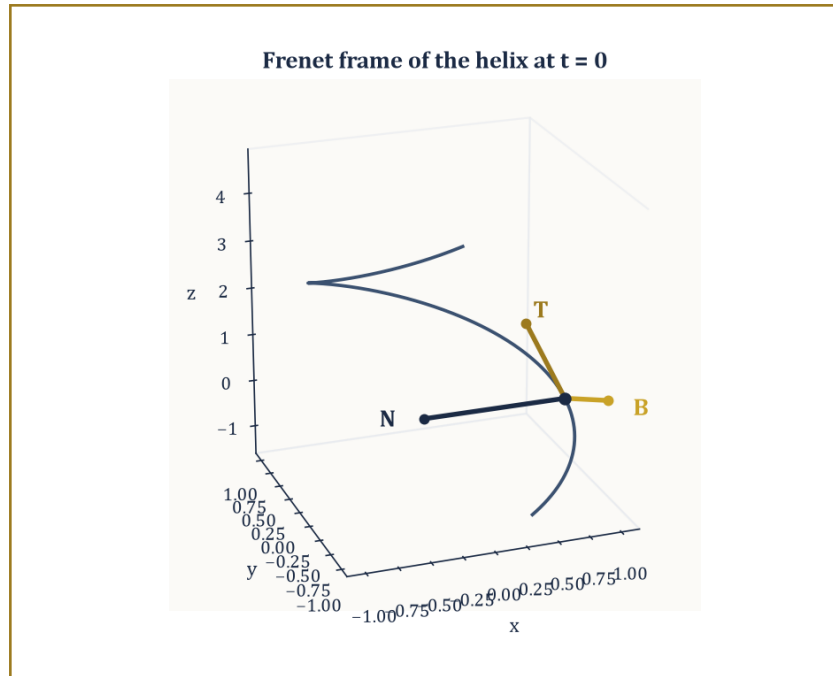
Numeric example — Arc length of a helix.

Numeric example — Circle and parabola.

Abstract example — Two circles, one shape.

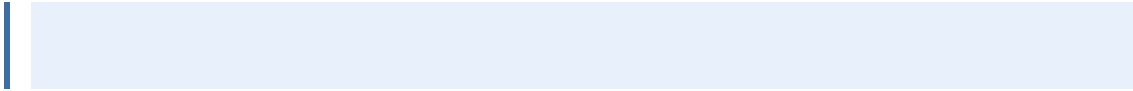
Room 31.3: The Frenet-Serret Apparatus

Three mutually perpendicular vectors, doing all the work

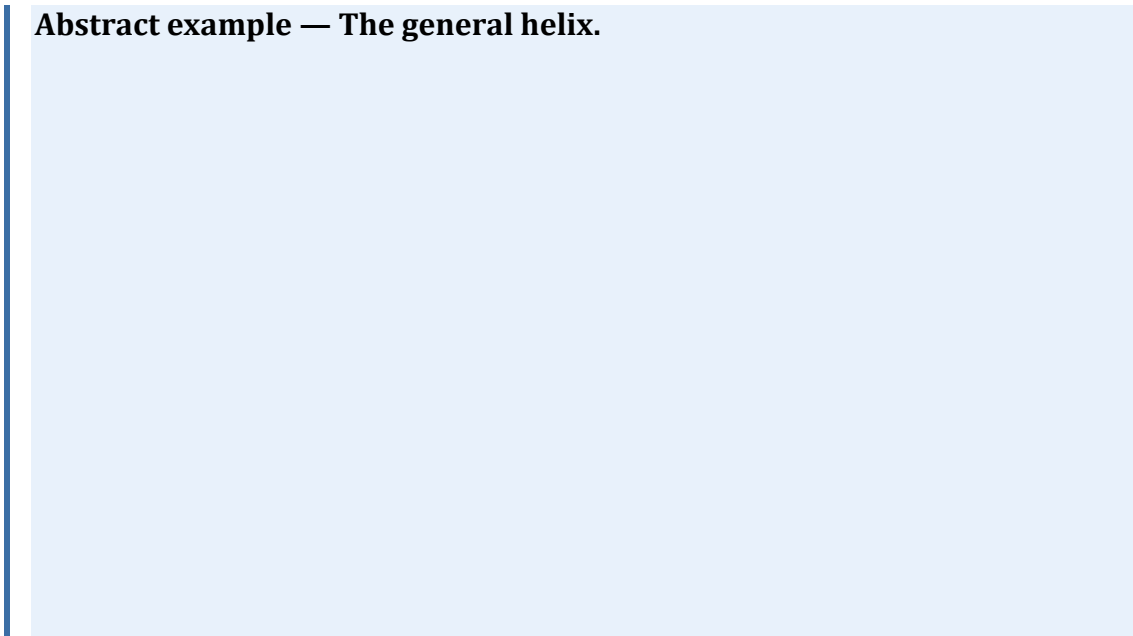


The Frenet frame $\{T, N, B\}$ of the helix at $t = 0$, computed in the worked example alongside it.

Numeric example — The Frenet frame of a helix.

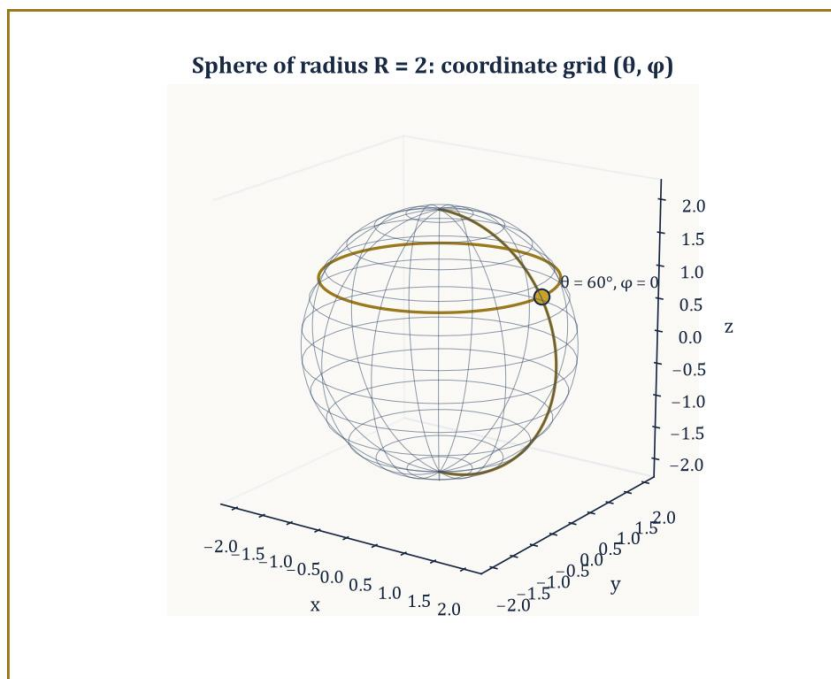


Abstract example — The general helix.



Room 31.4: Surfaces and the First Fundamental Form

Measuring a surface without ever admitting it lives in space

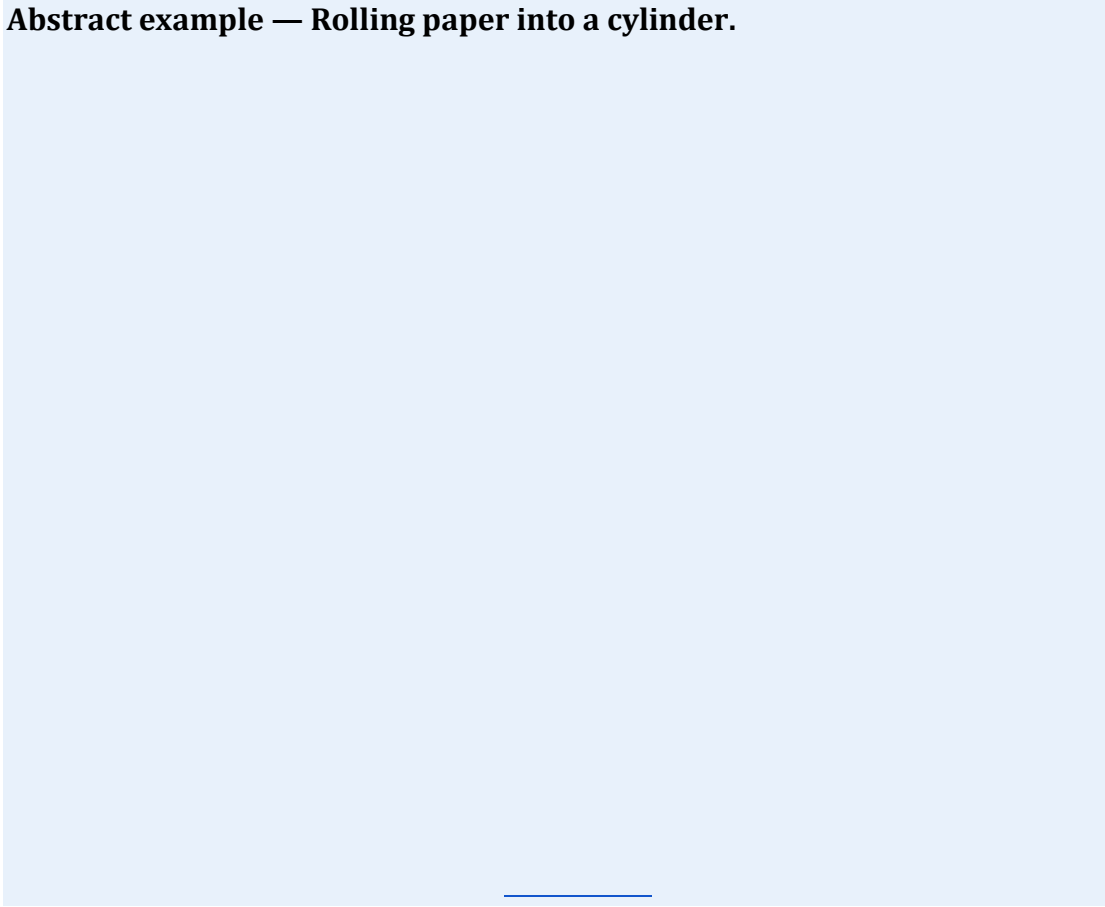


The (θ, φ) coordinate grid on a sphere of radius 2, with the meridian and latitude circle through the worked point $\theta = 60^\circ$, $\varphi = 0$.

Numeric example — The first fundamental form of a sphere.

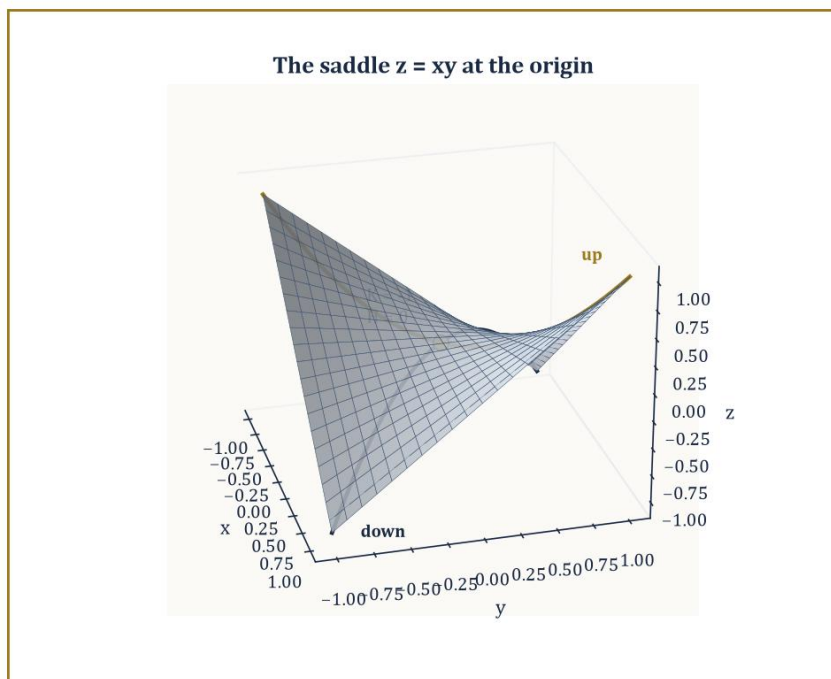
Numeric example — The area of a sphere, derived intrinsically.

Abstract example — Rolling paper into a cylinder.



Room 31.5: The Second Fundamental Form and Curvature

Where the surface finally admits it isn't flat



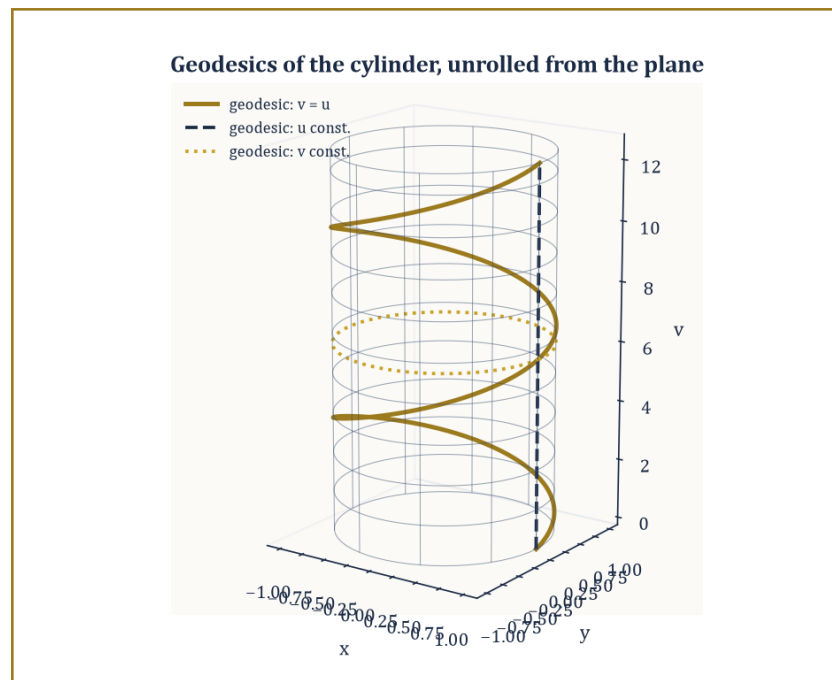
The saddle $z = xy$ at the origin: $K = -1$, $H = 0$ — curving upward along one principal direction and downward along the other.

Numeric example — Curvature of a sphere.

Numeric example — Curvature of a saddle.

Room 31.6: Geodesics

As straight as the surface will allow



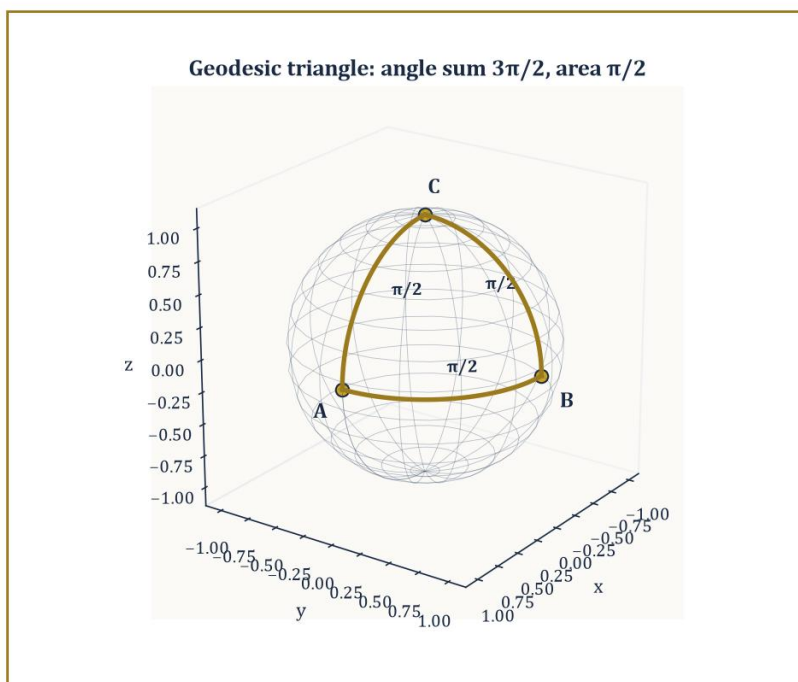
The geodesics of a cylinder are the images of straight lines under unrolling: helices, vertical lines, and circles.

Numeric example — The equator is a geodesic.

Abstract example — Unrolling the cylinder's geodesics.

Room 31.7: The Gauss-Bonnet Theorem

Where local bending finally has to answer to global shape



A right-angled octant triangle on the unit sphere: three $\pi/2$ angles summing to $3\pi/2$, exceeding π by the enclosed area.

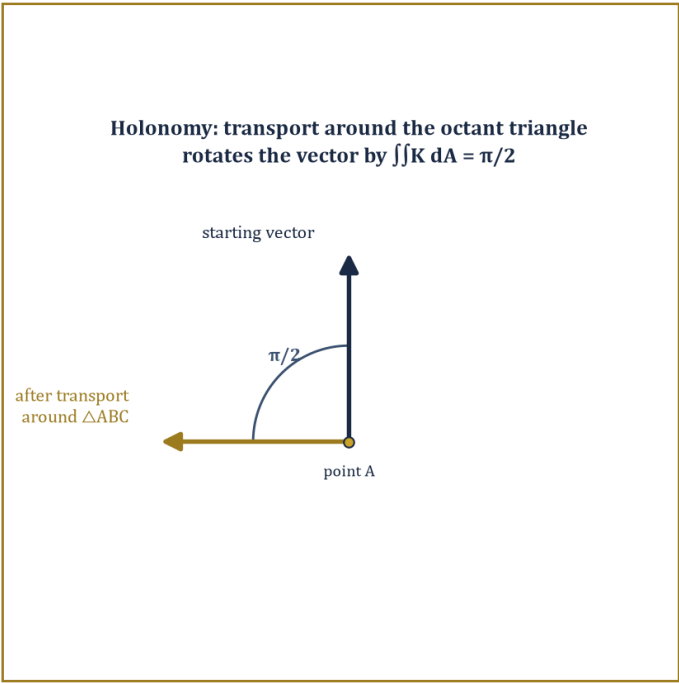
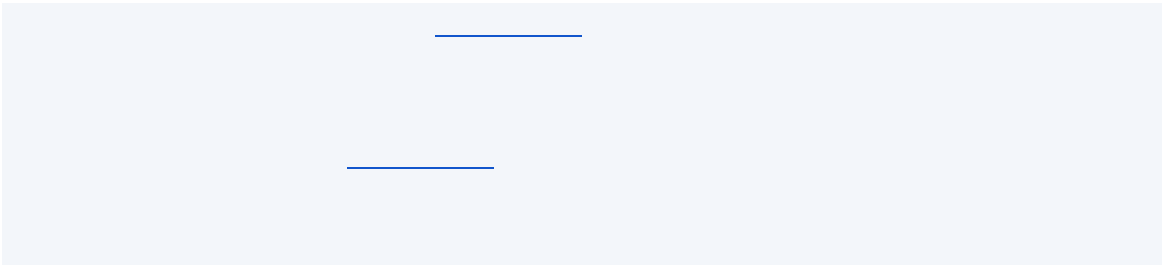
Numeric example — Total curvature of a sphere, any size.

Numeric example — A right-angled octant triangle.

Abstract example — The torus's curvature budget.

Room 31.8: Intrinsic Geometry and the Theorema Egregium

Gauss’s remarkable theorem, and why he had every right to name it that



Carried around the octant triangle of Room 31.7, a vector returns rotated by $\pi/2$ — holonomy made visible, and equal to the enclosed curvature.

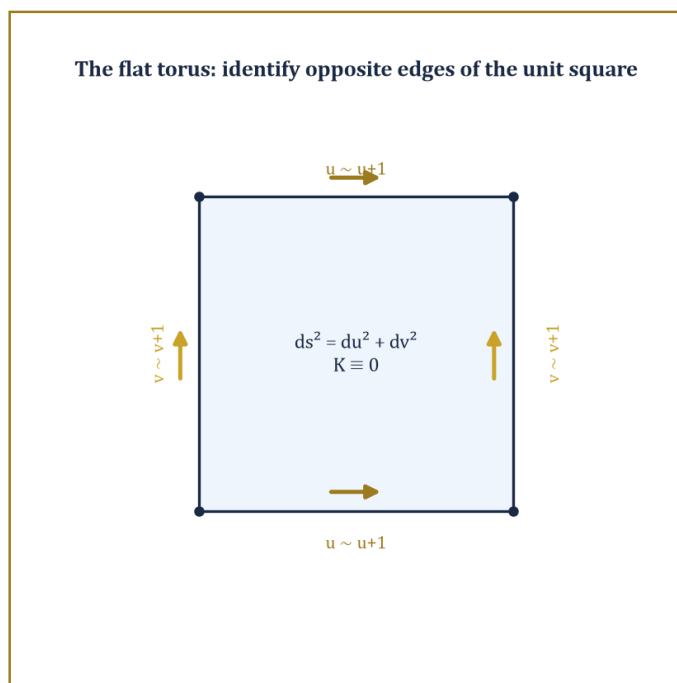


Numeric example — The cylinder's honest confession.

Abstract example — Parallel transport around the octant triangle.

Room 31.9: Looking Ahead: From Surfaces to Manifolds, Riemannian Geometry, and Relativity

The view from the top of the classical theory



The flat torus: glue opposite edges of a unit square and the flat metric $ds^2 = du^2 + dv^2$ survives intact, with $K \equiv 0$ everywhere.

Abstract example — Two toruses, one topology.

Check Your Understanding — Floor 31

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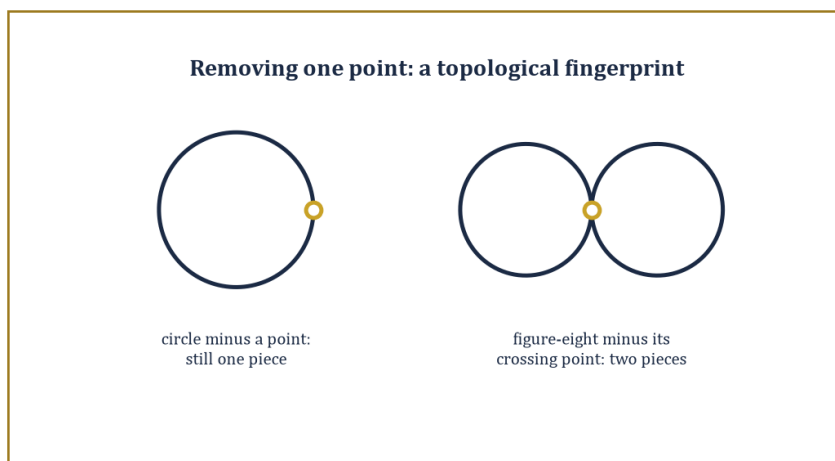
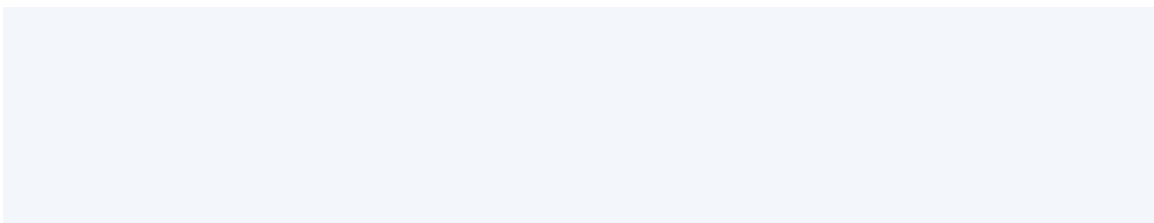
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Floor 32: Topology

Continuity, Connectedness, and the Art of Geometry Without Distance

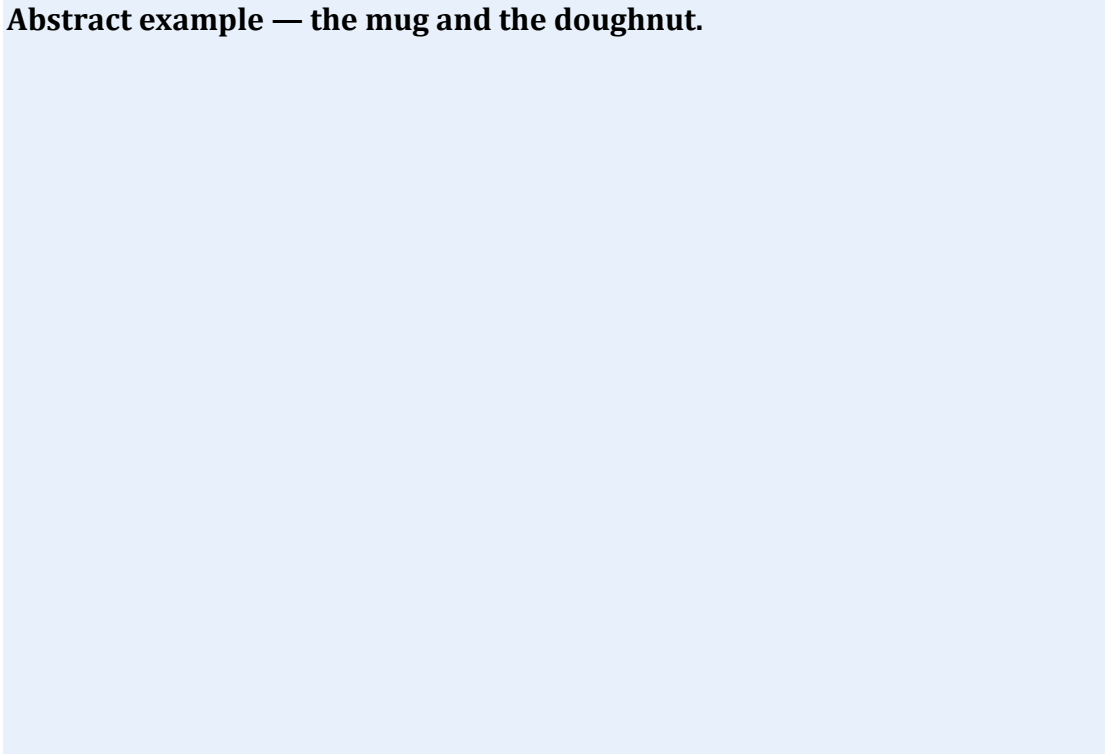
Room 32.1: What Topology Is

In Which We Confiscate the Ruler



Delete one point from a circle and it stays whole; delete the crossing point from a figure-eight and it falls into two pieces — a topologist's fingerprint, no ruler required.

Abstract example — the mug and the doughnut.

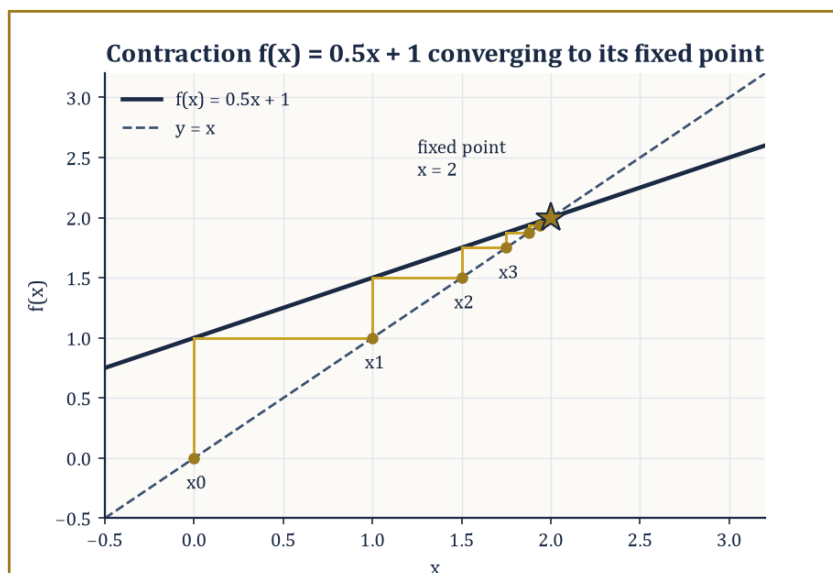


Abstract example — telling a circle from a disc.

Abstract example — a point-set argument, no algebra required.

Room 32.2: Metric Spaces and Continuity

Where Distance Still Behaves Itself, For Now



The contraction $f(x) = 0.5x + 1$, iterated from $x_0 = 0$, staircases its way straight into the fixed point $x = 2$ — as promised by a constant $L = 0.5$ comfortably under 1.

Numeric example — checking the axioms.

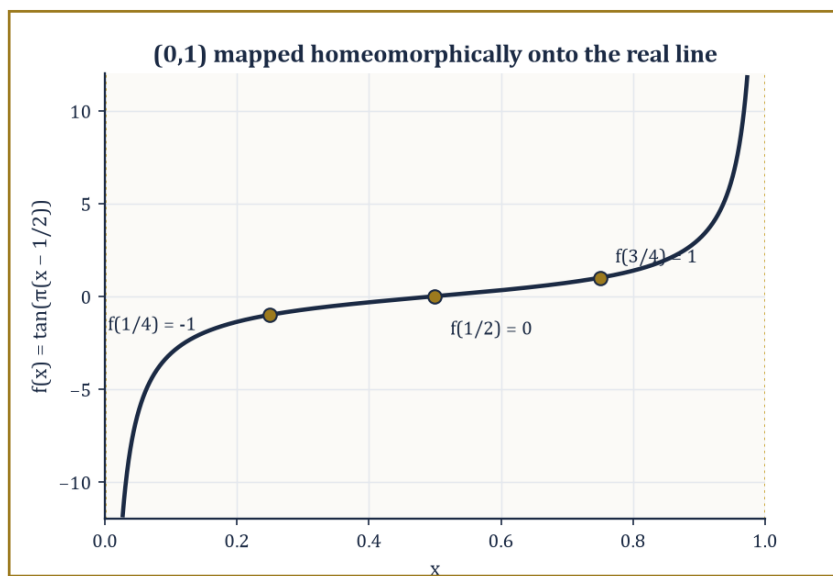
Numeric example — an ε - δ check.

Numeric example — the discrete metric.

Numeric example — a contraction in action.

Room 32.3: Topological Spaces and Continuous Maps

Open Sets, All the Way Down



$f(x) = \tan(\pi(x - 1/2))$ bends the bounded interval $(0,1)$ into the whole real line — proof that boundedness was only ever a fact about the ruler, not about the space.

Abstract example — a topology on three points.

Numeric example — interior, closure, boundary.

Abstract example — continuity between finite spaces.

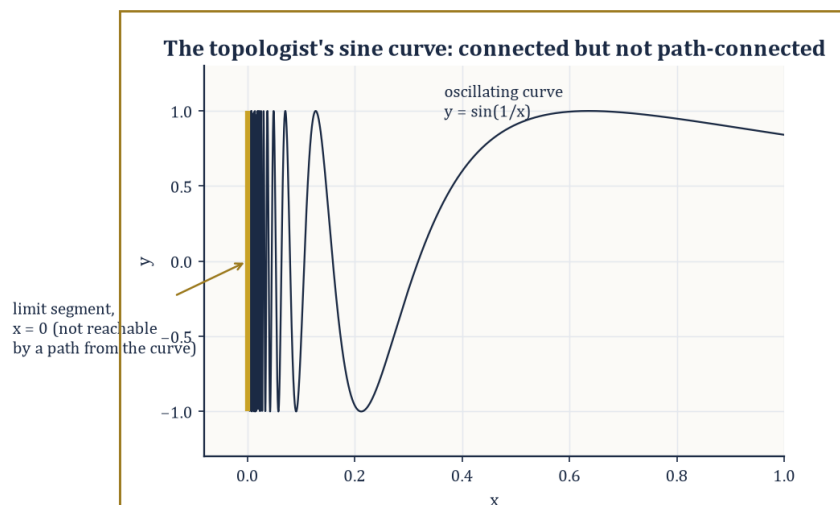
Worked example — a bounded interval is homeomorphic to the whole line.

Worked example — building a circle by gluing an interval's ends.

Numeric example — axioms in the cofinite topology.

Room 32.4: Connectedness and Path-Connectedness

On Being in One Piece



The topologist's sine curve: the oscillating graph and the vertical segment it accumulates onto form one connected set that no path can cross.

Numeric example — a disconnection and an intermediate value.

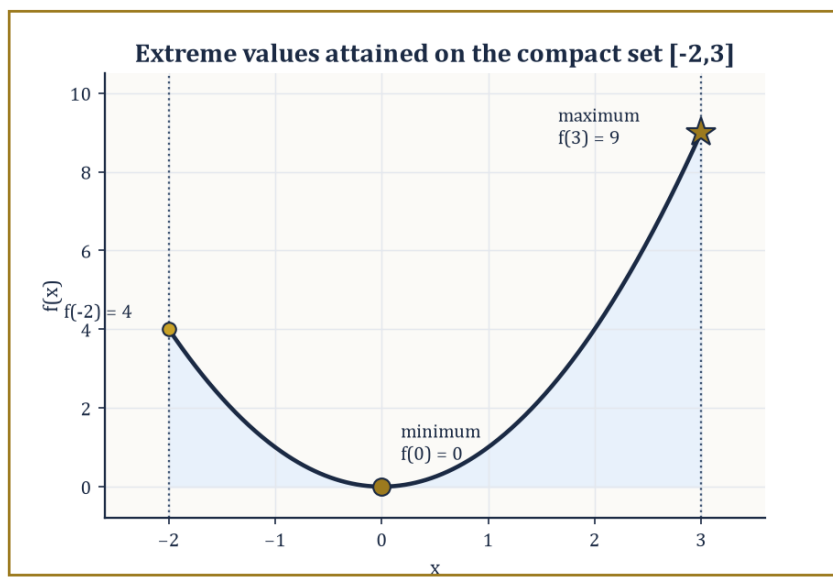
Numeric example — counting the pieces.

Numeric example — a straight-line path.

Abstract example — the topologist's sine curve.

Room 32.5: Compactness

A Kind of Finiteness, Politely Disguised



$f(x) = x^2$ on the compact interval $[-2, 3]$: the extreme value theorem's minimum and maximum are not just approached but actually attained, at $x = 0$ and $x = 3$.

Numeric example — two spaces that fail to be compact, and why.

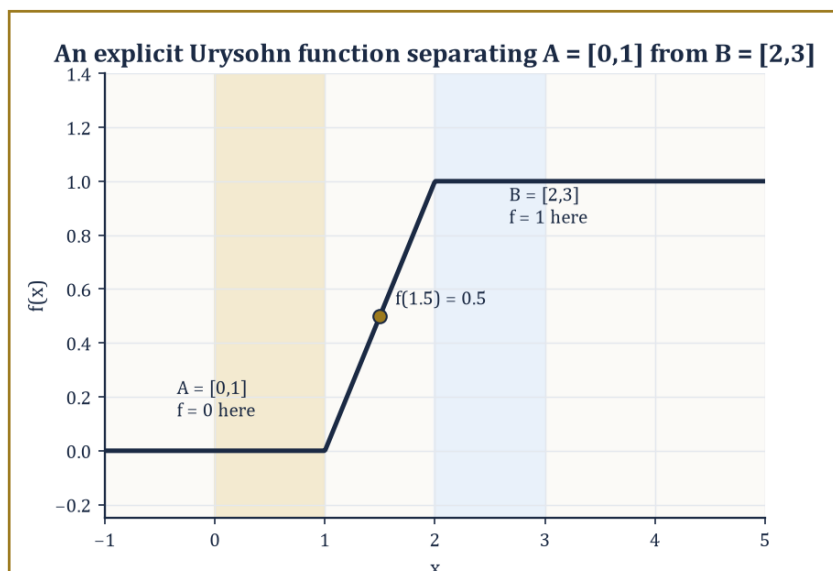
Numeric example — extreme values on a compact set.

Numeric example — a subsequence that behaves.

Abstract example — an infinite product that stays compact.

Room 32.6: Separation Axioms and Countability

How Civilised Is Your Space?



An explicit Urysohn function ramping linearly from 0 on $A = [0,1]$ to 1 on $B = [2,3]$ — the closed sets kept apart, the numbers kept honest.

Abstract example — T_0 without T_1 .

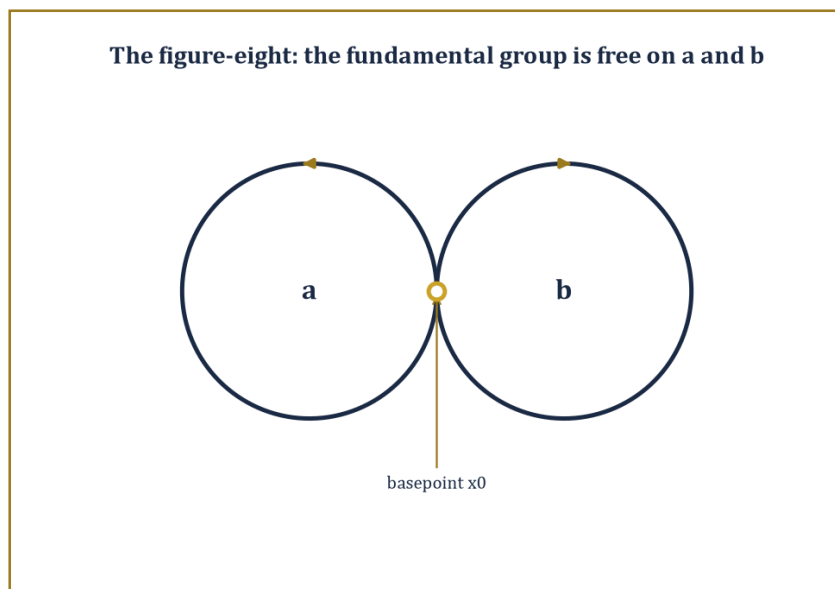
Abstract example — T_1 without Hausdorff.

Numeric example — $\mathbb{R}$ is second-countable and separable.

Numeric example — an explicit Urysohn function.

Room 32.7: Homotopy and the Fundamental Group

Counting Holes for a Living



The figure-eight's two lobes, a and b , generate its fundamental group freely — with no filled-in square behind them to let one loop slide past the other and force them to commute.

Numeric example — sliding one point to another, and shrinking a plane.

Numeric example — concatenating two loops.

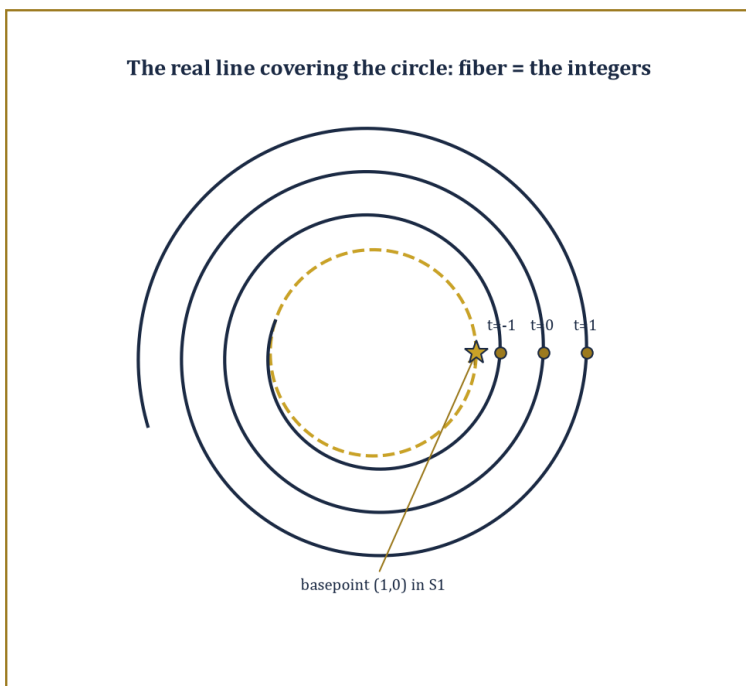
Numeric example — winding number two on the circle.

Abstract example — why the torus's two loops commute.

Abstract example — van Kampen on the figure-eight.

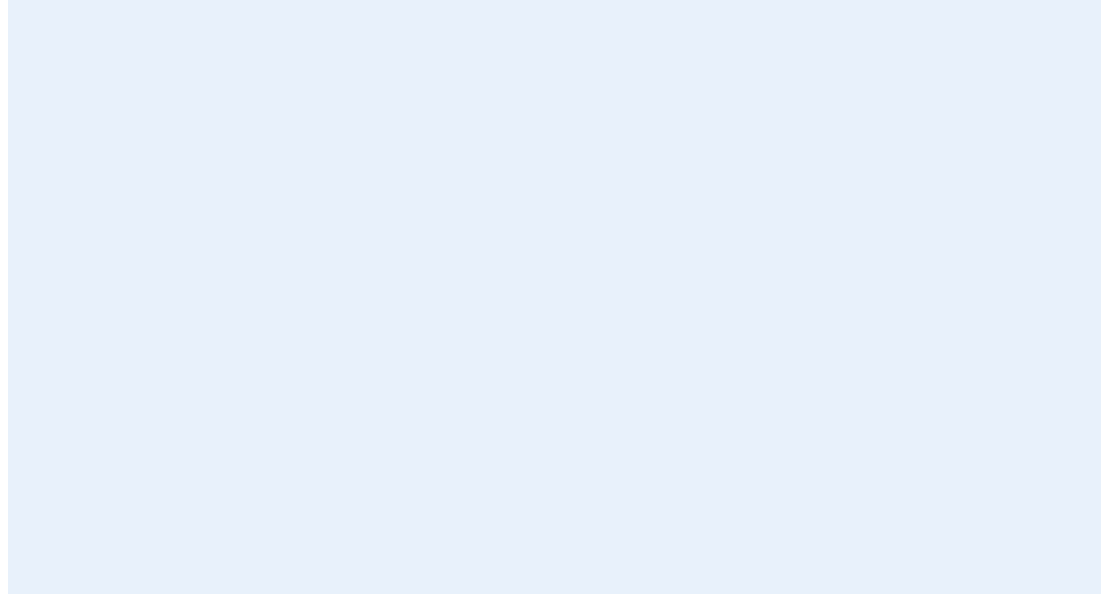
Room 32.8: Covering Spaces and Classification of Surfaces

The Complete List, No Exceptions

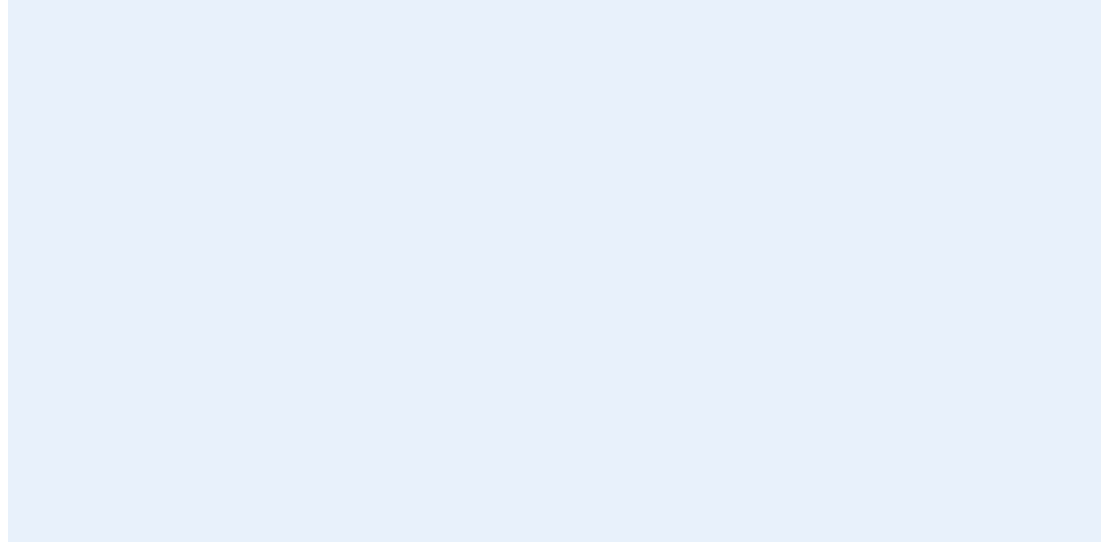


The real line spirals endlessly above the circle's basepoint $(1,0)$; each integer t marks one more sheet of the cover, and the whole tower unwinds the one loop the circle cannot shrink away.

Numeric example — the line covering the circle.



Abstract example — the double cover of the circle.



Numeric example — reading off genus from the Euler characteristic.

Room 32.9: Looking Ahead: From Basic Topology to Algebraic Topology and Manifolds

Onward, Upward, Still No Ruler



Abstract example — a fundamental group that misses something.

Abstract example — what homology forgets, on purpose.

Check Your Understanding — Floor 32

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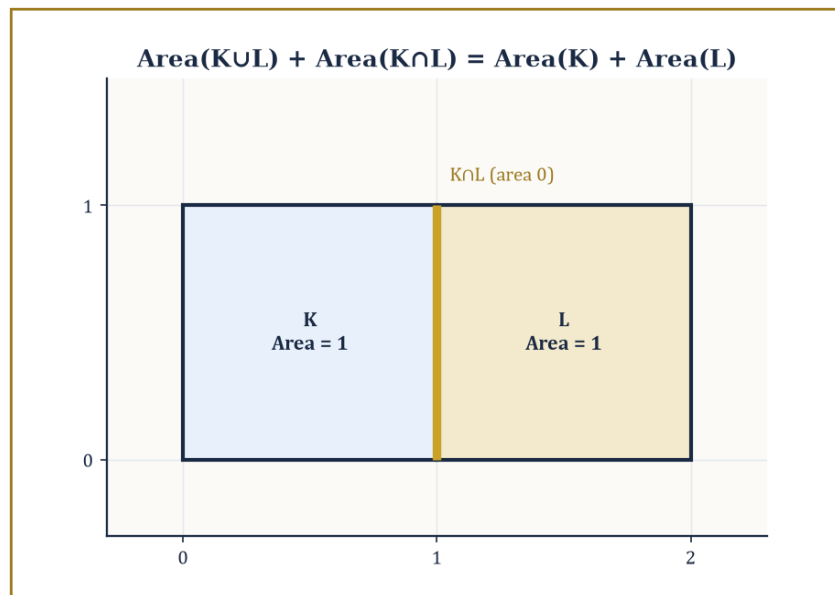
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Floor 33: Integral Geometry (Advanced)

Valuations, Kinematic Formulas, and the Art of Measuring Geometry with Groups

Room 33.1: From Classical Formulas to Modern Valuations

In which we stop computing averages and start classifying them



Two unit squares sharing an edge: the union and intersection contribute 2 and 0 respectively, so both sides of the valuation law come to 2.

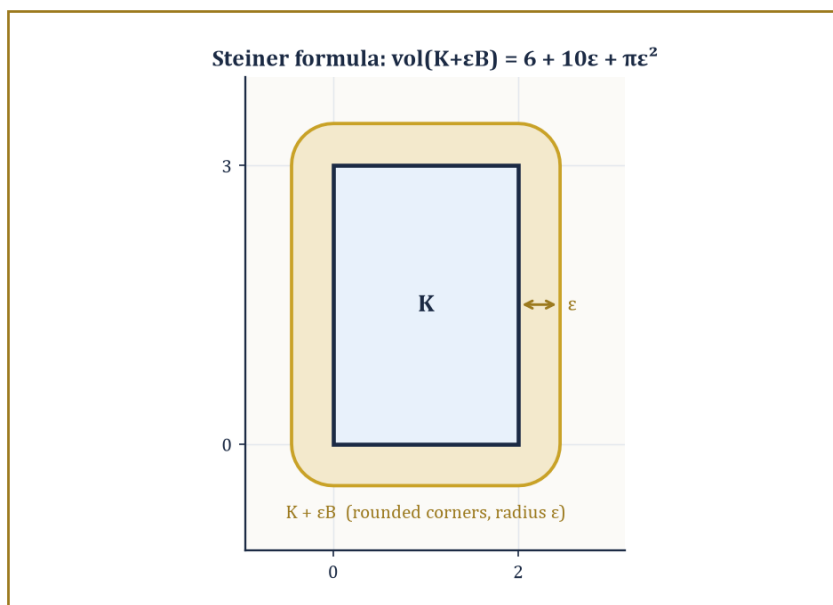
Numeric example — checking the valuation law on two rectangles.

Numeric example — inclusion–exclusion for three overlapping intervals.

Abstract example — why ‘union must be convex’ is the whole ballgame.

Room 33.2: Convex Bodies and Intrinsic Volumes

Steiner's formula, or: what happens when you inflate a shape with a hosepipe



The 2×3 rectangle fattened by ε : its rounded parallel body's area, $6 + 10\varepsilon + \pi\varepsilon^2$, reads off the three intrinsic volumes at a glance.

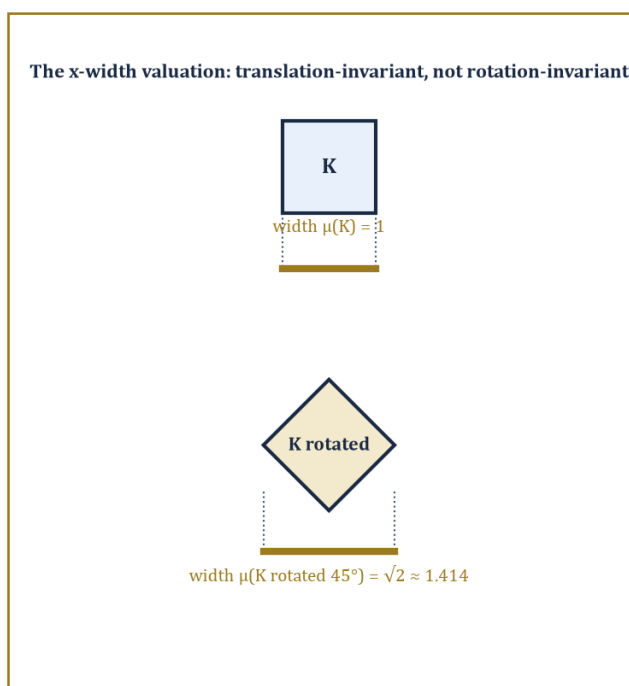
Numeric example — Steiner's formula for a 2×3 rectangle.

Numeric example — Steiner's formula for a ball in $\mathbb{R}^3$.

Numeric example — Cauchy's formula on a unit square.

Room 33.3: Continuous Valuations and Hadwiger's Theorem

The theorem that puts the whole subject out of a job, in the best possible way



*A square's x-shadow has width 1, but rotate it 45° and the shadow widens to $\sqrt{2}$ — the anisotropy
Hadwiger's theorem forbids once rotation-invariance is required.*

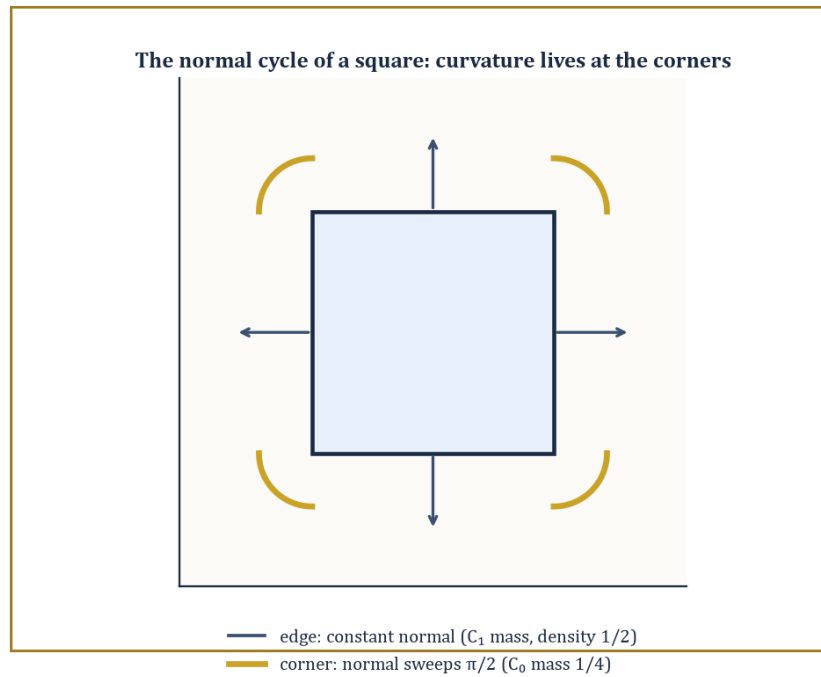
Numeric example — pinning down a valuation from its values on discs.

Abstract example — a functional that looks geometric but is not a valuation.

Numeric example — a valuation that survives translation but not rotation.

Room 33.4: Curvature Measures and the Normal Cycle

Smearing the intrinsic volumes across the boundary, one corner at a time

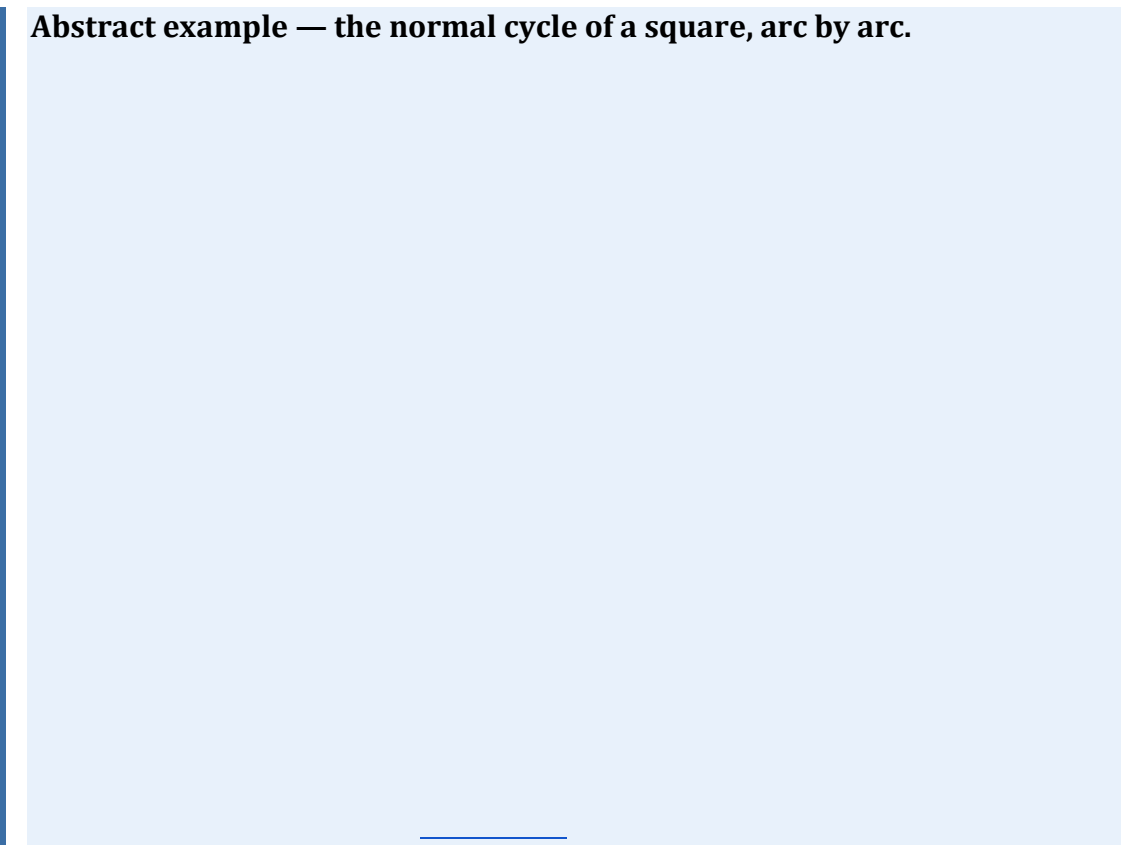


The normal cycle of a square: C_1 spreads evenly along the flat edges while C_0 concentrates entirely in the four $\pi/2$ turns at the corners.

Numeric example — the Euler curvature measure of a square, concentrated at its corners.

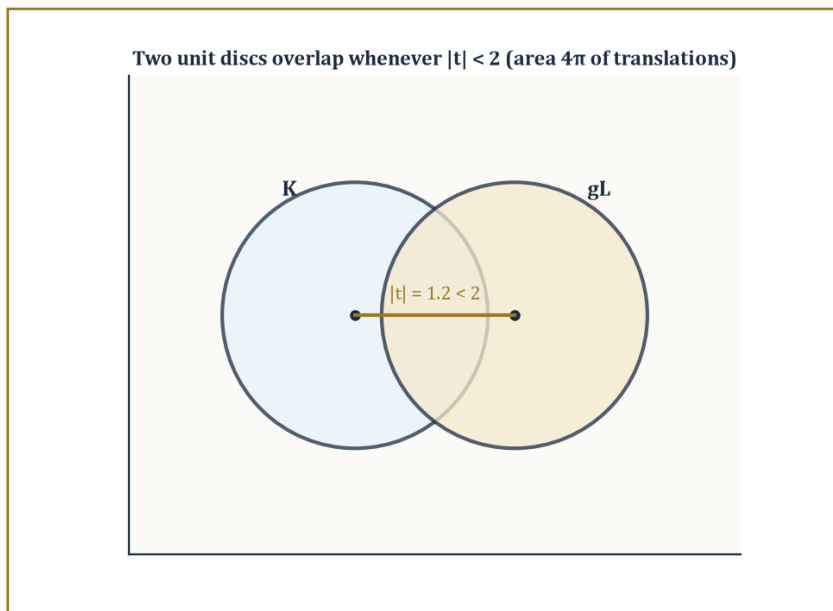
Numeric example — the C_1 curvature measure of a 2×3 rectangle.

Abstract example — the normal cycle of a square, arc by arc.



Room 33.5: The Principal Kinematic Formula and Its Relatives

Two shapes, one enormous group, and a very tidy answer



Two unit discs centred a distance $|t| = 1.2$ apart: they overlap for every $|t| < 2$, the disc of translations whose area 4π drives Santaló's formula.

Numeric example — the simplest possible case, two intervals on a line.

Numeric example — two unit discs in the plane.

Abstract example — Crofton's formula as a shrunk-down kinematic formula.

Room 33.6: The Algebra of Valuations

In which the kinematic formula turns out to be a multiplication table



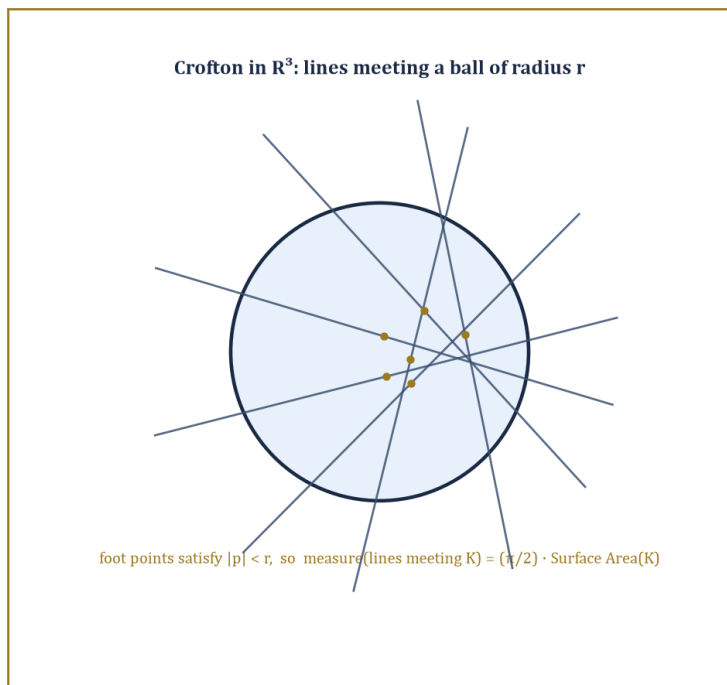
Numeric example — reading structure constants off Santaló's formula.

Abstract example — how small the algebra really is.

Abstract example — Poincaré duality, read off the same table.

Room 33.7: Integral Geometry on Spaces of Higher Rank

Lines were only ever the beginning



A ball in $\mathbb{R}^3$ with several random lines crossing it, foot point marked on each: only lines whose foot lies within radius r contribute, giving measure $(\pi/2) \cdot \text{Surface Area}(K)$.

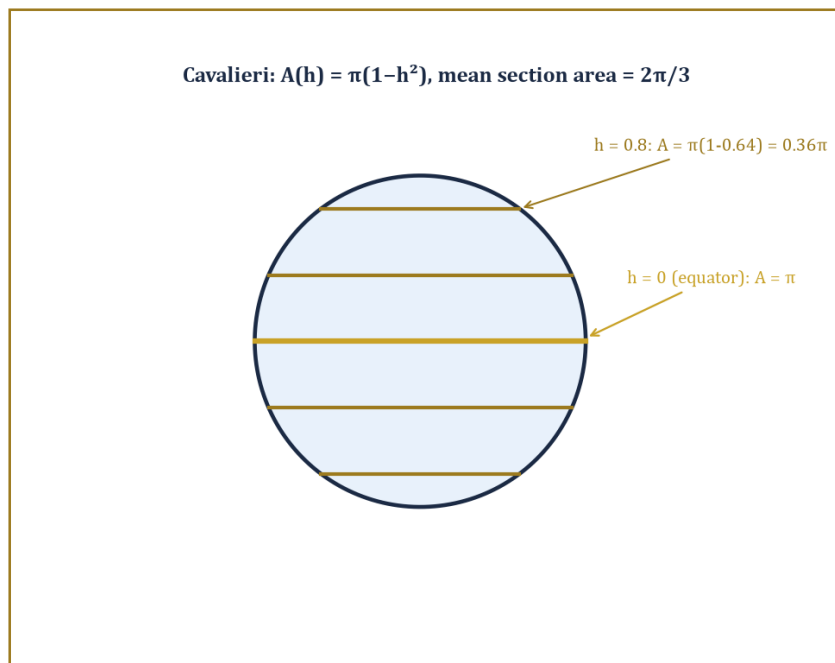
Numeric example — lines meeting a ball in $\mathbb{R}^3$.

Numeric example — planes meeting a ball in $\mathbb{R}^3$.

Abstract example — two 2-planes in $\mathbb{R}^4$, a genuinely higher-rank instance.

Room 33.8: Applications: Stereology, Tomography, and Geometric Probability

Where all this machinery finally earns its keep



Slicing the unit ball at several heights h : the cross-sectional area $\pi(1-h^2)$ shrinks away from the equator, and its average over all h recovers the volume by Cavalieri's principle.

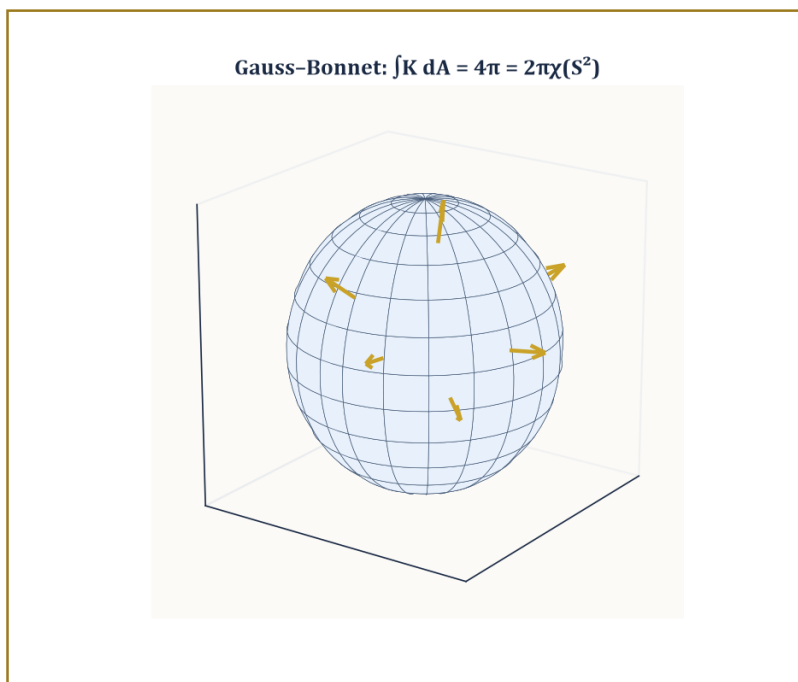
Numeric example — mean section area of a ball, and Cavalieri's check.

Numeric example — reconstructing a 2×2 grid from its projections.

Numeric example — the probability two unit discs overlap.

Room 33.9: Looking Ahead: Valuations on Manifolds and Algebraic Integral Geometry

The view from the thirty-third floor, before the lift doors close



The unit sphere, curving by exactly $K = 1$ at every point: integrate that curvature over the whole surface and Gauss-Bonnet hands back $4\pi = 2\pi\chi(S^2)$.

Numeric example — Gauss-Bonnet on the sphere, a first curved-space valuation.

Numeric example — an Euler integral of a simple constructible function.

—

Check Your Understanding — Floor 33

Answers are in the Answer Key at the back of the book — have a go before you look.

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See the answer →

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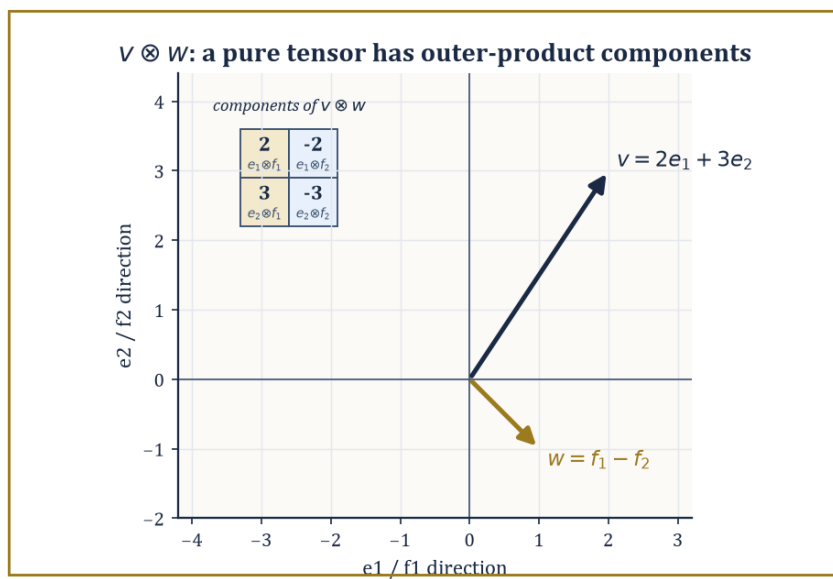
See the answer →

Floor 34: Tensors & Manifolds

Multilinear Algebra, Smooth Spaces, and the Art of Coordinate-Free Geometry

Room 34.1: Multilinear Algebra and Tensors

In which one vector at a time turns out to have been the beginner slope



The worked example $v \otimes w = 2e_1 \otimes f_1 - 2e_1 \otimes f_2 + 3e_2 \otimes f_1 - 3e_2 \otimes f_2$, with its components laid out as the outer product they are.

Numeric example — a linear map as a (1,1)-tensor.

Numeric example — an explicit tensor product.

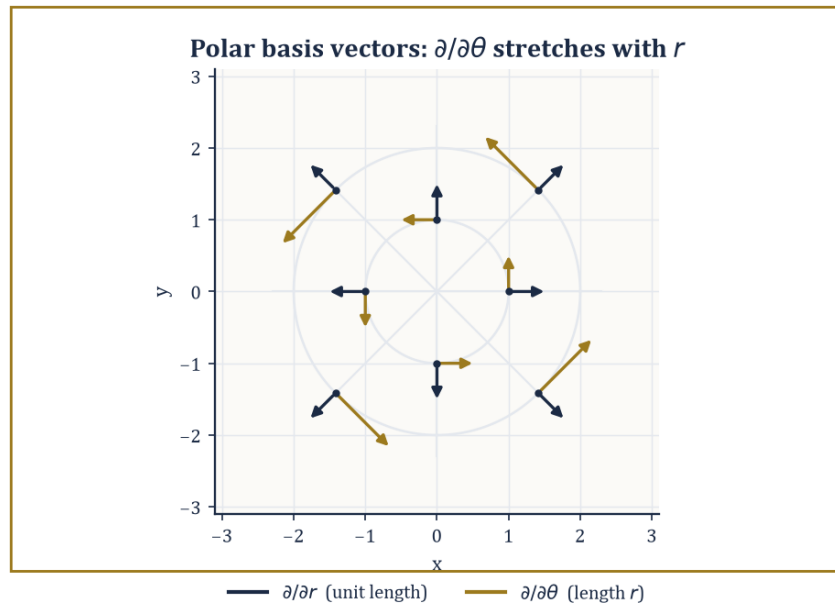
Numeric example — symmetrizing and alternating a bilinear form.

Numeric example — two transformation laws from one array of numbers.

Abstract example — the exterior square of $\mathbb{R}^3$.

Room 34.2: Tensor Fields and Transformation Laws

Or: how to tell a real tensor from an impostor with delusions



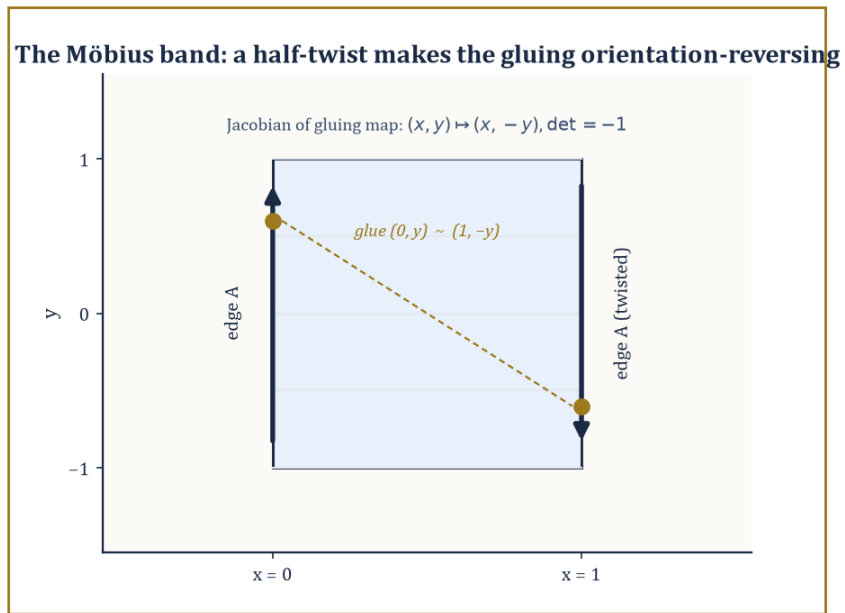
Why the Christoffel symbols are not tensor components: the polar basis vectors themselves rotate and stretch from point to point, even though the underlying plane is flat.

Numeric example — a covector field in polar coordinates.

Abstract example — why Christoffel symbols fail the test.

Room 34.3: Smooth Manifolds

A space that is locally boring and globally anything but



The half-twisted gluing $(0, y) \sim (1, -y)$ that makes the Möbius band the Tower's smallest counterexample to orientability.

Numeric example — the sphere via stereographic charts.

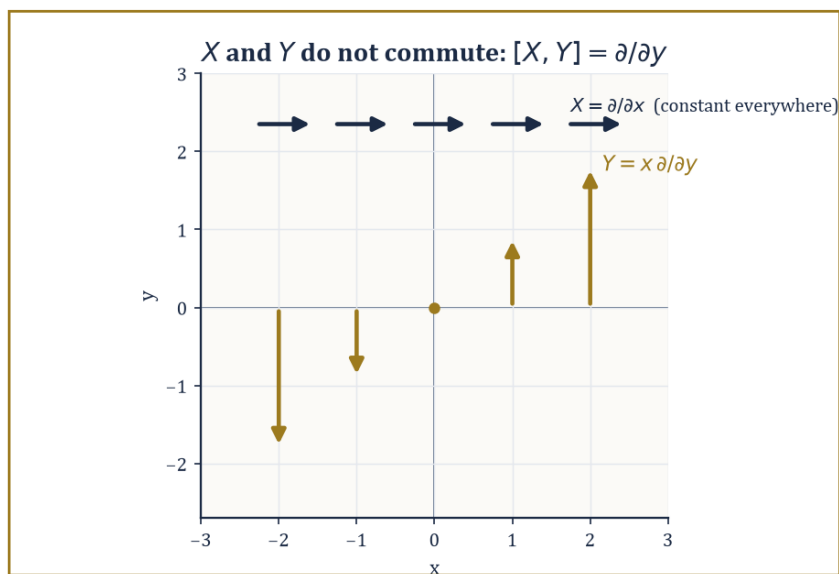
Abstract example — $GL(2, \mathbb{R})$ as an open manifold.

Numeric example — a diffeomorphism from an interval to the whole line.

Abstract example — the Möbius band fails to be orientable.

Room 34.4: Tangent Vectors, Vector Fields, and Flows

Three definitions of 'direction' that turn out to be secretly the same one



$X = \partial/\partial x$ is the same everywhere, but $Y = x \partial/\partial y$ grows and flips sign with x — which is why flowing along one and then the other is not the same as doing it in the opposite order.

Numeric example — two definitions of tangent vector agree.

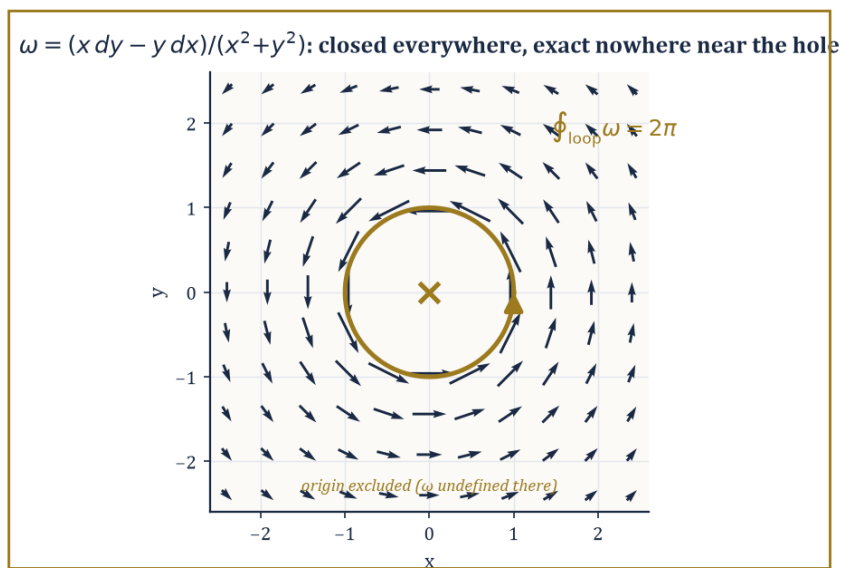
Numeric example — computing a Lie bracket.

Numeric example — a flow by direct solution.

Numeric example — dimension count for a tensor bundle.

Room 34.5: Differential Forms and Exterior Calculus

The totally antisymmetric tensors get their own weather system



The angle form ω , closed everywhere it is defined, circulating endlessly around the one point where it isn't — the puncture is the entire reason it fails to be exact.

Numeric example — computing an exterior derivative.

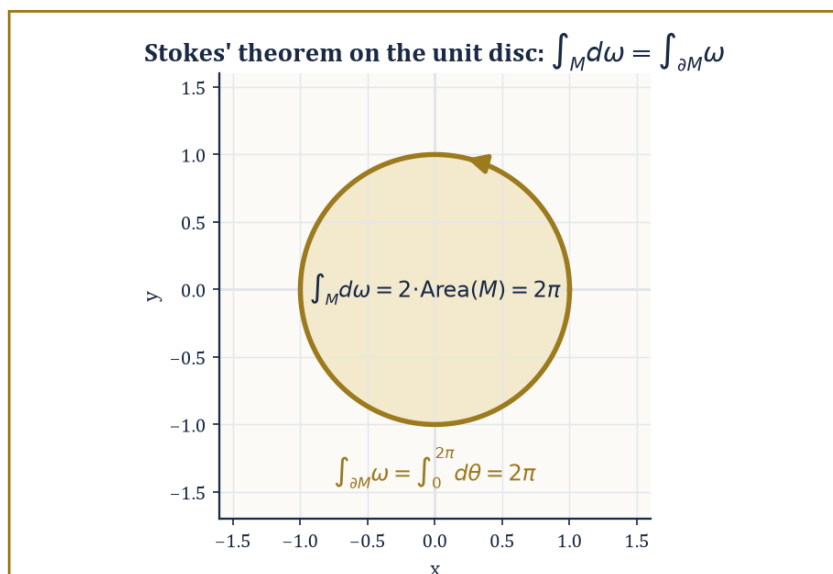
Abstract example — closed but not exact — the angle form.

Numeric example — pulling a form back along a circle.

Numeric example — checking Cartan's magic formula.

Room 34.6: Integration on Manifolds and Stokes' Theorem

Green, Gauss, and Stokes turn out to have been the same theorem all along



Two routes to 2π on the unit disc — integrate $d\omega$ over the interior, or ω over the boundary — arriving, as Stokes insists they must, at the same number.

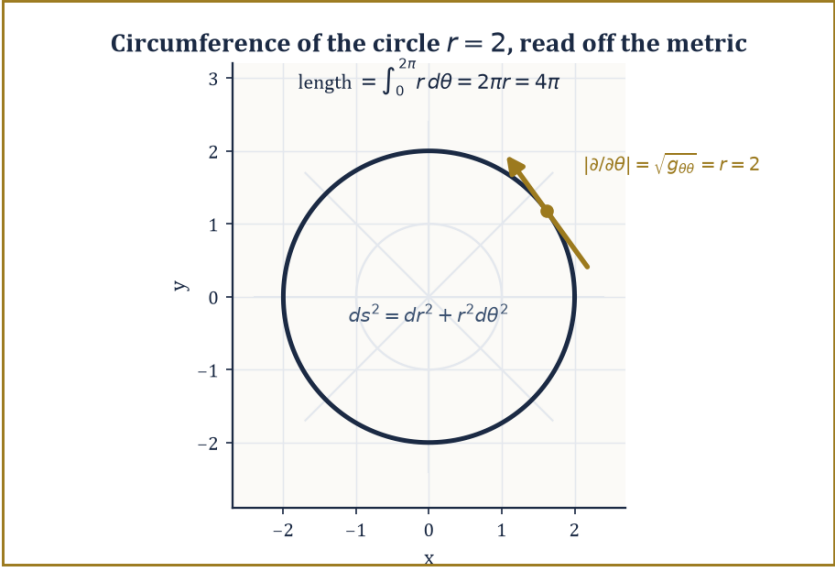
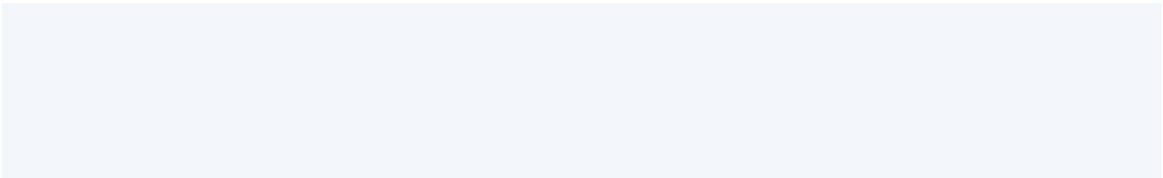
Numeric example — Stokes' theorem on the unit disc.

Numeric example — integrating over a boundaryless manifold.

Abstract example — what the hole obstructs.

Room 34.7: Riemannian Metrics

Where the manifold finally learns to measure itself



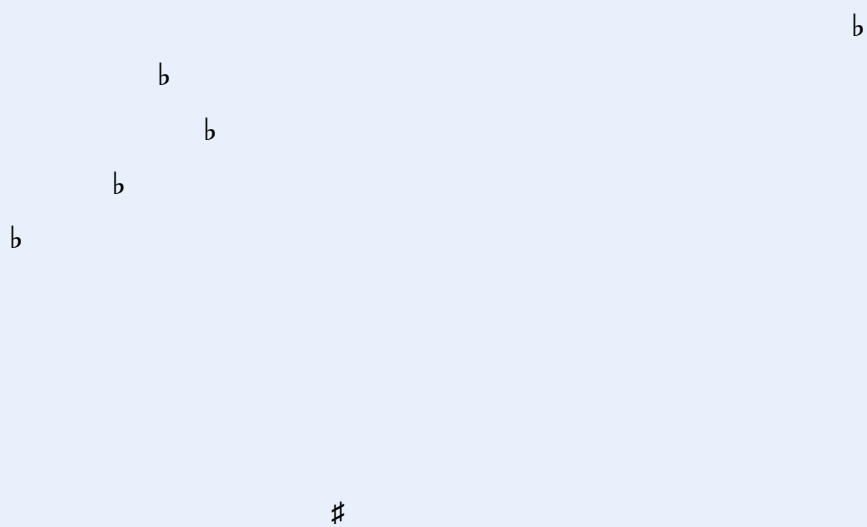
The circle $r=2$, its tangent direction $\partial/\partial\theta$ stretched to length r by the metric $g = \text{diag}(1, r^2)$, integrating out to the circumference 4π you already trusted.

b

#

Numeric example — circumference from the metric.

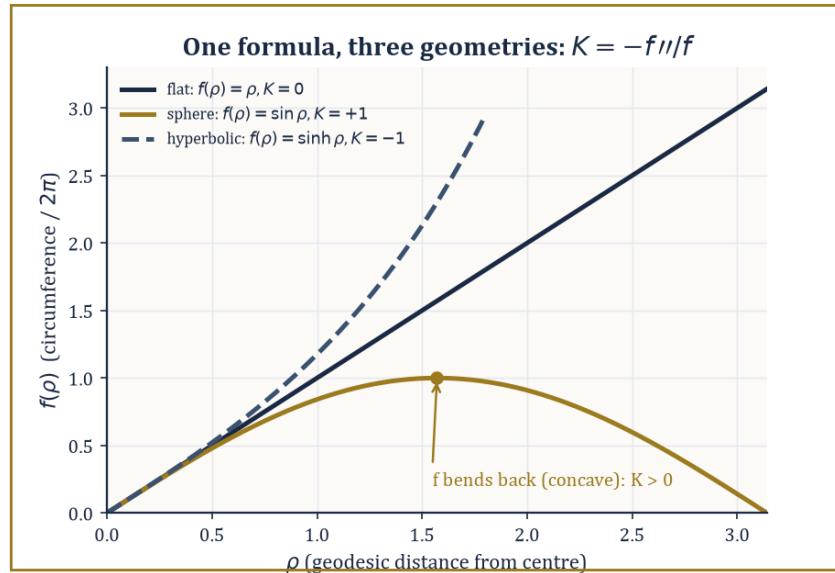
Numeric example — lowering an index with the musical isomorphism.



Numeric example — Christoffel symbols of the flat plane in polar coordinates.

Room 34.8: Connections, Covariant Derivatives, and Curvature

The room in which flatness finally gets a precise definition, and the sphere fails it magnificently



$K = -f''/f$, plotted for the same $f(\rho)$ in all three geometries at once: a straight line stays straight, a sine bends back on itself, a sinh runs away — zero, positive, and negative curvature, read directly off the shape of one function.

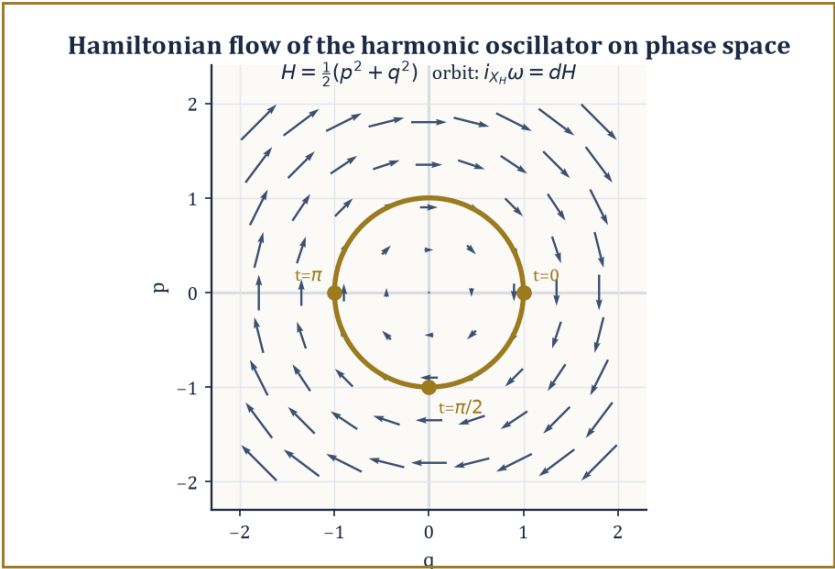
Numeric example — the flat plane really is flat, Christoffel symbols notwithstanding.

Numeric example — the plane, the sphere, and the saddle, unified by one formula.

Numeric example — holonomy on the unit sphere, via Gauss–Bonnet.

Room 34.9: Looking Ahead: From Manifolds to Riemannian Geometry, Gauge Theory, and Relativity

The grammar is finished; the essays are due next



The harmonic oscillator's orbit in phase space, riding the Hamiltonian vector field defined by $i_{X_H}\omega = dH$ — classical mechanics, revealed as one closed curve inside a symplectic swirl.



Numeric example — a round sphere under Ricci flow.

Abstract example — electromagnetism as curvature.

Numeric example — Hamilton's equations from a symplectic form.

Check Your Understanding — Floor 34

Answers are in the Answer Key at the back of the book — have a go before you look.

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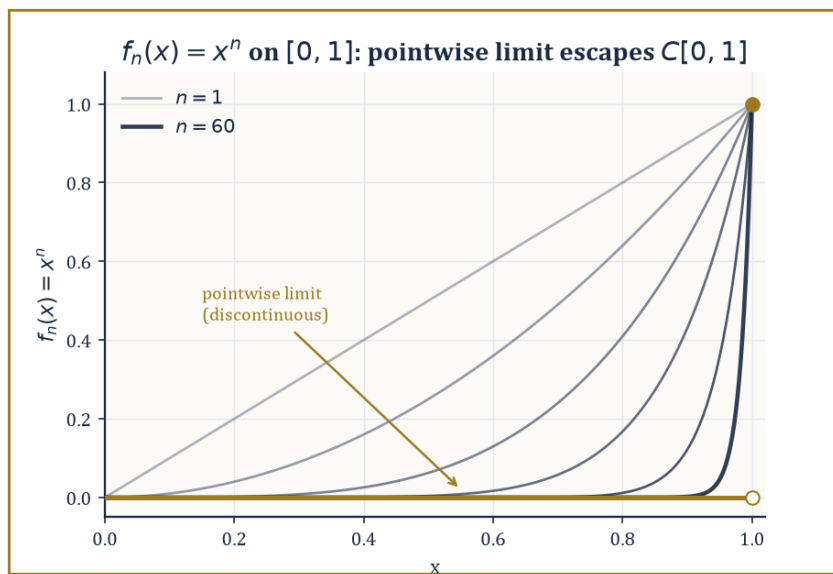
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Floor 35: Functional Analysis

Infinite-Dimensional Spaces, Operators, and the Art of Analysis in Function Spaces

Room 35.1: What Functional Analysis Is

In Which the Word 'Space' Stops Meaning What You Think It Means



Same sequence x^n , two topologies, two verdicts: pointwise it collapses to a jump at $x=1$; uniformly it never converges at all.

Abstract example — Why the topology is not optional.

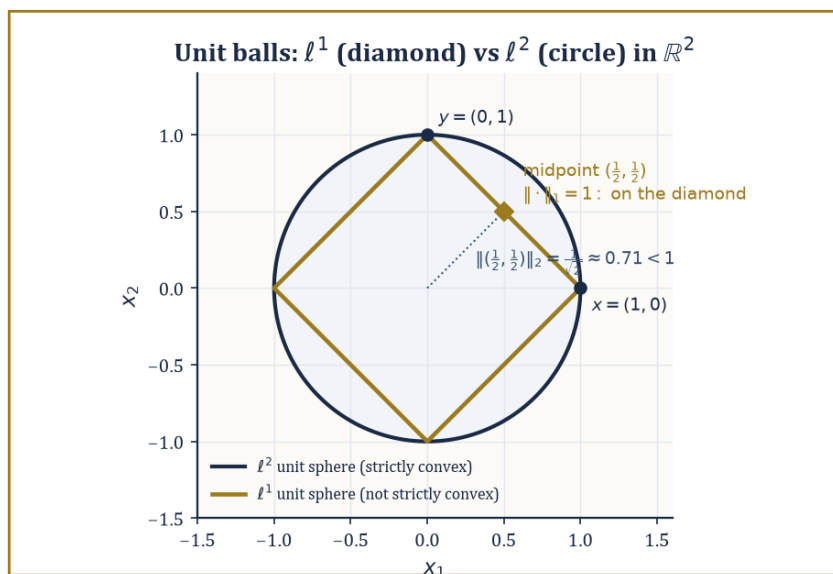
Abstract example — The same rule, legal in one space and meaningless in another.

Numeric example — A first citizen of the dual space.

Numeric example — A functional from the calculus of variations.

Room 35.2: Banach Spaces and Hilbert Spaces

Completeness: The One Thing You Cannot Do Without



The ℓ^1 diamond lets a whole edge sit on the unit sphere; the ℓ^2 circle bows a midpoint strictly inside — one figure, two verdicts on strict convexity.

Numeric example — What incompleteness feels like.

Numeric example — A norm computed in ℓ^2 .

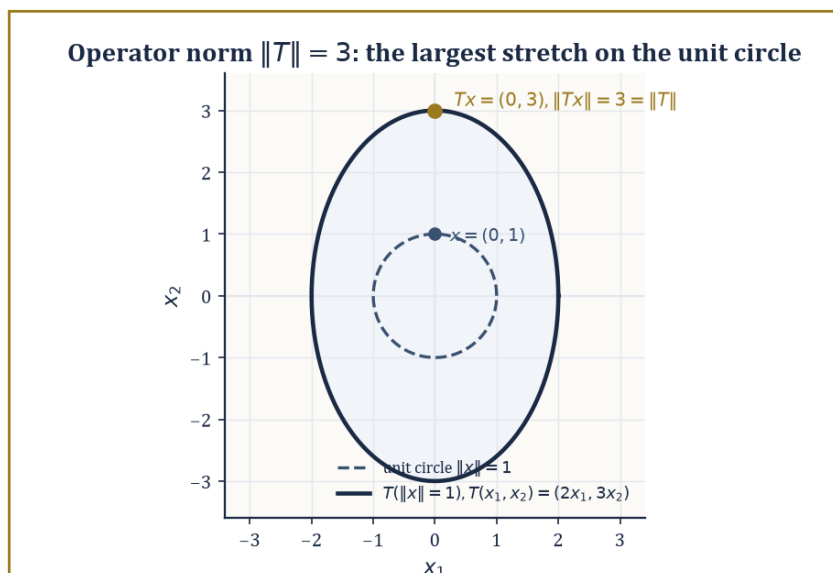
Numeric example — Riesz representation, finite-dimensional dress rehearsal.

Numeric example — ℓ^1 fails the parallelogram law.

Numeric example — ℓ^1 fails to be strictly convex; ℓ^2 succeeds.

Room 35.3: Linear Operators and Duality

Matrices, All Grown Up and Slightly Feral



T stretches the unit circle into an ellipse, and the operator norm is caught in the act at the one point that gets stretched the most.

Numeric example — Operator norm of a diagonal map.

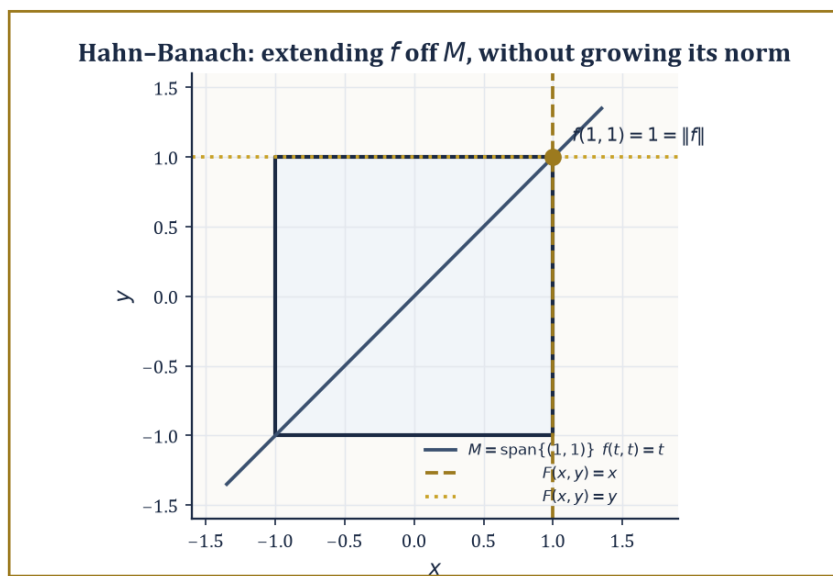
Abstract example — ℓ^1 is not reflexive, and it is ℓ^∞ 's fault.

Abstract example — The shift operator: a spectrum with no eigenvalues in it.

Numeric example — The bidual gap for c_0 , made concrete.

Room 35.4: The Fundamental Theorems

Three Theorems and a Quiet Funeral for Your Remaining Intuitions



In the sup-norm square, f 's norm-1 verdict on the diagonal M extends two different ways — $F(x, y) = x$ and $F(x, y) = y$ both hug the same corner without ever growing past it.

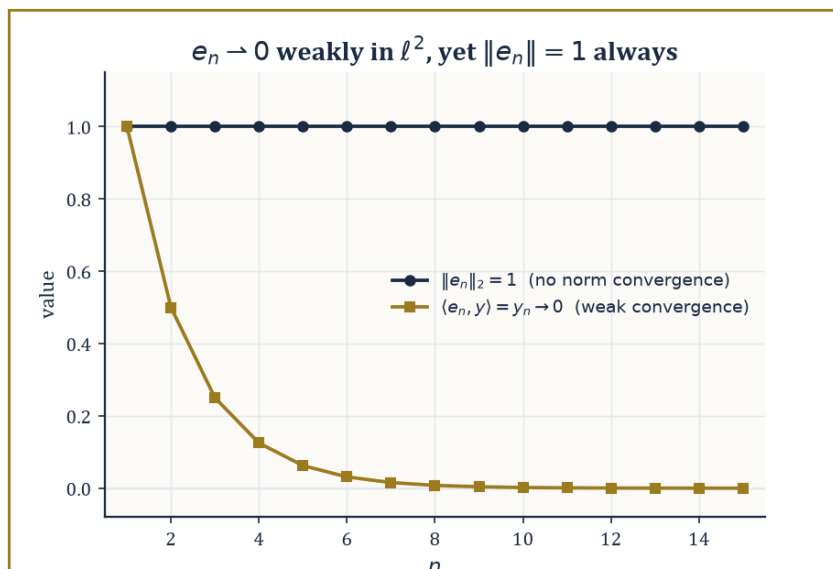
Numeric example — An extension, worked in full.

Abstract example — Banach-Steinhaus, run in reverse.

Abstract example — A graph that is not closed, and why that is fine.

Room 35.5: Weak Topologies and Reflexivity

Convergence, Now with Adjustable Standards



e_n never budes in norm, yet every honest pairing against it dies down to zero — the two faces of weak convergence, plotted side by side.

Numeric example — Weak convergence without norm convergence.

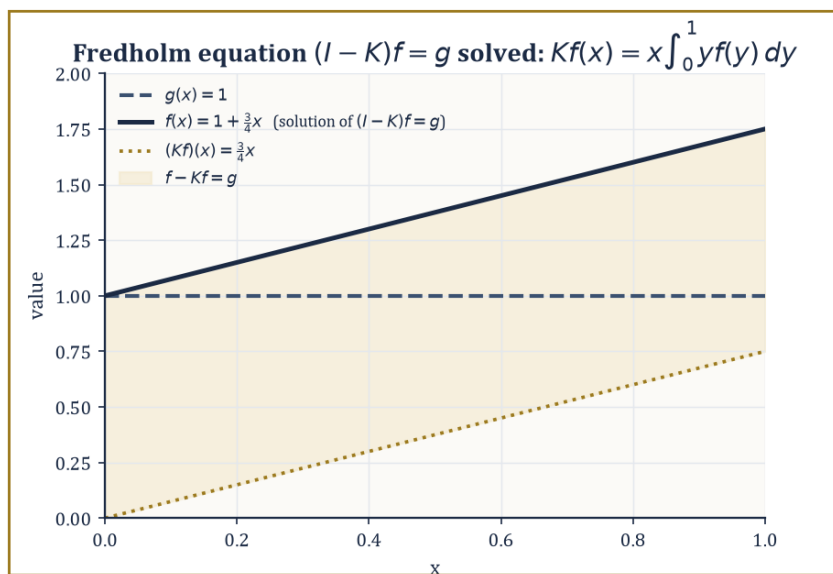
Numeric example — Unbounded norm disqualifies weak convergence outright.

Abstract example — Alaoglu in action.

Abstract example — ℓ^1 again: this time, failing to be reflexive on purpose.

Room 35.6: Compact Operators and the Fredholm Alternative

Where Infinite Dimensions Grudgingly Behave Like Matrices Again



The exact worked case $g \equiv 1$: solving $(I - K)f = g$ by hand hands back the straight line $f(x) = 1 + \frac{3}{4}x$, and the shaded strip is the gap K quietly closes.

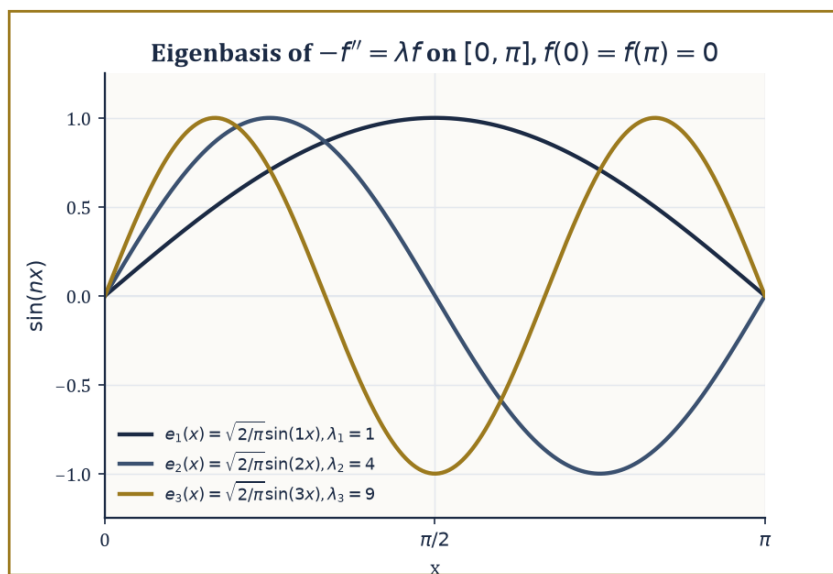
Numeric example — A rank-one integral operator and its eigenvalue.

Numeric example — Compactness of K , verified from the definition.

Numeric example — Solving $(I - K)f = g$ explicitly.

Room 35.7: Spectral Theory of Compact Self-Adjoint Operators

Diagonalization, Rescued from the Wreckage



The first three eigenfunctions of $-f'' = \lambda f$ on $[0, \pi]$ — the ordinary Fourier sine basis, caught in the act of being an eigenbasis.

Numeric example — Diagonalizing the rank-one operator from Room 35.6.

Abstract example — Why compact self-adjoint eigenvalues must be real.

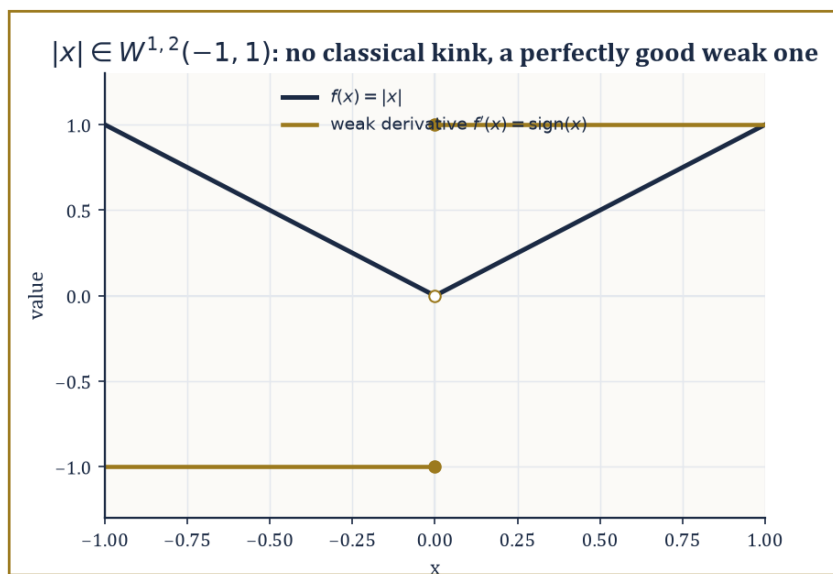
Numeric example — The classical Fourier sine basis as an eigenbasis.

Numeric example — Checking the Rayleigh quotient at the eigenvector.

Numeric example — Checking the second rung of the ladder.

Room 35.8: Unbounded Operators and Sobolev Spaces

Differentiation Refuses to Be Tamed, So We Negotiate Instead



$|x|$ has a kink no classical derivative can survive, but $\text{sign}(x)$ is standing by as its perfectly respectable weak derivative.

Numeric example — Differentiation is unbounded.

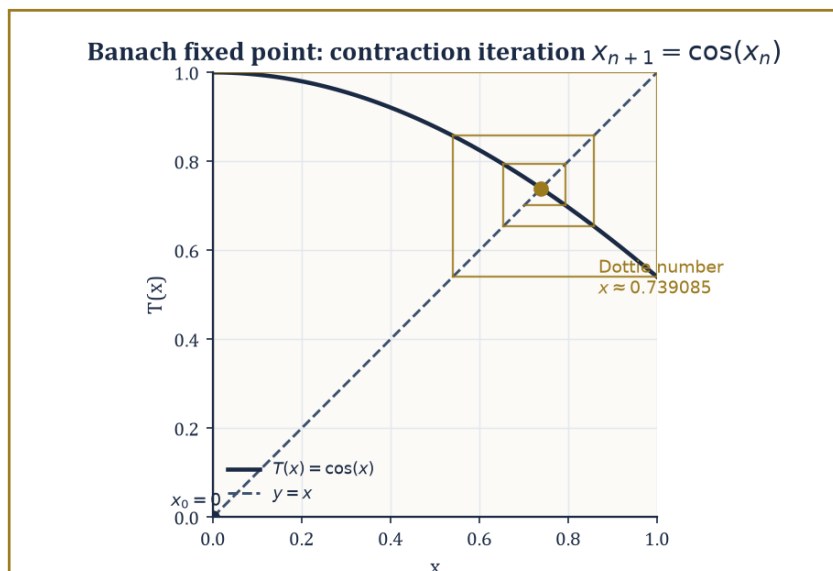
Abstract example — Symmetric but not self-adjoint: momentum on an interval.

Numeric example — The weak derivative of $|x|$, checked against a test function.

Numeric example — A compact embedding, watched in real time.

Room 35.9: Looking Ahead: From Functional Analysis to Operator Algebras, Distributions, and PDEs

The Tower Keeps Building on What You Just Learned to Trust



The cobweb of $x_0=0$ under $T(x)=\cos(x)$, spiraling obediently into the Dottie number — completeness and a contraction doing all the work by themselves.

Numeric example — The C*-identity, checked on a 2×2 matrix.

Abstract example — A type I factor you already know.

Numeric example — The derivative of a jump is a spike.

Numeric example — A contraction, iterated by hand.

Check Your Understanding — Floor 35

Answers are in the Answer Key at the back of the book — have a go before you look.

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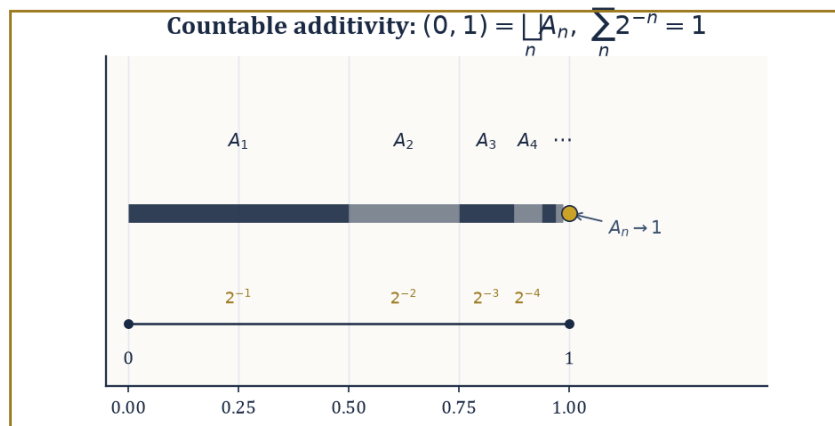
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Floor 36: Measure Theory

Length, Area, Volume, and the Art of Integrating Almost Everywhere

Room 36.1: What Measure Theory Is

In Which Length Grows Up and Gets a Proper Job



The interval $(0,1)$ split into countably many shrinking pieces $A_n = [1-2^{-(n-1)}, 1-2^{-n})$, whose lengths 2^{-n} sum, with countable additivity's blessing, to exactly 1.

Numeric example — where Riemann gives up.

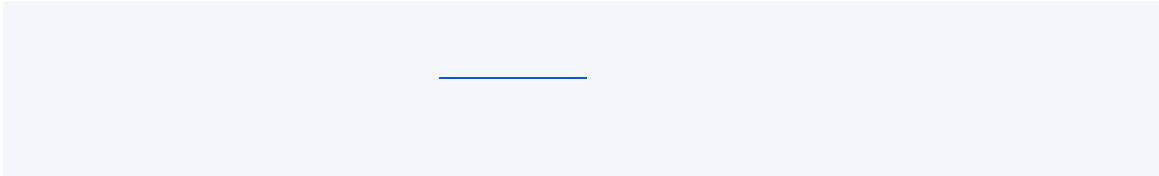
Numeric example — countable additivity in action.

Numeric example — agreement on a continuous function.

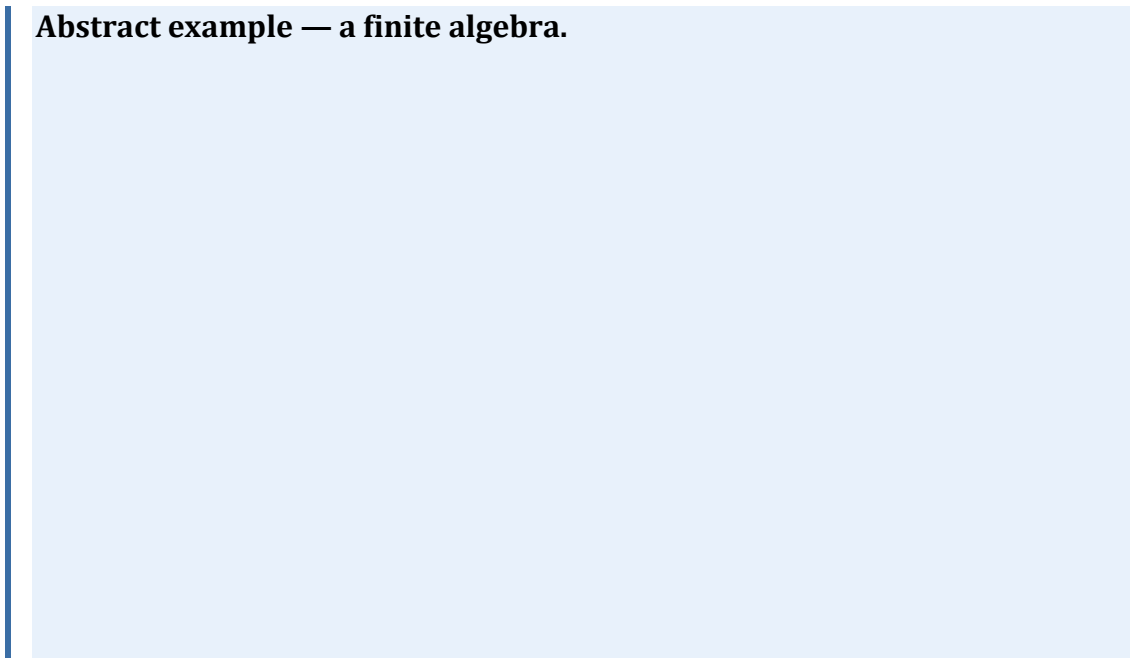
Numeric example — probability as measure, expectation as integral.

Room 36.2: Algebras, σ -Algebras, and Measures

Bureaucracy for Sets, and Why the Countable Version Is the One That Matters



Abstract example — a finite algebra.



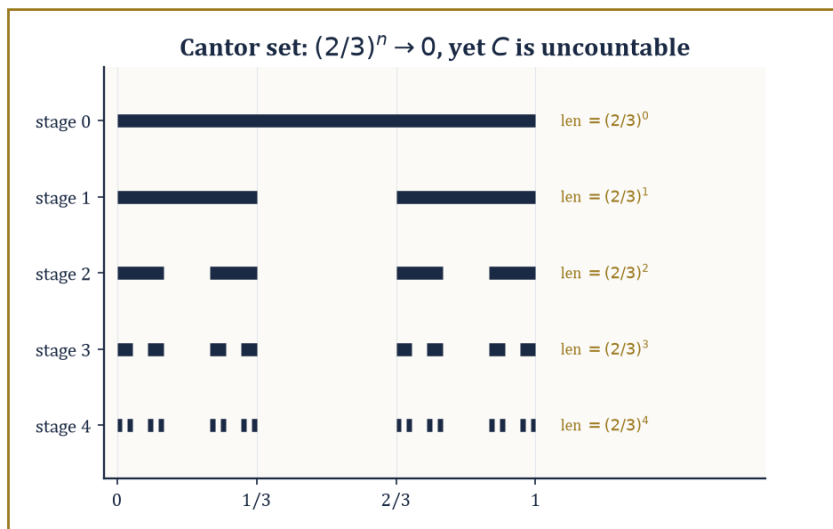
Abstract example — an algebra that is not a σ -algebra.

Numeric example — three flavours of measure.

Abstract example — why cumulative distribution functions determine measures.

Room 36.3: Outer Measure and Lebesgue Measure

Building Size From the Outside In, and Meeting the One Set That Refuses to Play



Five stages of deleting middle thirds: the remaining length shrinks geometrically to zero, yet what survives in the limit is an uncountable set — measure zero and cardinality of the continuum, comfortably at once.

Numeric example — a finite set has outer measure zero.

Abstract example — the Vitali set is not measurable.

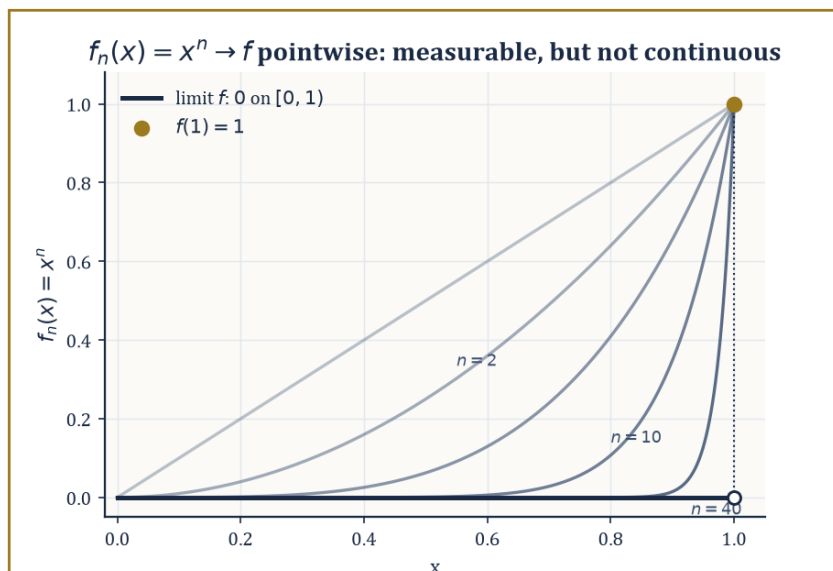
Numeric example — measuring a rectangle.

Numeric example — the Cantor set has measure zero.

—

Room 36.4: Measurable Functions

The Minimum Standard of Behaviour Required Before Integration Is Even Discussed



The continuous curves $f_n(x)=x^n$ pile up, pointwise, into a limit with a genuine jump at $x=1$ — proof that measurability survives a limit far more graciously than continuity ever does.

Numeric example — checking measurability directly.

Numeric example — a discontinuous pointwise limit that is still measurable.

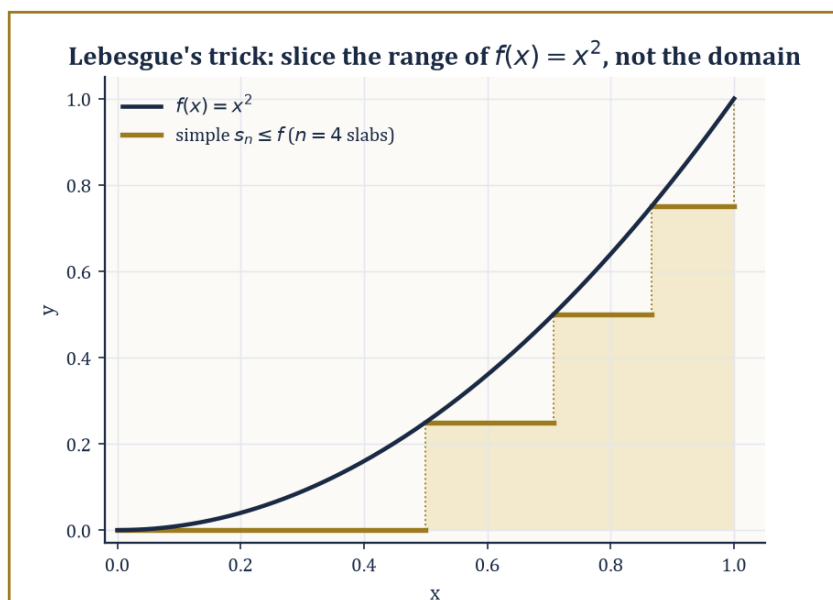
Numeric example — approximating from below with a simple function.

Abstract example — sums of measurable functions are measurable.

Numeric example — essential supremum ignores a single bad point.

Room 36.5: The Lebesgue Integral

Slicing the Range Instead of the Domain, Which Is Cleverer Than It Sounds



Four horizontal slabs approximate $f(x)=x^2$ from below — Lebesgue's entire innovation rendered as a staircase hugging the curve from underneath, rather than Riemann's vertical strips.

Numeric example — integrating a simple function directly.

Numeric example — recovering a familiar integral by slicing the range.

Numeric example — splitting into positive and negative parts.

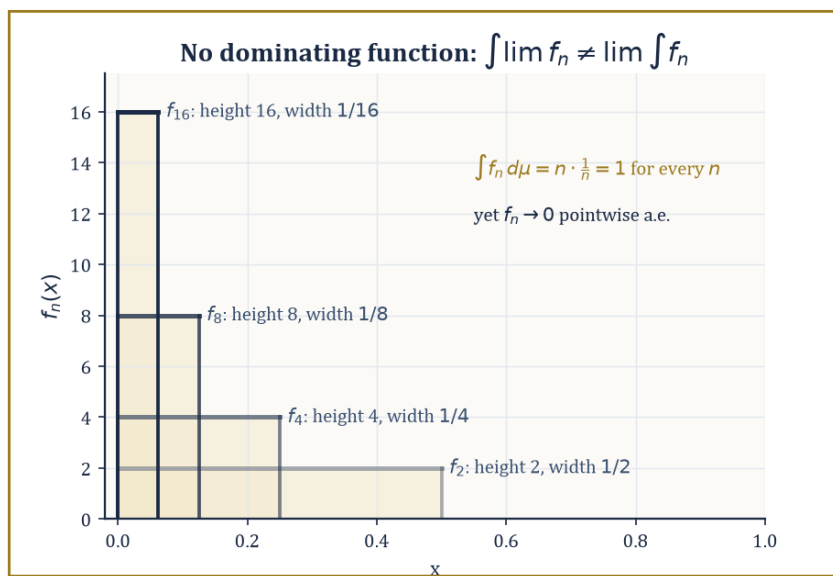
Numeric example — resolving the Dirichlet function at last.

Abstract example — why absolute integrability matters.

Numeric example — linearity of the integral confirmed directly.

Room 36.6: Convergence Theorems

The Actual Point of the Whole Exercise



Spikes $f_n = n \cdot 1_{[0, 1/n]}$ all carry integral exactly 1 while collapsing pointwise to 0 — the standard portrait of a sequence with no integrable function large enough to dominate it.

Numeric example — monotone convergence in action.

Numeric example — Fatou's lemma with strict inequality (the moving bump).

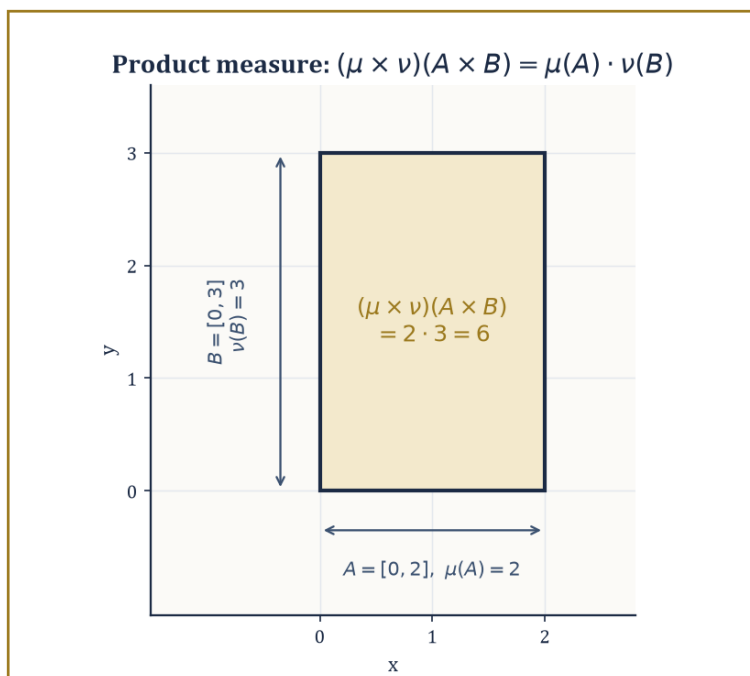
Numeric example — dominated convergence resolving a limit cleanly.

Numeric example — no dominating function, and the limit fails.

Abstract example — convergence in measure without pointwise convergence anywhere.

Room 36.7: Product Measures and Fubini's Theorem

Changing the Order of Integration Without Getting Sued



The measurable rectangle $[0,2] \times [0,3]$: product measure turns out to be nothing more mysterious than the ordinary area you already knew how to compute.

Numeric example — a product measure recovers ordinary area.

Numeric example — Fubini's theorem confirmed directly.

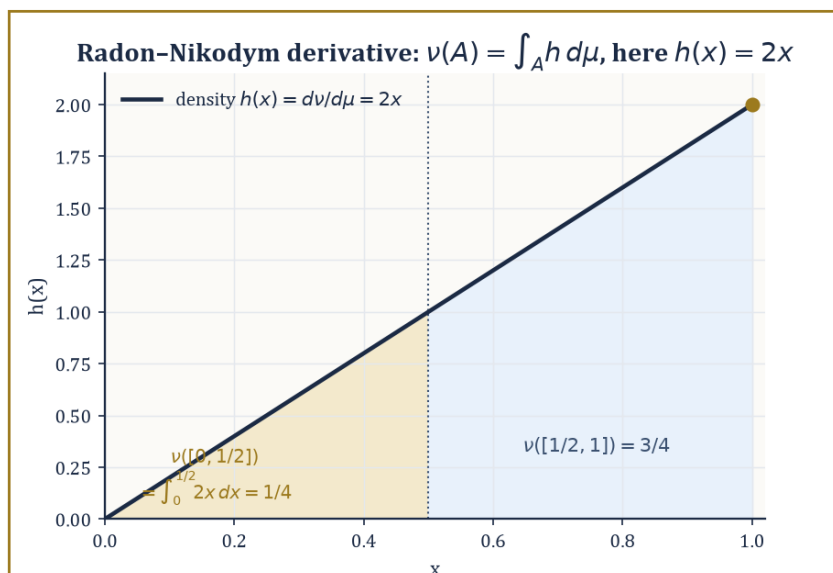
Numeric example — Tonelli with an infinite answer.

Numeric example — a function where order of integration matters.

Numeric example — an infinite product measure for coin flips.

Room 36.8: L^p Spaces and the Radon–Nikodym Theorem

In Which the Dinner Guests From the Opening Finally Receive Their Papers



The density $h(x)=2x$ weights the upper half of $[0,1]$ three times as heavily as the lower half — $\nu([0, \frac{1}{2}])=\frac{1}{4}$ against $\nu([\frac{1}{2}, 1])=\frac{3}{4}$, read straight off the areas under the line.

Numeric example — three norms of the same function.

Numeric example — checking Hölder and Minkowski on a concrete pair.

Numeric example — an explicit dual pairing.

Numeric example — a density and its Radon–Nikodym derivative.

Abstract example — decomposing a measure with an atom.

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Check Your Understanding — Floor 36

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1.

See the answer →

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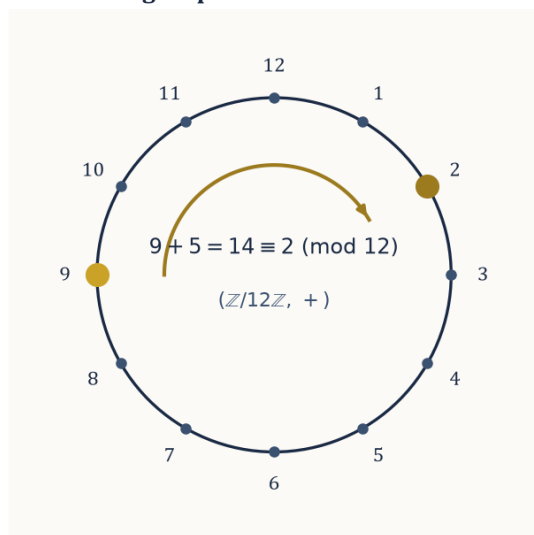
Floor 37: Abstract Algebra

Groups, Rings, Fields, and the Art of Structure for Its Own Sake

Room 37.1: What Abstract Algebra Is

In Which We Stop Caring What the Symbols Stand For

Clock arithmetic: a group of twelve costumes for one skeleton



Nine o'clock plus five hours is two o'clock — the same three-line checklist that governs the integers, wearing a wristwatch.

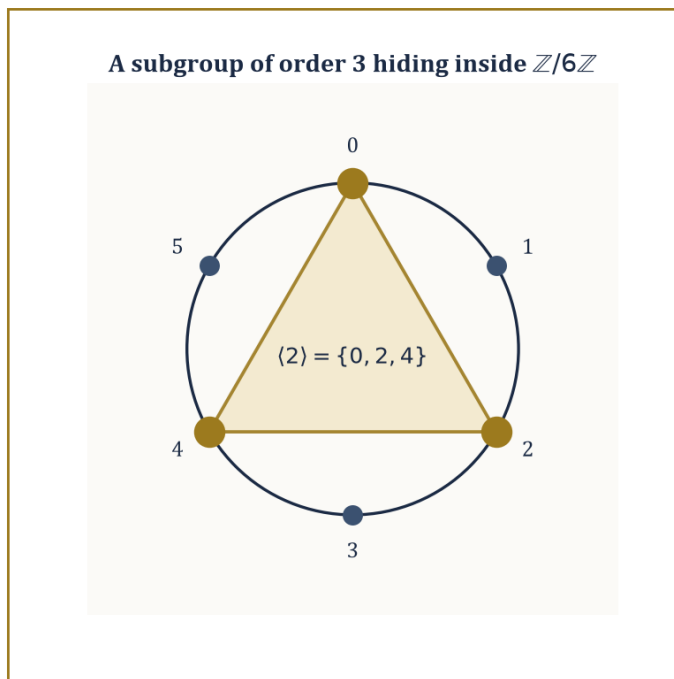
Abstract example — Two systems, one skeleton.

Abstract example — Same axioms, different costumes.

Numeric example — Spotting a disguise.

Room 37.2: Groups and Homomorphisms

Addition, Symmetry, and Rubik's Cube, Unified by Spite

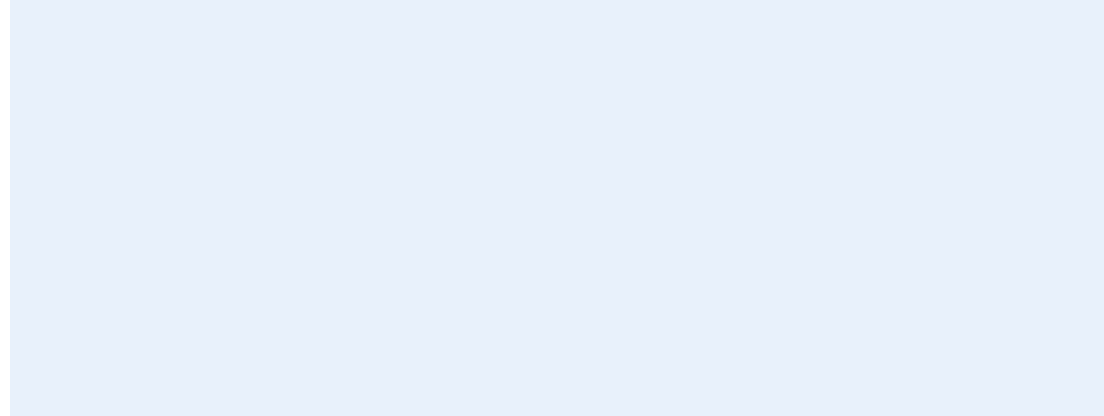


The element 2 doesn't generate all of $\mathbb{Z}/6\mathbb{Z}$ — it generates exactly the golden triangle, a subgroup of order 3, because 3 divides 6.

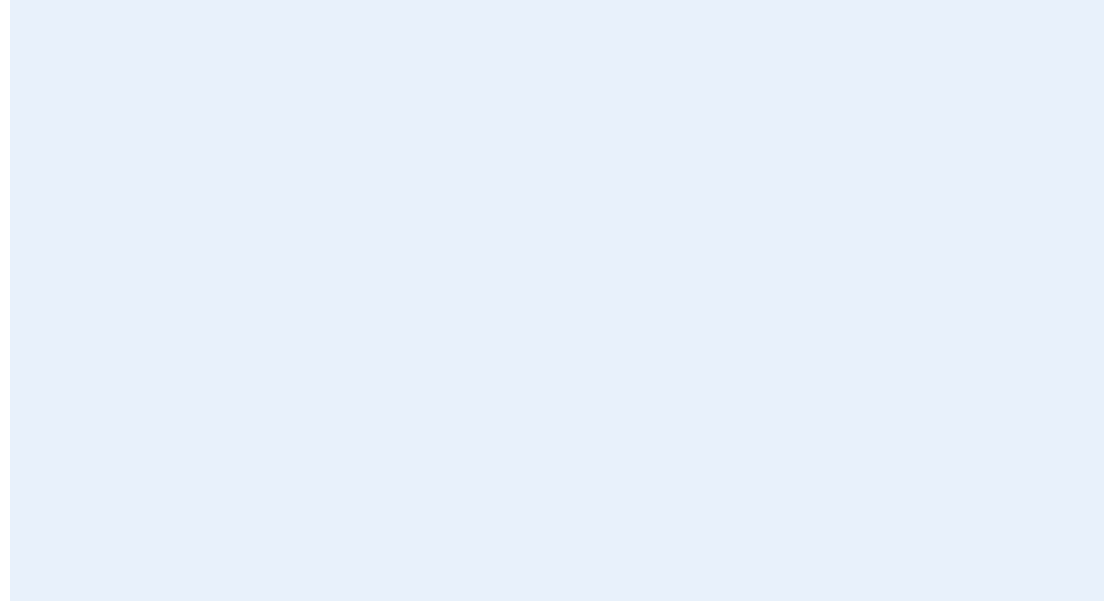
Numeric example — The smallest genuinely non-abelian group.

Numeric example — Reduction mod 5.

Numeric example — A subgroup hiding inside a clock.

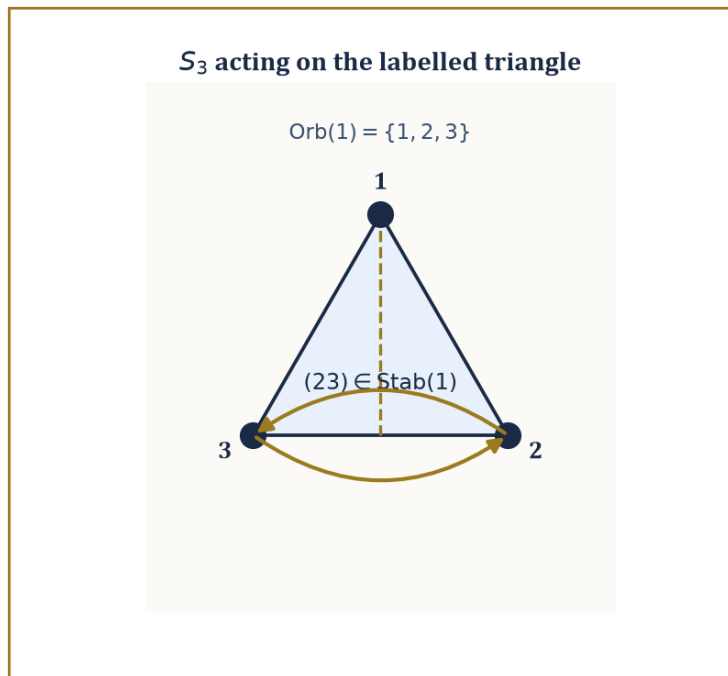


Numeric example — What Lagrange forbids.



Room 37.3: Group Actions, Cosets, and Quotient Groups

Groups Doing Things to Sets, and the Bookkeeping That Follows



The reflection (23) is the one symmetry that leaves vertex 1 where it found it — everything $\text{Stab}(1)$ means, drawn as a single fold of the triangle.

Numeric example — S_3 moving three points.

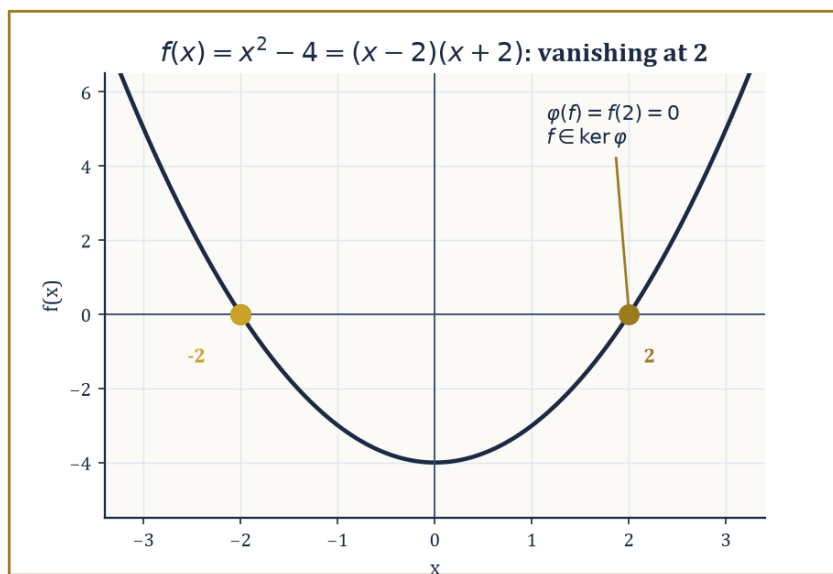
Numeric example — Counting orbit and stabilizer in S_3 .

Numeric example — Building $\mathbb{Z}/n\mathbb{Z}$, and a quotient inside S_3 .

Abstract example — The smallest non-abelian simple group.

Room 37.4: Rings, Ideals, and Quotient Rings

Two Operations Are Twice the Fun and Considerably More Than Twice the Bookkeeping



$x^2 - 4$ lands in the kernel of 'evaluate at 2' for the most literal reason possible: its graph passes right through the point.

Numeric example — Distributivity, checked by hand.

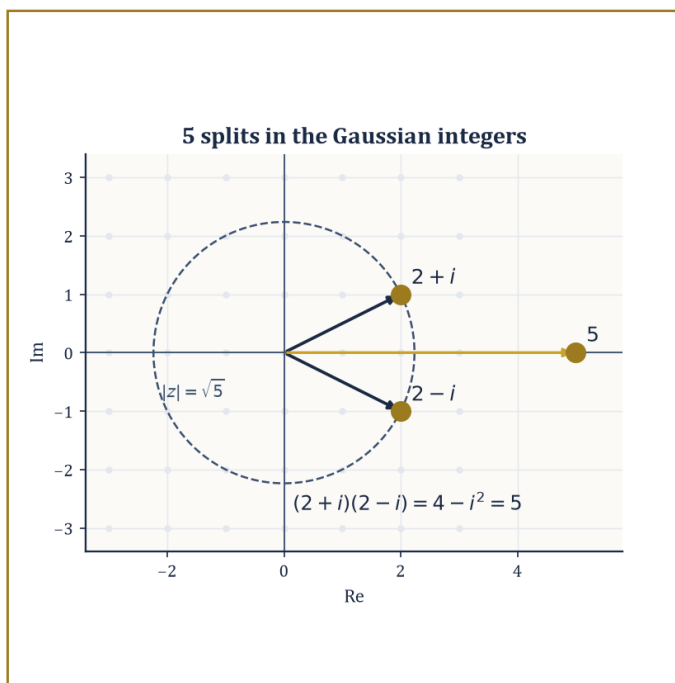
Numeric example — $3\mathbb{Z}$ absorbs everything you throw at it.

Numeric example — 5 is prime, 6 is not, and their quotients know it.

Numeric example — Evaluating a polynomial at $x=2$.

Room 37.5: Factorization in Integral Domains

Where Unique Factorization Turns Out to Be a Privilege, Not a Right



5 looks stubbornly prime from the real line, but step into the complex plane and it visibly splits into $2+i$ and its mirror image.

Numeric example — Cancellation succeeds and fails.

Numeric example — The Euclidean algorithm, the mechanism underneath it all.

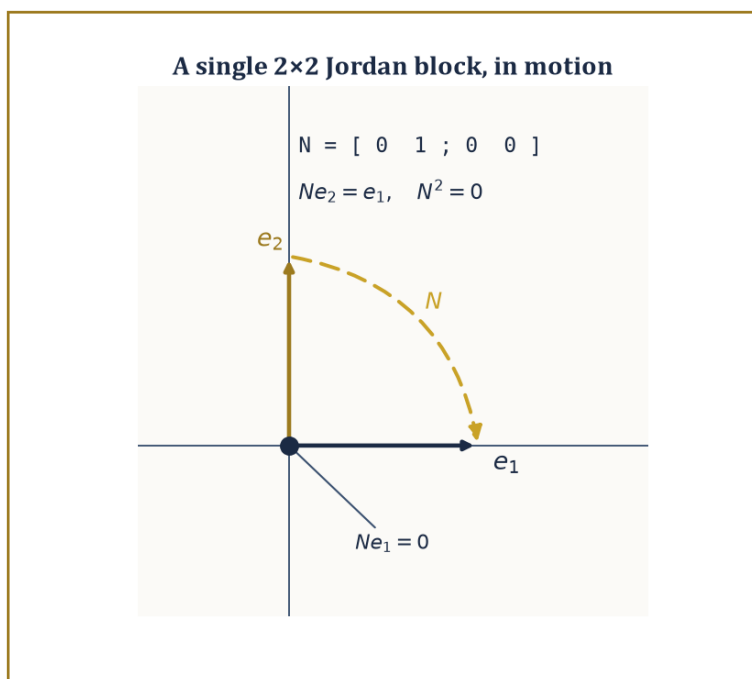
Numeric example — 5 stops being prime in $\mathbb{Z}[i]$.

Numeric example — Two different factorizations of 6.

Numeric example — $x^3 - 2$ is irreducible, by Eisenstein.

Room 37.6: Modules over Rings

Vector Spaces That Lost Their Field and Gained a Ring Instead



N pushes e_2 down onto e_1 and e_1 down onto nothing — one step from an eigenbasis, which is why a Jordan block is needed at all.

Numeric example — The clock as a module over the integers.

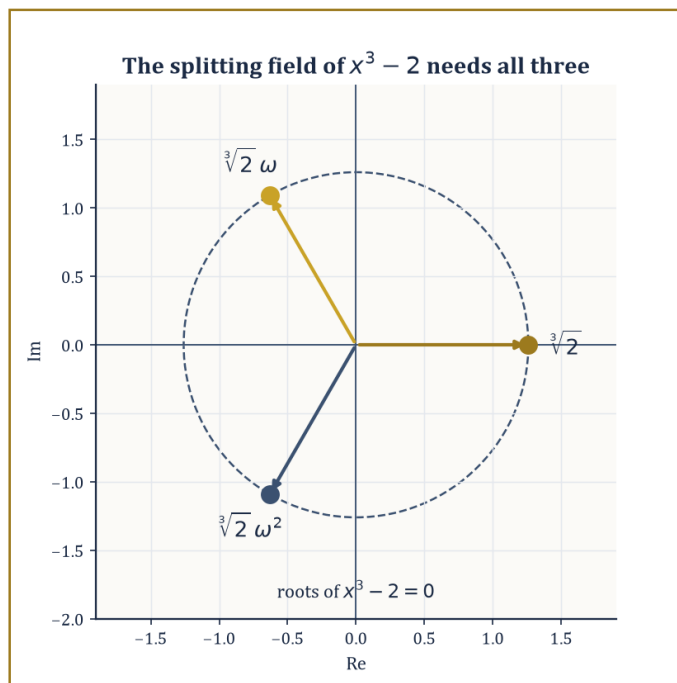
Worked example — A short exact sequence you can hold in your hand.

Numeric example — Every abelian group of order 12, listed exhaustively.

Numeric example — A 2×2 Jordan block, made concrete.

Room 37.7: Field Extensions

Building Bigger Number Systems by Politely Insisting a Root Exists



$\mathbb{Q}(\sqrt[3]{2})$ alone catches only one of these three roots — the splitting field has to reach for the other two, out in the complex plane, at a cost of degree 6.

Numeric example — $[\mathbb{Q}(\sqrt{2}):\mathbb{Q}] = 2$, basis and all.

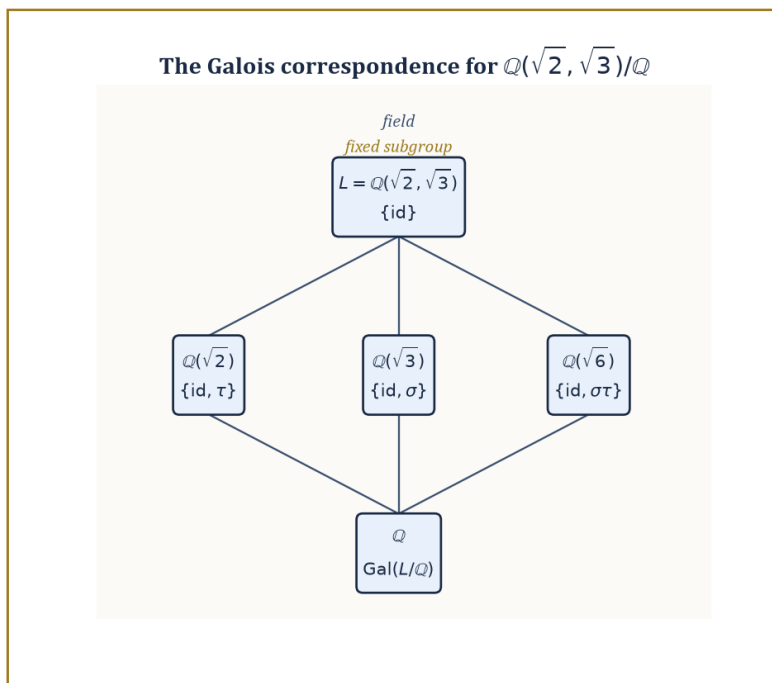
Numeric example — Building $\mathbb{Q}(\sqrt[3]{2})$ to order.

Numeric example — Two splitting fields, one small and one not.

Numeric example — The tower law, brick by brick.

Room 37.8: Introduction to Galois Theory

In Which Symmetry Groups Reveal Which Equations Can Be Solved



Bigger field, smaller subgroup: the lattice of intermediate fields and the lattice of subgroups mirror each other exactly, one node per node.

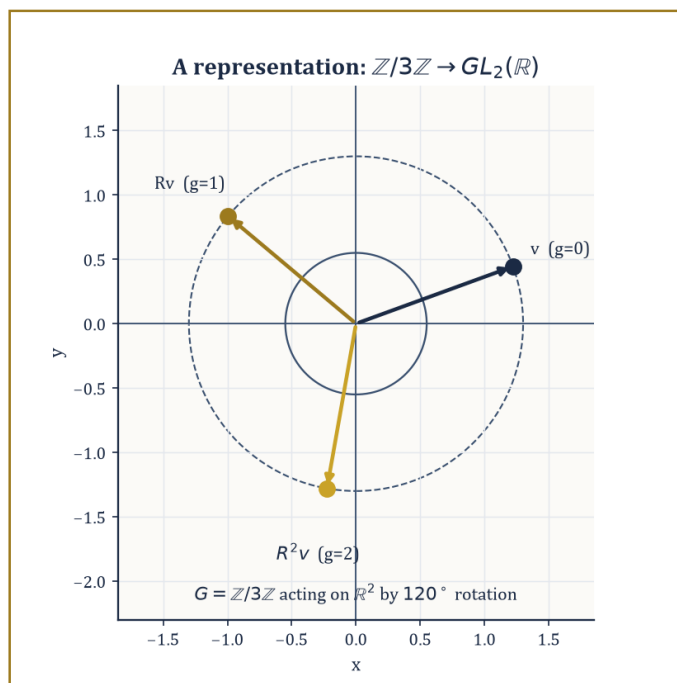
Numeric example — The Galois group of $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$, found by hand.

Numeric example — Four automorphisms, three intermediate fields, one Klein four-group.

Abstract example — Why S_5 refuses to cooperate.

Room 37.9: Looking Ahead: From Abstract Algebra to Algebraic Geometry, Number Theory, and Representation Theory

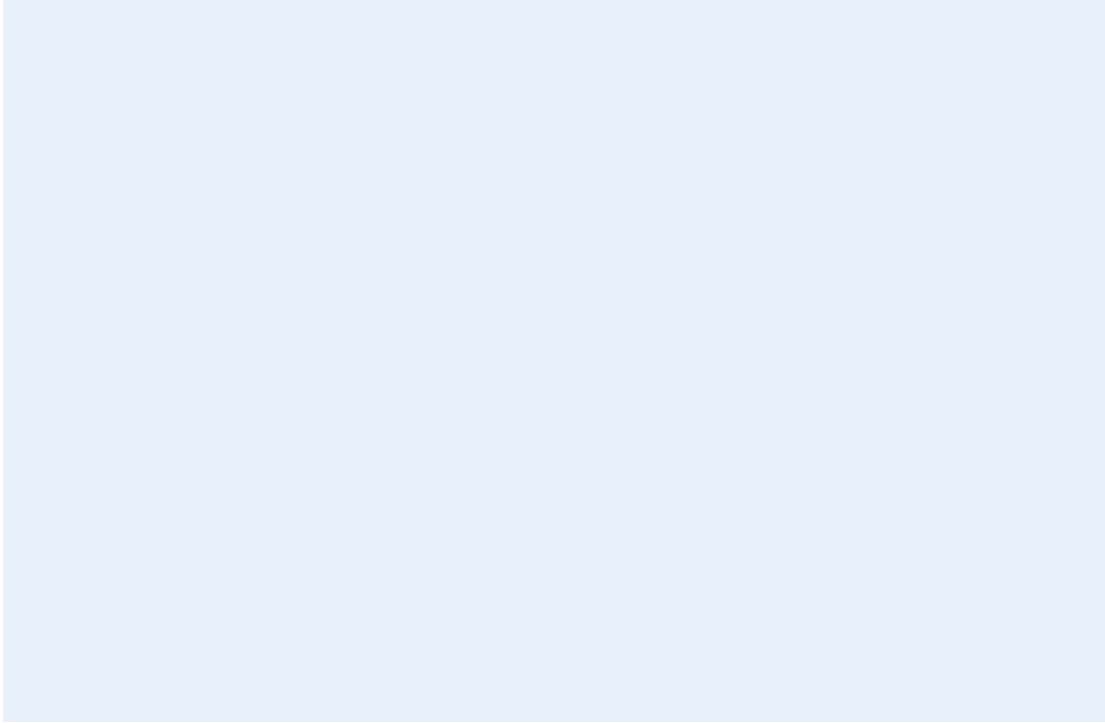
Where the Structure We Just Built Goes to Work



The abstract group $\mathbb{Z}/3\mathbb{Z}$, made concrete: three applications of a 120° rotation return a vector to where it started.

Abstract example — The ring of integers as a bridge to geometry.

Numeric example — Rotating the plane by a third of a turn.



Check Your Understanding — Floor 37

Answers are in the Answer Key at the back of the book — have a go before you look.

1.

See the answer →

2.

See the answer →

3.

See the answer →

4.

See the answer →

Epilogue — Pausing on the Landing

Answer Key

Answers to every “Check Your Understanding” question, grouped by floor and in the same order they were asked.

Floor 26: Differential and Integro-Differential Equations

1.

[← Back to question](#)

2.

[← Back to question](#)

3.

[← Back to question](#)

4.

[← Back to question](#)

Floor 27: Partial Differential Equations

1.

[← Back to question](#)

2.

[← Back to question](#)

3.

[← Back to question](#)

4.

[← Back to question](#)

Floor 28: Optimization & Calculus of Variations

1.

[← Back to question](#)

2.

[← Back to question](#)

3.

[← Back to question](#)

4.

[← Back to question](#)

Floor 29: Numerical Methods

1.

[← Back to question](#)

2.

[← Back to question](#)

3.

[← Back to question](#)

4.

[← Back to question](#)

Floor 30: Optimization

1.

[← Back to question](#)

2.

[← Back to question](#)

3.

[← Back to question](#)

4.

[← Back to question](#)

Floor 31: Differential Geometry

1.

[← Back to question](#)

2.

[← Back to question](#)

3.

[← Back to question](#)

4.

[← Back to question](#)

Floor 32: Topology

1.

[← Back to question](#)

2.

[← Back to question](#)

3.

[← Back to question](#)

4.

[← Back to question](#)

Floor 33: Integral Geometry (Advanced)

1.

[← Back to question](#)

2.

[← Back to question](#)

3.

[← Back to question](#)

4.

[← Back to question](#)

Floor 34: Tensors & Manifolds

1.

[← Back to question](#)

2.

[← Back to question](#)

3.

[← Back to question](#)

4.

[← Back to question](#)

Floor 35: Functional Analysis

1.

[← Back to question](#)

2.

[← Back to question](#)

3.

[← Back to question](#)

4.

[← Back to question](#)

Floor 36: Measure Theory

1.

[← Back to question](#)

2.

[← Back to question](#)

3.

[← Back to question](#)

4.

[← Back to question](#)

Floor 37: Abstract Algebra

1.

[← Back to question](#)

2.

[← Back to question](#)

3.

[← Back to question](#)

4.

[← Back to question](#)

Symbol & Notation Glossary

Notation particular to, or redefined within, each floor — organised by where it first appears rather than strict alphabetical order, since the same symbol can mean different things on different floors.

Floor 26: Differential and Integro-Differential Equations

$\Phi(t)$ —

e^{At} —

$\dot{V}(x)$ —

$\lambda_1 = \min_y R[y]$ —

$G(x, \xi)$ —

$K * u$ —

$R(t, s)$ —

Floor 27: Partial Differential Equations

u_t, u_{xx}, u_{xy} —

$B^2 - 4AC$ —

λ —

$\Phi(x, t)$ —

Δ —

$$G(\mathbf{x}, \xi) \text{ —}$$

Floor 28: Optimization & Calculus of Variations

$$\nabla f(\mathbf{x}^*) = 0 \text{ —}$$

$$H(\mathbf{x}^*) \text{ —}$$

$$\nabla f = \lambda \nabla g \text{ —}$$

$$\lambda \cdot g(\mathbf{x}) = 0 \text{ —}$$

$$d/dx(\partial L/\partial y') = \partial L/\partial y \text{ —}$$

$$S = \int L \, dt \text{ —}$$

Floor 29: Numerical Methods

$$O(h^n) \text{ —}$$

$$\xi \text{ —}$$

$$r = \Delta t / \Delta x^2 \text{ —}$$

$$a\Delta t / \Delta x \text{ (Courant number) —}$$

$$PA = LU \text{ —}$$

$$J(\mathbf{x}) \text{ —}$$

Floor 30: Optimization

$\nabla f(\mathbf{x})$ —

$\mathbf{H}(\mathbf{x})^{-1} \nabla f(\mathbf{x})$ —

$L(\mathbf{x}, \lambda)$ —

$\text{prox}_{\mathbf{h}}(\mathbf{v})$ —

$g(\lambda)$ —

$\partial F(\mathbf{x})$ —

Floor 31: Differential Geometry

$\kappa(\mathbf{s})$ —

$\boldsymbol{\tau}$ —

$\{\mathbf{T}, \mathbf{N}, \mathbf{B}\}$ —

$\mathbf{E}, \mathbf{F}, \mathbf{G}$ —

$\mathbf{K}$ —

κ_g —

χ —

Floor 32: Topology

$B(\mathbf{x}, r)$ —

τ —

$f^{-1}(V)$ —

$\pi_1(X,x_0)$ —

$*$ —

χ —

Floor 33: Integral Geometry (Advanced)

$V_i(K)$ —

$K + \varepsilon B$ —

$\chi(K)$ —

$C_i(K, \cdot)$ —

$\mu \star \nu$ (Alesker product) —

$\int \ldots dg$ —

Floor 34: Tensors & Manifolds

$V \otimes W$ —

$$\Lambda^k V^* \text{ (and } \wedge) \text{ —}$$

$$[X,Y] \text{ —}$$

$$d \text{ (and } d^2 = 0) \text{ —}$$

$$v_b / \omega_{\#} \text{ —}$$

b
 $\#$

$$\nabla_X Y \text{ —}$$

$$R(X,Y)Z \text{ —}$$

Floor 35: Functional Analysis

$$\rightarrow \text{ —}$$

$$\sigma(T) \text{ —}$$

$$X^* \text{ —}$$

$$\cong \text{ —}$$

$$\langle \cdot, \cdot \rangle \text{ —}$$

$$W^{\{k,p\}} \text{ —}$$

Floor 36: Measure Theory

$$\sigma\text{-algebra } (\Sigma \text{ or } \mathcal{A}) \text{ —}$$

$$m^*(A) \text{ —}$$

a.e. (almost everywhere) —

$\operatorname{ess\,sup} f$ —

$\int f \, d\mu$ —

$\nu \ll \mu$ and $d\nu/d\mu$ —

Floor 37: Abstract Algebra

$\cong$ —

$\langle g \rangle$ —

$H \trianglelefteq G$ —

G/H —

$\mathbb{Z}[i]$ —

$[L:K]$ —

$\operatorname{Gal}(L/K)$ —

Appendix — Further Reading

The Lower Floors — Arithmetic to Calculus (Floors 1–12)

The Middle Floors — Multivariable to Foundations (Floors 13–25)

The Upper Floors — Equations to Abstract Algebra (Floors 26–37)

The Penthouse — Category Theory to the Frontier (Floors 38–50)

If you read only one

Subject Index

[illegible]

This image shows a single sheet of white paper with horizontal blue lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper is slightly off-center against a black background.

Also in This Series

The Mathematics Tower, Volume 1: The Lower Floors

Floors 1–12: From Arithmetic to Calculus

The Mathematics Tower, Volume 2: The Middle Floors

Floors 13–25: From Multivariable Calculus to Set Theory & Logic

The Mathematics Tower, Volume 4: The Penthouse

Floors 38–50: From Category Theory to the Frontier